Solutions to: John B. Conway - Functions of One Complex Variable

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Introduction (Please Read)

This contains all of my solutions to all of the exercises in Conway's Complex Analysis book. Many of these problems were initially assigned as homework for a course I audited in Graduate Complex Analysis (MATH210A and MATH210B) at the University of California, Riverside, and I went back and solved the rest of them. This is not meant to be a "solution guide" to any of the problems, but rather as a way for admissions officers to see I'm not a total moron to be applying to graduate school in mathematics with a bachelor's in Business Administration. These solutions are **NOT** meant to be copied. There is a good chance that well over 90 percent of my solutions are incorrect. I think you will know soon enough that the introduction will not be the only time I'll forgo formalities. I'll use this introduction to be a way to just communicate with you before you inevitably disregard my warnings about copying down my answers. I can't control you :/. I'll come back to this every so often and see how writing the answers is going to go. This is actually my first time using a proper LaTeX editor for homework (not the first time I've done graduate mathematics though). I've been having formatting issues with the first question, which I can't bother to go back and fix... So it's clear to see that my formatting is improving over time. A few months into writing this paper I finally found out that $\backslash varnothing$ is a better than $\backslash varemptyset$ to represent the emptyset $\varnothing$.

1 Chapter 1: The Complex Number System

1.1 The real numbers

There are no problems here. Look at you! You already completed a whole subsection of the book with no effort :) Take a break, you earned it!

1.2 The field of complex numbers

Now, this is where the meat and potatoes are.

1. Find the real and imaginary parts of each of the following:

$$\tfrac{1}{z};\ \tfrac{z-a}{z+a}(a \in \mathbb{R});\ z^3;\ \tfrac{3+5i}{7i+1};\ (\tfrac{-1+i\sqrt{3}}{2})^3;$$
$$(\tfrac{-1-i\sqrt{3}}{2})^6;\ i^n;\ (\tfrac{1+i}{\sqrt{2}})^n \text{ for } 2 \leq n \leq 8$$

All solutions containing z as a free variable will assume that $z = x + iy$

1. $\frac{1}{z}$

   Recall: $\frac{1}{z} = \frac{\overline{z}}{|z|^2}$

   Thus, $\frac{1}{z} = \frac{x-iy}{x^2+y^2}$

   $\text{Re}(\frac{1}{z}) = \frac{x}{x^2+y^2}$, $\text{Im}(\frac{1}{z}) = -\frac{y}{x^2+y^2}$

2. $\frac{z-a}{z+a}(a \in \mathbb{R})$

   Multiply the numerator and the denominator by the conjugate of the denominator:

   $\frac{z-a}{z+a}=\frac{(x-a)+iy}{(x+a)+iy}\frac{(x+a)-iy}{(x+a)-iy}=\frac{(x-a)(x+a)-y^2+i((x+a)y-(x-a)y)}{(x+a)^2+y^2}$

   $\text{Re}(\frac{z-a}{z+a})=\frac{x^2-a^2-y^2}{(x+a)^2+y^2}$, $\text{Im}(\frac{z-a}{z+a})=\frac{2ay}{(x+a)^2+y^2}$

3. z^3

   Expand the cubic: $z^3 = (x + iy)^3 = x^3 + 3x^2iy - 3xy^2 - iy^3$

   $\text{Re}(z^3)=x^3 - 3xy^2$, $\text{Im}(z^3)=3x^2y - y^3$

4. $\frac{3+5i}{7i+1}$

Multiply the numerator and the denominator by the conjugate of the denominator:

$\frac{3+5i}{7i+1}\frac{7i-1}{7i-1}=\frac{(3+5i)(1-7i)}{50}$

$\text{Re}(\frac{3+5i}{7i+1})=\frac{19}{25}$, $\text{Im}(\frac{3+5i}{7i+1})=-\frac{8}{25}$

5. $(\frac{-1+i\sqrt{3}}{2})^3$

   While the "cis" function and De Moivre's theorem are not introduced in this chapter, you won't catch me dead expanding the cubic by hand

   $(\frac{-1+i\sqrt{3}}{2})^3=\text{cis}(\frac{2\pi}{3})^3=\text{cis}(2\pi)$

   $\text{Re}((\frac{-1+i\sqrt{3}}{2})^3)=1$, $\text{Im}((\frac{-1+i\sqrt{3}}{2})^3)=0$

   See how easy that was?

6. $(\frac{-1-i\sqrt{3}}{2})^6$

   $(\frac{-1-i\sqrt{3}}{2})^6=\text{cis}(-\frac{2\pi}{3})^6=\text{cis}(-4\pi)$

   $\text{Re}((\frac{-1-i\sqrt{3}}{2})^6)=1$, $\text{Im}((\frac{-1-i\sqrt{3}}{2})^6)=0$

7. i^n

   $i^n=\text{cis}(\frac{\pi}{2})^n=\text{cis}(\frac{n\pi}{2})$

   $\text{Re}(i^n)=\cos(\frac{n\pi}{2})$, $\text{Im}(i^n)=\sin(\frac{n\pi}{2})$

8. $(\frac{1+i}{\sqrt{2}})^n$

   $(\frac{1+i}{\sqrt{2}})^n=\text{cis}(\frac{\pi}{4})^n=\text{cis}(\frac{n\pi}{4})$

   $\text{Re}((\frac{1+i}{\sqrt{2}})^n))=\cos(\frac{n\pi}{4})$, $\text{Im}(i^n)=\sin(\frac{n\pi}{4})$

2. Find the absolute value and conjugate of each of the following.

$$-2+i;-3;(2+i)(4+3i);\frac{3-i}{\sqrt{2}+3i};\frac{i}{i+3};(1+i)^6;i^{17}$$

All solutions containing z as a free variable will assume that $z = x + iy$
By definition, for $z = x + iy$, then $\overline{z} = x - iy$ and $|z| = \sqrt{x^2 + y^2}$

1. $z = -2 + i$

   $\overline{z} = -2 - i$; $|z| = \sqrt{(-2)^2 + (1)^2} = \sqrt{5}$

2. $z = -3$

   $\overline{z} = -3$; $|z| = 3$

3. $z = (2 + i)(4 + 3i)$

   Evaluate the product

   $z = (2 + i)(4 + 3i) = 5 + 10i$

   $\overline{z} = 5 - 10i$; $|z| = \sqrt{(5)^2 + (10)^2} = 5\sqrt{5}$

4. $z = \frac{3-i}{\sqrt{2}+3i}$

   Use the quotient rule for the modulus and conjugate:

   $\overline{z} = \frac{3+i}{\sqrt{2}-3i}$; $|z| = \frac{|3-i|}{|\sqrt{2}+3i|} = \frac{\sqrt{10}}{\sqrt{11}}$

5. $z = \frac{i}{i+3}$

   Multiply the numerator and the denominator by the conjugate of the denominator

   $z = \frac{i}{i+3}\frac{i-3}{i-3} = \frac{1+3i}{10}$

   $\overline{z} = \frac{1-3i}{10}$; $|z| = \frac{|1+3i|}{|10|} = \frac{\sqrt{10}}{10}$

6. $z = (1 + i)^6$

   $z = (1 + i)^6 = \sqrt{2}^6 \text{cis}(\frac{\pi}{4})^6 = 8\text{cis}(\frac{3\pi}{2}) = -8i$

   $\overline{z} = 8\text{cis}(\frac{-3\pi}{2}) = 8i$; $|z| = 8$

7. $z = i^{17}$

$i^{17} = i^{4 \cdot 4+1} = i$

$\overline{z} = -i; |z| = 1$

3. Show that z is a real number if and only if $z = \overline{z}$

To show $z \in \mathbb{R} \iff z = \overline{z}$, show $z \in \mathbb{R} \Rightarrow z = \overline{z}$ and conversely; $z = \overline{z} \Rightarrow z \in \mathbb{R}$

Let $z = x + iy$ for $x, y \in \mathbb{R}$

If $z \in \mathbb{R}$, then for $z = x + iy \Rightarrow y = 0$ which means $z = x$ and $\overline{z} = x$ and therefore $z = \overline{z}$

If $z = \overline{z}$, then $x + iy = x - iy \Rightarrow y = -y$ meaning $y = 0$ and thus $z = x$ and z has no real component

Therefore $z \in \mathbb{R} \iff z = \overline{z}$ □

4. If z and w are complex numbers, prove the following equations:

$$|z + w|^2 = |z|^2 + 2\text{Re}z\overline{w} + |w|^2$$
$$|z - w|^2 = |z|^2 - 2\text{Re}z\overline{w} + |w|^2$$
$$|z - w|^2 + |z + w|^2 = 2(|z|^2 + |w|^2)$$

1. $|z + w|^2 = |z|^2 + 2\text{Re}z\overline{w} + |w|^2$

   Using theorem **2.1**

   $|z + w|^2 = (z + w)(\overline{z + w}) = z\overline{z} + z\overline{w} + w\overline{z} + w\overline{w}$

   Using theorem **2.2**

   $|z|^2 + z\overline{w} + w\overline{z} + |w|^2 = |z|^2 + 2\text{Re}z\overline{w} + |w|^2$ □

2. $|z - w|^2 = |z|^2 - 2\text{Re}z\overline{w} + |w|^2$

   Solution similar to problem 1

   $|z - w|^2 = (z - w)(\overline{z - w}) = z\overline{z} - z\overline{w} - w\overline{z} + w\overline{w}$

   $|z|^2 - z\overline{w} - w\overline{z} + |w|^2 = |z|^2 - 2\text{Re}z\overline{w} + |w|^2$ □

3. $|z-w|^2+|z+w|^2=2(|z|^2+|w|^2)$

   Use the results from proofs 1 and 2

   $|z+w|^2+|z-w|^2=|z|^2+2\mathrm{Re}z\overline{w}+|w|^2+|z|^2-2\mathrm{Re}z\overline{w}+|w|^2$

   $=2(|z|^2+|w|^2)$ □

Use induction to prove that for $z=z_1+...+z_n$**;** $w=w_1+w_2+...+w_n$**:** $|w|=|w_1|...|w_n|$**;** $\overline{z}=\overline{z_1}+...+\overline{z_n}$**;** $\overline{w}=\overline{w_1}\overline{w_2}...\overline{w_n}$

Use theorems **2.3** and **2.4**. All three proofs are very similar

$$\overline{(z+w)}=\overline{z}+\overline{w};\ \overline{(zw)}=\overline{z}\ \overline{w};\ |z||w|=|zw|$$

1. $\overline{z}=\overline{z_1}+...+\overline{z_n}$

$$\overline{z}=\sum_{k=1}^{n}\overline{z_k}$$

   Base case $n=1$
   $\overline{z_1}=\overline{z_1}$ trivially

   Inductive step: Assume the claim holds for some $n\geq 1$ Show it's true for $n+1$

$$\overline{z'}=\overline{\sum_{k=1}^{n}z_k+z_{n+1}}=\sum_{k=1}^{n}\overline{z_k}+\overline{z_{n+1}}=\sum_{k=1}^{n+1}\overline{z_k}$$

   Thus the statement is true for $n+1$ and by induction, holds for every n □

2. $|w|=|w_1||w_2|...|w_n|$

$$|w|=\prod_{k=1}^{n}|w_k|$$

   Base case $n=1$
   $|w_1|=|w_1|$ trivially

Inductive step: Assume the claim holds for some $n \geq 1$ Show it's true for $n+1$

$$|w|' = |\prod_{k=1}^{n} w_k||w_{n+1}| = \prod_{k=1}^{n} |w_k||w_{n+1}| = \prod_{k=1}^{n+1} |w_k|$$

Thus the statement is true for $n+1$ and by induction, holds for every n □

3. $\overline{w} = \overline{w_1 w_2}...\overline{w_n}$

$$\overline{w} = \prod_{k=1}^{n} \overline{w_k}$$

Base case $n = 1$
$\overline{w_1} = \overline{w_1}$ trivially

Inductive step: Assume the claim holds for some $n \geq 1$ Show it's true for $n+1$

$$\overline{w'} = \overline{\prod_{k=1}^{n} w_k \cdot w_{n+1}} = \prod_{k=1}^{n} \overline{w_k} \cdot \overline{w_{n+1}} = \prod_{k=1}^{n+1} \overline{w_k}$$

Thus the statement is true for $n+1$ and by induction, holds for every n □

Let $R(z)$ be a a rational function of z. Show that $\overline{R(z)} = R(\overline{z})$ if all the coefficients in $R(z)$ are real.

By definition

$$R(z) = \frac{P(z)}{Q(z)}, P(z) = \sum_{k=0}^{n} a_k z^k, Q(z) = \sum_{k=0}^{m} b_k z^k$$

where $a_k, b_k \in \mathbb{R}$ and thus $\overline{a_k} = a_k$

$$\overline{P(z)} = \overline{\sum_{k=0}^{n} a_k z^k} = \sum_{k=0}^{n} \overline{a_k z^k} = \sum_{k=0}^{n} a_k \overline{z^k} = \sum_{k=0}^{n} a_k \overline{z}^k = P(\overline{z})$$

$$\overline{R(z)} = \frac{\overline{P(z)}}{\overline{Q(z)}} = \frac{P(\overline{z})}{Q(\overline{z})} = R(\overline{z})$$

□

1.3 The complex plane

1. Prove (3.4) and give necessary and sufficient conditions for equality.

3.4 $||z| - |w|| \leq |z - w|$ for $z, w \in \mathbb{C}$

From a previous exercise, we proved the following

$|z - w|^2 = |z|^2 - 2\text{Re}(z\overline{w}) + |w|^2$

Theorem 3.2: $\text{Re}(z\overline{w}) \leq |z||w|$

$$|z - w|^2 \geq |z|^2 - 2|z||w| + |w|^2$$
$$|z - w|^2 \geq (|z| - |w|)^2$$
$$|z - w| \geq |z| - |w|$$

Sufficient conditions for equality occur $\iff \text{Re}(z\overline{w}) = |z||w|$

$$\iff z\overline{w} \in [0, \infty) \subset \mathbb{R}$$

□

2. Show that equality occurs in (3.3) if and only if $\frac{z_k}{z_l} \geq 0$ for any integers k and l, $1 \leq k$, $l \leq n$, for which $z_l \neq 0$.

3.3 States the following

$$|z_1 + z_2 + ... + z_n| \leq |z_1| + |z_2| + ... + |z_n|$$

The "vibe" behind the proof is as follows:

Equality holds when all of our numbers "point" in the same direction, or all their arguments are the same. Therefore, if $\frac{z_k}{z_l}$ is a

real number greater than or equal to zero, the $e^{i\theta}$ term must have canceled out; this is the guide to our proof.

Proof:
Recall that all $z \in \mathbb{C}$ can be rewritten as $re^{i\theta}$ for $r \geq 0$ and $\theta \in [0, 2\pi)$

$$\frac{z_k}{z_l} \in [0, \infty) \forall k, l \Rightarrow \frac{r_k e^{i\theta_k}}{r_l e^{i\theta_l}} = \frac{r_k e^{i(\theta_k - \theta_l)}}{r_1} \Rightarrow e^{i(\theta_k - \theta_l)} \in [0, \infty) \Rightarrow \theta_k - \theta_l = 0$$

Since r is always positive and $\theta_k = \theta_l$, then equality holds and we proved that

$$\frac{z_k}{z_l} \geq 0 \Rightarrow |z_1 + z_2 + ... + z_n| = |z_1| + |z_2| + ... + |z_n|$$

For the converse, assume equality holds. For the absolute value of the sum to equal the sum of the absolute values, all non-zero values must have the same argument. This means that for any two non-zero values, the ratio must be a positive real number. □

3. Let $a \in \mathbb{R}$ and $c > 0$ be fixed. Describe the set of points z satisfying

$$|z - a| - |z + a| = 2c$$

for every possible choice of a and c. Now let a be any complex number and, using a rotation of the plane, describe the locus of points satisfying the above equation.

If $z = x + iy$, then rewrite the equation as

$$\sqrt{(x-a)^2 + y^2} - \sqrt{(x+a)^2 + y^2} = 2c$$

After doing a CRAPTON of simplification and squaring, we get the following: (source: Trust me, bro)

$$\frac{x^2}{c^2} - \frac{y^2}{a^2 - c^2} = 1$$

A hyperbola for real numbers a and c. However, $|z-a|-|z+a| = 2c$ tells us that a must satisfy the following: $|z - a| > |z + a|$, thus z is closer to a than $-a$ and must be only one branch of a hyperbola.

For $a \in \mathbb{C}$:
let $a = |a|e^{i\theta}$ and define $w = ze^{-i\theta}$ and rewrite the equation:

$$|z-a| - |z+a| = 2c \;\Rightarrow\; \left|we^{i\theta} - |a|\, e^{i\theta}\right| - \left|we^{i\theta} + |a|\, e^{i\theta}\right| = 2c$$

Factor out $|e^{i\theta}|$

$$|w - |a|| - |w + |a|| = 2c$$

Which is the same as before in terms of w and a real number. Thus, we get a branch of a hyperbola but rotated by θ

□

1.4 Polar representation and roots of complex numbers

1. Find the sixth roots of unity.

The nth roots of unity can be described with the following:

$$z^n = 1 = e^{i2\pi k} \Rightarrow z = e^{\frac{i2\pi k}{n}}, k = 0, 1, 2, ..., n-1$$

for $n = 6$, $z = \text{cis}(\frac{\pi k}{3}))$ thus:

$$z = 1, \frac{1}{2} + \frac{\sqrt{3}}{2}i, -\frac{1}{2} + \frac{\sqrt{3}}{2}i, -1, -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \frac{1}{2} - \frac{\sqrt{3}}{2}i$$

□

2. Calculate the following

(a) The square roots of i

(b) The cube roots of i

(c) The square roots of $\sqrt{3} + 3i$

1. The square roots of i

$$z^2 = i = e^{i\frac{\pi}{2} + 2\pi k} \Rightarrow z = e^{i\frac{\pi}{4} + \pi k}$$

$$z = \tfrac{1}{\sqrt{2}}(1+i), -\tfrac{1}{\sqrt{2}}(1+i)$$

2. The cube roots of i

$$z^3 = i = e^{i\frac{\pi}{2} + 2\pi k} \Rightarrow z = e^{i\frac{\pi}{6} + \frac{2\pi}{3}k}$$

$$z = (\tfrac{\sqrt{3}}{2} + \tfrac{1}{2}i), \tfrac{-\sqrt{3}}{2} + \tfrac{1}{2}i, -i$$

3. The square roots of $\sqrt{3} + 3i$

$$z^2 = \sqrt{3}+3i = 2\sqrt{3}(\frac{1}{2}+\frac{\sqrt{3}}{2}i) = 2\sqrt{3}\cdot e^{i\frac{\pi}{3}+2\pi k} \Rightarrow z = \sqrt{(2\sqrt{3})}e^{i\frac{\pi}{6}+\pi k}$$

$$z = \sqrt{(2\sqrt{3})}(\tfrac{\sqrt{3}}{2} + \tfrac{1}{2}i), \sqrt{(2\sqrt{3})}(-\tfrac{\sqrt{3}}{2} - \tfrac{1}{2}i)$$

3. Show that if a and b are nth and mth roots of unity, respectively, then ab is a kth root of unity for some integer k. What is the smallest possible value of k?

Show that if

$$a^n = 1, b^m = 1, (n, m) \in \mathbb{N}$$

then ab is the kth root of unity for some integer k.

Choose $k := \mathbf{lcm}(n, m)$. Thus

$$k = nr = ms, (r, s) \in \mathbb{N}$$

$$a^k = a^{nr} = 1^r = 1 \quad b^k = a^{ms} = 1^s = 1$$

Proof $\mathbf{lcm}(n, m)$ is the smallest value of k
Let $\zeta_n := e^{\frac{2\pi i}{n}}$ and $\zeta_m := e^{\frac{2\pi i}{m}}$ and assume there exists k such that $(\zeta_n\zeta_m)^k = 1$. Then taking the arguments:

$$k(\frac{1}{n} + \frac{1}{m}) \equiv 0 \ \mathbf{mod}\ 1 \Rightarrow k(m + n) \equiv 0 \ (\mathbf{mod}\ nm)$$

Therefore $\mathbf{lcm}(n, m)$ must be the smallest value of k

If a and b are non-primitive roots of unity
Then we can similarly take the arguments again:

$$k(\frac{u}{n} + \frac{v}{m}) \equiv 0 \ \mathbf{mod}\ 1 \Rightarrow k(mu + nv) \equiv 0 \ (\mathbf{mod}\ nm)$$

$$k_{min} = \frac{nm}{\gcd(nm, mu + nv)}$$

□

4. Use the binomial equation

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

where

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

and compare the real and imaginary parts of each side of de Moivre's formula to obtain the formulas:

$$\cos n\theta = \cos^n \theta - \binom{n}{2} \cos^{n-2} \sin^2 \theta + \binom{n}{4} \cos^{n-4} \sin^2 \theta - ...$$

$$\sin n\theta = \binom{n}{1} \cos^{n-1} \sin \theta - \binom{n}{3} \cos^{n-3} \sin^3 \theta + ...$$

Proof:

Moivre's theorem states the following

$$(\cos\theta + i\sin\theta)^n = (e^{i\theta})^n = e^{i\theta n} = \cos n\theta + i\sin n\theta$$

Using the binomial theorem, we get the following.

$$\cos n\theta + i\sin n\theta = \sum_{k=0}^{n} \binom{n}{k} (\cos^{n-k}\theta)(i^k \sin^k \theta)$$

We can see that if k is even, the term of the sum is real, and if k is odd, the term is complex. Therefore, we can split the sum into two separate sums.

$$\cos n\theta + i\sin n\theta =$$
$$\sum_{k=0}^{n} \binom{n}{2k} (\cos^{n-2k}\theta)(\sin^k \theta)(-1)^k + i\sum_{k=1}^{n} \binom{n}{2k-1} (\cos^{n+1-2k}\theta)(\sin^{2k-1}\theta)$$

Splitting the sums into real and imaginary parts

$$\cos n\theta = \sum_{k=0}^{n} \binom{n}{2k} (\cos^{n-2k}\theta)(\sin^k \theta)(-1)^k$$

$$\sin n\theta = \sum_{k=1}^{n} \binom{n}{2k-1} (\cos^{n+1-2k}\theta)(\sin^{2k-1}\theta)$$

□

5. Let $z = \mathbf{cis}(\frac{2\pi}{n})$ for an integer $n \geq 2$ show that $1 + z + ...z^{n-1} = 0$

By definition if $z = \mathbf{cis}(\frac{2\pi}{n})$, then $z^n = 1$. The geometric series can be rewritten as the following for $z \neq 1$:

$$1 + z + ...z^{n-1} = \frac{z^n - 1}{z - 1} = \frac{1 - 1}{z - 1} = 0$$

n must also be greater than 1 because $z = \mathbf{cis}(\frac{2\pi}{1}) = 1$ □

6.Show that $\phi(t) = \mathbf{cis}(t)$ is a group homomorphism of the additive group $\mathbb{R}$ onto the multiplicative group $T = \{z : |z| = 1\}$

1. **The Homomorphism property**: $\phi(t_1 + t_2) = \phi(t_1)\phi(t_2)$for all $t_1, t_2 \in \mathbb{R}$
   Recall that $\text{cis}(t) = e^{it} = \cos(t) + i\sin(t)$
   $$\phi(t_1 + t_2) = e^{i(t_1+t_2)} = e^{it_1}e^{it_2} = \phi(t_1)\phi(t_2)$$
2. **Surjectivity**: Every element in T can be written as $\phi(t)$ for some $t \in \mathbb{R}$
   For some $z \in T$, take the argument of z, we take $t = \theta$ to get $z = e^{i\theta}$. This shows every element of T has a preimage under ϕ. □

1.5 Lines and half planes in the complex plane

1. Let C be the circle $\{z : |z - c| = r\}, r > 0$; let $a = c + r\mathbf{cis}(\alpha)$ and put

$$L_\beta = \left\{ z : \mathbf{Im}(\frac{z - a}{b}) = 0 \right\}$$

where $b = \mathbf{cis}(\beta)$. Find necessary and sufficient conditions in terms of β that L_β be tangent to C at a.

We are given a circle centered at c with radius r and a fixed point a on the circle. Our line must be orthogonal to the vector from c to a, which is described as α. Orthogonality in $\mathbb{C}$ is defined if $\text{Re}(u\overline{v}) = 0$. Recall that we can then parameterize L_β as all points z such that:

$$z = a + bt, \{-\infty \leq t \leq \infty\}$$

Thus, b is our direction vector and a is our initial point. Our line is at angle β. Our line must be orthogonal to the vector from c to a, which is described as α. Orthogonality in $\mathbb{C}$ is defined if $\mathrm{Re}(u\overline{v}) = 0$. So

$$\mathrm{Re}(\mathrm{cis}(\beta)\cdot r\mathrm{cis}(-\alpha)) = 0 \Rightarrow r\cos(\beta-\alpha) = 0 \Rightarrow \beta-\alpha = \frac{\pi}{2}+k\pi, k \in \mathbb{Z}$$

Therefore, accounting for symmetries of the line:

$$\beta \equiv \alpha + \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$$

□

1.6 The extended plane and its spherical representation

Give the details in the derivation of (6.7) and (6.8)

The theorems state the following:
(6.7)

$$d(z, z') = \frac{2|z - z'|}{\sqrt{[(1 + |z|^2)(1 + |z'|^2)]}}$$

(6.8)

$$d(z, \infty) = \frac{2}{\sqrt{(1 + |z|^2)}}$$

Derivation of (6.7):
Use theorem (6.5)

$$d(z, z')^2 = (x_1 - x_1')^2 + (x_2 - x_2')^2 + (x_3 - x_3')^2$$

Since Z, Z' lie on S, $x_1^2 + x_2^2 + x_3^2 = 1$, this simplifies to

$$d(z, z')^2 = 2 - 2(x_1x_1' + x_2x_2' + x_3x_3')$$

Substitute x_1, x_2, x_3 using the stereographic formulas.

$$d(z, z')^2 = 2-2(\frac{(z + \overline{z})(z' + \overline{z'}) - (z - \overline{z})(z' - \overline{z'}) + (|z|^2 - 1)(|z'|^2 - 1)}{(1 + |z|^2)(1 + |z'|^2)})$$

Expand and simplify the numerator.

$$d(z, z')^2 = 2 - 2(\frac{4\mathrm{Re}(z\overline{z'}) + |z|^2|z'|^2 - |z|^2 - |z'|^2 + 1}{(1 + |z|^2)(1 + |z'|^2)})$$

Combine terms and factor

$$d(z,z')^2 = \frac{4|z-z'|^2}{(1+|z|^2)(1+|z'|^2)}$$

Take the square root of both sides

$$d(z,z') = \frac{2|z-z'|}{\sqrt{(1+|z|^2)(1+|z'|^2)}}$$

□

Derivation of (6.8):
Use theorem Euclidean metric

$$d(z,\infty)^2 = (x_1-0)^2 + (x_2-0)^2 + (x_3-1)^2$$

Substitute x_1, x_2, x_3 using the stereographic formulas.

$$d(z,\infty)^2 = (\frac{2x}{|z|^2+1})^2 + (\frac{2y}{|z|^2+1})^2 + (\frac{|z|^2-1}{|z|^2+1}-1)^2$$

Simplify each term

$$d(z,\infty)^2 = \frac{4x^2+4y^2}{(|z|^2+1)^2} + \frac{4}{(|z|^2+1)^2} = \frac{4|z|^2+4}{(|z|^2+1)^2} = \frac{4}{|z|^2+1}$$

Take the square root of both sides

$$d(z,\infty) = \frac{2}{\sqrt{|z|^2+1}}$$

□

2. For each of the following points in $\mathbb{C}$, give the corresponding point of S

$$0, 1+i, 3+2i$$

These are elementary computations, so I'll just provide my solutions and the substitutions below

$$x_1 = \frac{z+\overline{z}}{|z|^2+1}, x_2 = \frac{z-\overline{z}}{i|z|^2+1}, x_3 = \frac{|z|^2-1}{|z|^2+1}$$

$$0 \mapsto (0,0,-1), 1+i \mapsto (\frac{2}{3},\frac{2}{3},\frac{1}{3}), 3+2i \mapsto (\frac{3}{7},\frac{2}{7},\frac{6}{7})$$

3. Which subsets of S correspond to the real/imaginary axes in $\mathbb{C}$?

(Again, a very easy problem)

Using the same substitutions

Real axis $(y = 0)$:

$$S_{\text{Re}} = \{(x_1, 0, x_3) \in S\}$$

Imaginary axis $(y = 0)$:

$$S_{\text{Re}} = \{(0, x_2, x_3) \in S\}$$

4. Let A be a circle lying in S. Then there is a unique plane P in $\mathbb{R}^3$ such that $P \cap S = A$. Recall from analytic geometry that

$$P = \{(x_1, x_2, x_3) : x_1\beta_1 + x_2\beta_2 + x_3\beta_3 = l\}$$

where $(\beta_1, \beta_2, \beta_3)$ is a vector orthogonal to P and l is some real number. It can be assumed that $\beta_1^2 + \beta_2^2 + \beta_3^2 = 1$. Use this information to show that if A contains the point N, then its projection on $\mathbb{C}$ is a straight line. otherwise, A projects onto a circle

Also, a very extremely easy problem (joking)

If A contains $N = (0, 0, 1)$, substituting gives

$$0 \cdot \beta_1 + 0 \cdot \beta_2 + 1 \cdot \beta_3 = l \Rightarrow \beta_3 = l$$

The plane equation becomes

$$(x_1 - 0)\beta_1 + (x_2 - 0)\beta_2 + (x_3 - 1)\beta_3 = 0$$

Performing substitution with the stereographic projection formulas and multiplying by $|z|^2 + 1$, we get the following

$$(z + \overline{z})\beta_1 + i(z - \overline{z})\beta_2 - 2\beta_3 = 0$$

Which is a linear equation in z, thus we get a straight line if $N \in A$ Similarly, if the plane passes through N, then the plane follows the stereographic projection. Then the projection of the plane P on the complex plane is the same as the intersection, we get a line. $\square$

If A does not contain N, we perform the stereographic substitution and simplify:

$$(\frac{z+\overline{z}}{|z|^2+1})\beta_1 + (\frac{z-\overline{z}}{i|z|^2+1})\beta_2 + (\frac{|z|^2-1}{|z|^2+1})\beta_3 = l$$

$$(z+\overline{z})\beta_1 - i(z-\overline{z})\beta_2 + (|z|^2-1)\beta_3 = |z|^2 l + l$$

Distribute and combine z terms

$$(\beta_1 - i\beta_2)z + (\beta_1 + i\beta_2)\overline{z} + (\beta_3 + l)|z|^2 = \beta_3 l$$

Let Z and Z' be points on S corresponding to z and z' respectively. Let W be the point on S corresponding to $z+z'$. Find the coordinates of W in terms of the coordinates of Z and Z'. Do the same exercise for multiplication.

For $w = z + z'$

Express z and z' in terms of Z and Z'

$$z = \frac{x_1 + ix_2}{1 - x_3}, \qquad z' = \frac{x_1' + ix_2'}{1 - x_3'}$$

If $W = (w_1, w_2, w_3)$ is the preimage of $z + z'$, then use projection substitutions as shown

$$w_1 = \frac{(z+z') + \overline{(z+z')}}{|(z+z')|^2+1}, w_2 = \frac{(z+z') - \overline{(z+z')}}{i|(z+z')|^2+1}, w_3 = \frac{|(z+z')|^2-1}{|(z+z')|^2+1}$$

Finally, we do one gnarly substitution

$$w_1 = \frac{(\frac{x_1+ix_2}{1-x_3} + \frac{x_1'+ix_2'}{1-x_3'}) + \overline{(\frac{x_1+ix_2}{1-x_3} + \frac{x_1'+ix_2'}{1-x_3'})}}{|(\frac{x_1+ix_2}{1-x_3} + \frac{x_1'+ix_2'}{1-x_3'})|^2+1}$$

$$w_2 = \frac{(\frac{x_1+ix_2}{1-x_3} + \frac{x_1'+ix_2'}{1-x_3'}) - \overline{(\frac{x_1+ix_2}{1-x_3} + \frac{x_1'+ix_2'}{1-x_3'})}}{i|(\frac{x_1+ix_2}{1-x_3} + \frac{x_1'+ix_2'}{1-x_3'})|^2+1}$$

$$w_3 = \frac{|(\frac{x_1+ix_2}{1-x_3} + \frac{x_1'+ix_2'}{1-x_3'})|^2-1}{|(\frac{x_1+ix_2}{1-x_3} + \frac{x_1'+ix_2'}{1-x_3'})|^2+1}$$

For $w = z \cdot z'$ The calculation is the same

$$w_1 = \frac{\left(\frac{x_1+ix_2}{1-x_3} \cdot \frac{x_1'+ix_2'}{1-x_3'}\right) + \overline{\left(\frac{x_1+ix_2}{1-x_3} \cdot \frac{x_1'+ix_2'}{1-x_3'}\right)}}{\left|\left(\frac{x_1+ix_2}{1-x_3} \cdot \frac{x_1'+ix_2'}{1-x_3'}\right)\right|^2 + 1}$$

$$w_2 = \frac{\left(\frac{x_1+ix_2}{1-x_3} \cdot \frac{x_1'+ix_2'}{1-x_3'}\right) - \overline{\left(\frac{x_1+ix_2}{1-x_3} \cdot \frac{x_1'+ix_2'}{1-x_3'}\right)}}{i\left|\left(\frac{x_1+ix_2}{1-x_3} \cdot \frac{x_1'+ix_2'}{1-x_3'}\right)\right|^2 + 1}$$

$$w_3 = \frac{\left|\left(\frac{x_1+ix_2}{1-x_3} \cdot \frac{x_1'+ix_2'}{1-x_3'}\right)\right|^2 - 1}{\left|\left(\frac{x_1+ix_2}{1-x_3} \cdot \frac{x_1'+ix_2'}{1-x_3'}\right)\right|^2 + 1}$$

2 Chapter 2: Metric Spaces and the Topology of $\mathbb{C}$

2.1 Definitions and examples of metric spaces

1. Show that each of the examples of metric spaces given in (1.2)-(1.6) is, indeed, a metric space. Example (1.6) is the only one likely to give any difficulty. Also, describe $B(x;r)$ for each of these examples. Many of the axioms are easy to show, and so any omitted axioms are trivial. The triangle inequality is the only hard one to prove.

For each function $d : X \times X \to [0, \infty)$ we must check the following

1. **Positivity:** $d(x, y) \geq 0$
2. **Symmetry:** $d(x, y) = d(y, x)$
3. **Zero Property:** $d(x, y) = 0 \iff x = y$
4. **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$

1.2 (Y, d) where $Y \subset \mathbb{R}$ and $d(x, y) = |x - y|$
All four axioms remain true by restriction.
Open Ball:

$$B_Y(x, r) = B_{\mathbb{R}}(x, r) \cap Y$$

1.3 $(\mathbb{C}, d)$ (Taxicab metric) where

$$d(x + iy, a + ib) = |x - a| + |y - b|$$

Rewrite and use the triangle inequality for real numbers. Define an intermediate point $v := u + ci$:

$$d(z, w) = d(x, a) + d(y, b)$$

$$d(z, w) \leq (d(x, u) + d(u, a)) + (d(y, c) + d(c, b))$$

$$d(z, w) \leq d(z, v) + d(v, w)$$

Open Ball:

$$B(z_0, r) = \{(x, y) : |x_0 - x| + |y_0 - y| < r\}$$

A diamond with radius r

1.4 $(\mathbb{C}, d)$ (Chebyshev metric) where

$$d(x+iy, a+ib) = max\{|x-a|, |y-b|\}$$

Rewrite

$$d(z,w) = max\{\mathrm{Re}(z-w), \mathrm{Im}(z-w)\}$$

$$d(z,w) = 0 \iff max\{\mathrm{Re}(z-w), \mathrm{Im}(z-w)\} = 0 \iff z = w$$

Use the triangle inequality for real numbers: Define an intermediate point $v := u + ci$

$$d(z,w) = max\{|x-a|, |y-b|\} \leq max\{|x-u| + |u-a|, |y-c| + |c-b|\}$$

$$\leq max\{|x-u|, |y-c|\} + max\{|u-a|, |c-b|\}$$

$$d(z,w) \leq d(z,v) + d(v,w)$$

Open Ball:

$$B(z_0, r) = \{(x,y) : max\{|x_0 - x|, |y_0 - y|\} < r\}$$

A square centered at z_0 with side length $2r$

1.5 (X, d)

$$d(x,y) = \begin{cases} 0, x = y \\ 1, x \neq y \end{cases}$$

Define an intermediate point v. If $d(x,y) = 0$, then triangle equality holds trivially. if $d(x,y) = 1 \iff x \neq y$ then $d(x,y) \leq d(x,v) + d(x,v)$ Because $d(x,v)$ and $d(x,v)$ can't both be zero because then $x = u = v$ which would result in a contradiction.

Open Ball:

$$B(x,r) = \begin{cases} X, r > 1 \\ x, r \leq 1 \end{cases}$$

1.6 $(\mathbb{R}^n, d)$ for $x = (x_1, ..., x_n), y = (y_1, ..., y_n)$ in $\mathbb{R}^n$

$$d(x,y)^2 = \sum_{j=1}^{n} (x_j - y_j)^2$$

$d(x,y) = 0 \iff (x_j - y_j) = 0$ for all $1 \leq j \leq n$ Prove the triangle inequality:

$$d(x,y)^2 + d(y,z)^2 = \sum_{j=1}^{n}(x_j - y_j)^2 + \sum_{j=1}^{n}(y_j - z_j)^2$$

Using Minkowski's Inequality:

$$\geq \sum_{j=1}^{n}(x_j - y_j + y_j - z_j)^2 = \sum_{j=1}^{n}(x_j - z_j)^2 = d(x,z)^2$$

Thus

$$d(x,z)^2 \leq d(x,y)^2 + d(y,z)^2$$

□

2. Which of the following subsets of $\mathbb{C}$ are open and which are closed?

(a) $\{z : |z| < 1\}$

(b) The real axis $\{\mathbb{R} \subset \mathbb{C}\}$

(c) $\{z : z^n = 1, n \in \mathbb{N}\}$ for some $n \geq 1$

(d) $\{z \in \mathbb{C} : z \text{ is real and } 0 \leq z < 1\}$

(e) $\{z \in \mathbb{C} : z \text{ is real and } 0 \leq z \leq 1\}$

Definitions:
A subset $X \subset \mathbb{C}$ is open if for every $z \in \mathbb{C}$ there exists $\varepsilon > 0$ such that

$$B(z,\varepsilon) = \{w : |w - z| < \varepsilon\} \subset X$$

A subset is closed if its complement is open or it contains its limit points

1. $X = \{z : |z| < 1\}$
   For $z \in X$ let $\varepsilon := \frac{1-|z|}{2} > 0$. Then

   $$B(z,\varepsilon) = \left\{w : |w - z| < \frac{1 - |z|}{2}\right\} \subset X$$

   Therefore, X **is an open set**.

2. $X = \{\mathbb{R} \subset \mathbb{C}\}$
   Consider the compliment: $X^C = \{\mathbb{C} - \mathbb{R}\}$. For $z \in X^C$, let $\varepsilon := \frac{|\text{Im}(z)|}{2}$ Then

$$B(z, \varepsilon) = \left\{ w : |w - z| < \frac{|\text{Im}(z)|}{2} \right\} \subset X$$

   Therefore X^C is open and thus X **is closed**

3. $X = \{z : z^n = 1, n \in \mathbb{N}\}$ for some $n \geq 1$
   Take $z_0 = 1 \in X$. For any $\varepsilon > 0$ the point $w = 1 + \frac{\varepsilon}{2}$ satisfies

$$|w - z_0| = \frac{\varepsilon}{2} \leq \varepsilon$$

   X **can't be open** because for any $\varepsilon > 0$, the point $w^n \notin X$

   Recall we can represent it as the following for some irreducible $\frac{p}{q} \in \mathbb{Q}$

$$z^q = \cos(2\pi p) + i \sin(2\pi p) = 1$$

   Now fix $x \in \mathbb{R} - \mathbb{Q}$ to be an irrational number such that $z^x \notin X$ and let $\{x_n\}$ be a rational sequence that converges to x. Choose some $k \in \mathbb{N}$ which means $z^k = 1$ and thus $\sin(2\pi kx) = 0$. Thus $kx \in \mathbb{Z}$ and we thus constructed a sequence in X that converges to a point not in X. Thus X is **not closed**. □

4. $X = \{z \in \mathbb{C} : z \text{ is real and } 0 \leq z < 1\}$
   The sequence $z_n = 1 - \frac{1}{n}$ approaches $1 \notin X$ so X **is not closed**.

   Every ball $B(z, \varepsilon)$ contains the point $z + i\frac{\varepsilon}{2} \notin X$. So **X is not open** either.

5. $X = \{z \in \mathbb{C} : z \text{ is real and } 0 \leq z \leq 1\}$
   Consider the compliment: $X^C = \{\mathbb{C} - [0, 1]\}$. For $z \in X^C$, let $\varepsilon := \frac{|\text{Im}(z)|}{2}$ Then

$$B(z, \varepsilon) = \left\{ w : |w - z| < \frac{|\text{Im}(z)|}{2} \right\} \subset X$$

   Therefore X^C is open and thus X **is closed**

3. If (X, d) is any metric space show that every open ball is, in fact, an open set. Also, show that every closed ball is a closed set.

We show for every $x_0 \in X$ and for some $r > 0$, then

$$B(x_0, r) = \{x \in X : d(x, x_0) < r\} \qquad \overline{B(x_0, r)} = \{x \in X : d(x, x_0) \leq r\}$$

Show $B(x_0, r)$ is open: Choose some $y \in B(x_0, r)$ then $d(x_0, y) < r$ then let

$$\varepsilon := r - d(x_0, y) > 0$$

For any $z \in B(x_0, r) \Rightarrow d(y, z) < \varepsilon$ then, using triangle inequality, we get the following

$$d(x_0, z) \leq d(x_0, y) + d(y, z) < d(x_0, y) + \varepsilon = d(x_0, y) + (r - d(x_0, y)) = r$$

Thus, $B(y, \varepsilon) \subset B(x_0, r)$ there is a neighborhood for every y. Therefore $B(x_0, r)$ is open.

Show $\overline{B(x_0, r)}$ is closed: Consider the compliment $\overline{B(x_0, r)}^C = X - \overline{B(x_0, r)}$ and take some $y \in \overline{B(x_0, r)}^C$ then $d(x_0, y) > r$ and let

$$\varepsilon := d(x_0, y) - r > 0$$

Using inverse triangle inequality we get

$$d(x_0, z) \geq d(x_0, y) - d(y, z) > d(x_0, y) - \varepsilon = d(x_0, y) - (d(x_0, y) - r) = r$$

Thus $d(x_0, z) > r \iff z \in \overline{B(x_0, r)}^C$. Meaning $\overline{B(x_0, r)}^C$ is open and thus $\overline{B(x_0, r)}$ is closed. □

4. Give the details of the proof of (1.9c)

1.9: If $\{G_j : j \in J\}$ is a collection of open sets in X, J any indexing set, then $G = \bigcup\{G_j : j \in J\}$ is also open. (where (X, d) is a metric space)

Choose some arbitrary $x \in G$. There must exists a $j_0 \in J$ such that $x \in G_{j_0}$
Since G_{j_0} is open, that means there exists $\varepsilon > 0$ such that

$$B(x, \varepsilon) := \{y \in X : d(x, y) < \varepsilon\} \subset G_{j_0}$$

Thus $B(x, \varepsilon) \subset G_{j_0} \subseteq G$. Since x was arbitrary, for every point $x \in G$, there exists an open ball contained in G. Therefore G is open. □

5. Prove Proposition 1.11

1.11 Proposition. Let (X, d) be a metric space. Then:

1. The sets X and $\varnothing$ are closed;
2. If $F_1, ..., F_n$ are closed sets in X, then so is:
$$\bigcup_{k=1}^{n} F_k$$
3. If $\{F_j : j \in J\}$ is a collection of closed sets in X, J any indexing set, then $F = \bigcap\{F_j : j \in J\}$ is also closed.

Proofs:

1. By definition, a set $F \subseteq X$ is closed if its compliment is open Proposition (1.9) states X and $\varnothing$ are both open. Thus their compliments, $\varnothing$ and X, are closed
2. Using De Morgan law
$$X\backslash(\bigcup_{k=1}^{n} F_k) = \bigcap_{k=1}^{n}(X\backslash F_k)$$
Thus, the compliment of the union of closed sets is open, thus $\bigcup_{k=1}^{n} F_k$ is closed
3. Let $\bigcap_{j\in J} F_j$ and consider the compliment.
$$X\backslash(\bigcap_{j\in J} F_j) = \bigcup_{j\in J}(X\backslash F_j)$$
Where each $X\backslash F_j$ is open and thus F is closed □

6. Prove that a set $G \subset X$ is op[en if and only if $X - G$ is closed

By definition of a closed set:
If G is open, then $X\backslash G$ is closed

Using the definition again:
If $X\backslash G$ is closed, then $X\backslash(X\backslash G) = G$ is open. □

7. Show that $(\mathbb{C}_\infty, d)$ where d is given by (1.6.7) and (1.6.8) is a metric space.

1.6.8:
$$d(z, z') = \frac{2|z - z'|}{\sqrt{(1 + |z|^2)(1 + |z'|^2)}}$$

1.6.8:
$$d(z, \infty) = \frac{2}{\sqrt{(1 + |z|^2)}}$$

1. **Positivity.**
   For all $z, z' \in \mathbb{C}$, the numerator for $d(z, z')$ is $|z - z'| \geq 0$ and equality holds if and only if $z = z'$ the denominator is always positive. If $z = \infty$ then $d(z, \infty) = 0$.

2. **Symmetry**
   Trivial to show
   The formula (1.6.7) is symmetric in z and z', and $d(z, \infty)$ is symmetric by definition.

3. **Triangle Inequality** (This is probably the first genuinely difficult one and it's a damn triangle inequality proof.) Show that
   $$d(z, z') \leq d(z, x) + d(x, z')$$
   Or
   $$\frac{2|z - z'|}{\sqrt{(1 + |z|^2)(1 + |z'|^2)}} \leq \frac{2|z - x|}{\sqrt{(1 + |z|^2)(1 + |x|^2)}} + \frac{2|x - z'|}{\sqrt{(1 + |x|^2)(1 + |z'|^2)}}$$
   Multiplying everything by $\sqrt{(1 + |z|^2)(1 + |z'|^2)(1 + |x|)}/2$ we must prove the following:
   $$|z - z'|\sqrt{1 + |x|^2} \leq |z - x|\sqrt{1 + |z'|^2} + |x - z'|\sqrt{1 + |z|^2}$$
   Start with a trivial derivation
   $$z - z' = z - z' \qquad z - z' = z - x + x - z'$$
   Multiply by $1 + |x|^2$
   $$(z - z')(1 + |x|^2) = (z - x)(1 + |x|^2) + (x - z')(1 + |x|^2)$$
   Rewrite $(z - x)(1 + |x|^2) + (x - z')(1 + |x|^2)$ as
   $(z - x)(1 + \overline{x}z') + (x - z')(1 + \overline{x}z)$
   $$(z - z')(1 + |x|^2) = (z - x)(1 + \overline{x}z') + (x - z')(1 + \overline{x}z)$$

Apply triangle inequality

$$|(z-z')(1+|x|^2)| \leq |(z-x)||(1+\overline{x}z)| + |(x-z')||(1+\overline{x}z)|$$

Use the following identities

$$|(1+\overline{x}z)| \leq \sqrt{1+|x|^2}\sqrt{1+|z|^2}$$

$$|(1+\overline{x}z')| \leq \sqrt{1+|x|^2}\sqrt{1+|z'|^2}$$

We get the following

$$|z-z'||1+|x|^2| \leq |z-x|\sqrt{1+|x|^2}\sqrt{1+|z|^2}+|x-z'|\sqrt{1+|x|^2}\sqrt{1+|z'|^2}$$

Divide everything by $\sqrt{1+|x|^2}$

$$|z-z'|\sqrt{1+|x|^2} \leq |z-x|\sqrt{1+|z'|^2} + |x-z'|\sqrt{1+|z|^2}$$

□

8. Let (X,d) be a metric space where $Y \subset X$. Suppose $G \subset X$ is open; show that $G \cap Y$ is open in (Y,d). Conversely, show that if $G_1 \subset Y$ is open in (Y,d), there is an open set $G \subset X$ such that $G_1 = G \cap Y$

Since G is open, then we have an $\varepsilon > 0$ that gives us an open ball in G such that:

$$B_Y(x_0,\varepsilon) = B_X(x_0,\varepsilon) \cap Y \subset (G \cap Y)$$

Set $G_1 = G \cap Y$ If $G \cap Y$ is open in (Y,d), then for every $x_0 \in G_1$, there exists $\varepsilon > 0$ such that:

$$B_Y(x_0,\varepsilon) = \{x \in G_1 : d(x,x_0) < \varepsilon\} \subseteq G_1$$

Thus, G_1 is open in Y.

Assuming the converse is true, Prove $G \cap Y = G_1$ by showing $(G \cap Y) \subseteq G_1$ and $(G \cap Y) \supseteq G_1$

If $z \in G \cap Y$, then $z \in Y \Rightarrow d(z,y) < \varepsilon_y \Rightarrow z \in G_1$ and thus $G \cap Y \subseteq G_1$
If $z \in G_1$ then $z \in B_X(z,\varepsilon_z) \subseteq G$ and $z \in Y \subseteq G$. Thus $z \in G_1 \Rightarrow z \in G \cap Y$

To prove the converse: (if $G_1 \subset Y$ is open in (Y, d) then there is a $G \subset X$ such that $G_1 = G \cap Y$) First, construct G:

$$G := \bigcup_{y \in G_1} B_X(y; \varepsilon_y)$$

For every $y \in G$ we choose $\varepsilon_y > 0$ such that $B_Y(y; \varepsilon_y) \subset G_1$
Since every $B_X(y; \varepsilon_y)$ is open in X; the union of the open sets is open, so G is open in X □

9. (Do exercise 8 with "closed" in place of "open") Let (X, d) be a metric space where $Y \subset X$. Suppose $G \subset X$ is closed; show that $G \cap Y$ is closed in (Y, d). Conversely, show that if $G_1 \subset Y$ is closed in (Y, d), there is a closed set $G \subset X$ such that $G_1 = G \cap Y$

Using the results from exercise 8. If G is closed in X, $X \backslash G$ is open in X and we get $(Y \backslash G) = (X \backslash G) \cap Y$ is open in Y. The compliment of $(X \backslash G) \cap Y$ is $G \cap Y$ is closed in (Y, d)

Since G_1 is closed in (Y, d), the compliment $Y \backslash G_1$ is open in (Y, d). Using results from exercise 8. There exists an open set $G_2 \subset X$ such that

$$Y \backslash G_1 = G_2 \cap Y$$

Let $G = X \backslash G_2$. Since G_2 is open in X, G is closed in X. Recall $G_1 = G \cap Y$ because

$$G \cap Y = (X \backslash G_2) \cap Y = Y \backslash (G_2 \cap Y) = Y \backslash (Y \backslash G_1) = G_1$$

□

Prove proposition 1.13

Let A and B be subsets of a metric space (X, d). Then:

(a) A is open if and only if $A = \text{int}A$

(b) A is closed if and only if $A = A^-$

(c) $\text{int}A = X - (X - A)^-; A^- = X - \text{int}(X - A); \partial A = A^- - \text{int}A$

(d) $(A \cup B)^- = A^- \cup B^-$

(e) $x_0 \in \text{int}A$ if and only if there is an $\varepsilon > 0$ such that $B(x, \varepsilon) \subset A$

(f) $x_0 \in A^-$ if and only if for every $\varepsilon > 0$, $B(x, \varepsilon) \cap A \neq \varnothing$

Definitions:
The interior of A, int$A$ is the union of all open sets in A

$$\text{int}A = \bigcup \{G : G \text{ is open and } G \subset A\}$$

The closure of A, is the intersection of all closed sets holding A

$$A^- = \bigcap \{F : F \text{ is closed and } F \supset A\}$$

The boundary of A, is denoted by ∂A and defined by $\partial A = A^- \cap (X - A)^-$

Proofs of 1.13

(a) A **is open if and only if** $A = \textbf{int}A$
By definition, $A \supseteq \text{int}A$
If A is open, then A is one of the open sets that defines int$A$. Thus $A \subseteq \text{int}A$ and therefore $A = \text{int}A$
Conversely, if $A = \text{int}A$, and int$A$ is the union of open sets, then A is open.

(b) A **is closed if and only if** $A = A^-$
By definition $A \subseteq A^-$ and thus $A = A^-$
If A is closed, it is one of the closed sets that defines A^-, so $A^- \subseteq A$
Conversely, if $A = A^-$, and A^- is the intersection of closed sets, then A is closed.

(c) $\textbf{int}A = X - (X - A)^-; A^- = X - \textbf{int}(X - A); \partial A = A^- - \textbf{int}A$
For some $x \in \text{int}A$, by definition, there exists an open set $x \in G \subseteq A$ and thus $G \cap (X - A) = \varnothing$. Therefore $x \notin (X - A)^-$.
$x \in X - (X - A)^- \iff x \in \text{int}A$

For some $x \in A^-$. By definition, x is in all of the closed sets that contain, A So $x \notin \text{int}(X - A)$. Thus, $A^- = X - \text{int}(X - A)$

By definition, if $z \in \partial A$, then $z \in A^-$ and $z \in (X - A)^-$ meaning $z \notin \text{int}A$ Thus $z \in \partial A \iff z \in A^- - \text{int}A$

(d) If $x \in (A \cup B)^-$, then every $B(x, \varepsilon)$ intersects $A \cup B$ and therefore $x \in A^-$ or $x \in B^-$. So, $x \in A^- \cup B^-$. Thus $(A \cup B)^- \subseteq A^- \cup B^-$. The converse is the exact same proof. Thus $(A \cup B)^- = A^- \cup B^-$.

(e) int$A$ is defined as the union of all open sets that are subsets of A. Thus any open ball centered at $x_0 \in \text{int}A$ is one of the open sets that defines int$A$. Conversely, if there is an open ball in A then there is a point x_0 in the ball in int$A$ Thus $x_0 \in$ **int**A **if and only if there is an** $\varepsilon > 0$ **such that** $B(x, \varepsilon) \subset A$

(f) if $x_0 \notin A^-$, then x_0 is in the open set $X - A^-$ and thus there is a ball that misses A hence if $x \in A^-$, then every ε ball must intersect A. If every $B(x_0, \varepsilon)$ intersects A, then x_0 can't exist in any open set disjoint from A since that would imply there is a small enough ε that removes the ball from A. So x_0 is in the closure of A and thus $x_0 \in A^-$ **if and only if for every** $\varepsilon > 0$**,** $B(x, \varepsilon) \cap A \neq \varnothing$

□(QED 6 times)

2.2 Connectedness

The purpose of this exercise is to show that a connected subset of $\mathbb{R}$ **is an interval.**

(a) Show that a set $A \subset \mathbb{R}$ is an interval if and only if for two points a and b in A with $a > b$, the interval $[a, b] \subset A$

Assume $A = (\alpha, \beta)$ is an interval with $\alpha < a < b < \beta$. By definition, for any $a, b \in A$ with $a < b$, then every x that satisfies $a < x < b$ is also in A.

Assume $A = [\alpha, \beta]$ is an interval with $\alpha \leq a < b \leq \beta$. Then every x that satisfies $a < x < b$ is also in A.

Assume $a, b \in A$ with $a < b, [a, b] \subset A$. If A is an interval, then A contains every point between any two points in A. For any x such that $a < x < b$ we have $x \in [a, b] \subseteq A$

(b) Use part (a) to show that if a set $A \subset \mathbb{R}$ is connected, then it is an interval.

Assume for contradiction that $A \subset \mathbb{R}$ is connected, but is not an interval. Then there must exist t such that $t \in [a, b] \backslash A$

By definition: A metric space is disconnected if it is union of two non-empty disjoint open sets. We define two sets as the following

$$A_1 := A \cap (-\infty, t), A_2 := A \cap (t, \infty)$$

Thus

$$A = A_1 \cup A_2$$

And thus A is not connected, a contradiction. Therefore if $A \subset \mathbb{R}$ then A is an interval. □

2. Show that the sets S and T in the proof of theorem 2.3 are open

$$S = \{s \in [0, 1] : sb + (1 - s)a \in A\}$$

$$T = \{t \in [0, 1] : tb + (1 - t)a \in B\}$$

S and T are paths in A that we can parameterize
Define a function $\phi : [0, 1] \to A$ such that

$$\phi(s) := sb + (1 + s)a, s \in [0, 1]$$

Since $A \subset \mathbb{C}$ is open and ϕ is a continuous path in A, the preimage $\phi^{-1}(A)$ is open in $\mathbb{R}$ and thus S is open.
The same argument can be made to show T is open. Thus S and T are open. □

Which of the following subsets X of $\mathbb{C}$ are connected; if X is not connected, what are its components?

(a)

$$X = \{z : |z| \leq 1\} \cup \{z : |z - 2| < 1\}$$

X is two disks of radius 1 with centers at $z = 0$ and $z = 2$ where they only at best meet where $z = 1$.
Define the two disks.

$$A = \{z : |z| \leq 1\}, B = \{z : |z - 2| < 1\}$$

To show that X is path connected, it's trivial to show that there exists a path between two points in A and two points in

B by using a straight line. The issue arrives where we have to connect a path between A and B. Consider the interval $[0, 2]$ that connects the centers of the circles together. Since $[0, 1] \subset A$ and $(1, 2] \subset B$ that means $[0, 1] \cup (1, 2] = [0, 2] \subset X$ and we can use it in the parameterization from any point in A to any point in B. Let $x \in A$, and $y \in B$. We can create a path from x to the center of A, walk it through to the center of B, and then back onto y with the following continuous map.

$$f(t) = \begin{cases} x(1-3t) & 0 \leq t \leq 1/3 \\ 2(3t-1) & 1/3 \leq t \leq 2/3 \\ y(3t-2) & 2/3 \leq t \leq 1 \end{cases}$$

Thus, X is path connected and thus connected.

(b)

$$X = [0, 1) \cup \{1 + \frac{1}{n} : n \geq 1\}$$

X is not connected. For some point $p_n = (1 + \frac{1}{n}) \in X$ consider $B(p_n, \varepsilon_n)$ where $\varepsilon_n := \frac{1}{2n(n+1)}$ and thus $B(p_n, \varepsilon_n)$ isolates each point p_n. Therefore each one of those points is an isolated point and X can't be connected. The components are $[0, 1) \cup \{\frac{3}{2}\} \cup \{\frac{4}{3}\} \cup \{\frac{5}{4}\} ... \cup \{1 + \frac{1}{n} : n \geq 1\}$

(c)

$$X = \mathbb{C} - (A \cup B)$$

Where $A = [0, \infty)$ and $B = \{z = r\text{cis}\theta : r = \theta, 0 \leq \theta \leq \infty\}$ X is not connected because it can be represented as a union of multiple disjoint open sets. The components are:

$$\{z = r\text{cis}\theta : .r < \theta, 0 \leq \theta \leq 2\pi\}$$

$$\{z = r\text{cis}\theta : \theta - 2\pi < r < \theta, 2\pi \leq \theta \leq 4\pi\}$$

$$\{z = r\text{cis}\theta : \theta - 2\pi k < r < \theta, 2\pi(k-1) \leq \theta \leq 2\pi k\}$$

Prove the following generalization of Lemma 2.6

If $\{D_j : j \in J\}$ is a collection of connected subset of X and if for each j and k in J, we have $D_j \cap D_k \neq \varnothing$ then $D = \bigcup\{D_j : j \in J\}$ is connected.

In essence, we need to prove that the union of connected sets, where the intersection between any two sets is nonempty, is connected. Assume for contradiction that D is not connected. Thus there exist nonempty sets $U, V \subset A$ such that

$$U \cup V = D \qquad U \cap V = \varnothing$$

Since every D_j is connected and $D_j \subset A$, we get the following

$$D_j = (U \cap D_j) \cup (V \cap D_j) \qquad (U \cap D_j) \cap (V \cap D_j) = \varnothing$$

$(U \cap D_j)$ or $(V \cap D_j)$ must be empty because this would contradict that D_j is connected, therefore D_j lies entirely within U or V. Put it in U without loss of generalization. The same can be said for D_k. Since $D_k \cap D_j \neq \varnothing$, D_k is also in U. Thus we can deduce that.

$$D = \bigcup_{j \in J} D_j \subset U \Rightarrow V = \varnothing$$

Which contradicts V being nonempty. Thus D is connected. □

5. Show that if $F \subset X$ is closed and connected, then for every pair of points a, b in F and each $\varepsilon > 0$ there are points $z_0, z_1, ..., z_n$ in F with $z_0 = a$, $z_n = b$ and $d(z_{k-1}, z_k) < \varepsilon$ for $1 \leq k \leq n$. Is the hypothesis that F can be closed needed? If F is a set which satisfies this property, then F is not necessarily connected, even if F is closed. Give an example to illustrate this.

Construct a set S_ε such that

$$S_\varepsilon = \{b \in F | \exists \{z_0, z_1, ..., z_n\} \subseteq F, z_0 = a, z_n = b, d(z_{k-1}, z_k) < \varepsilon\}$$

S_ε is the set of all points b that can be reached from a starting point a in n steps with length less than ε and the goal is to show that $S_\varepsilon = F$.

let $\{b_n\} \subset S_\varepsilon$ be a sequence that converges to $b \in \overline{S}_\varepsilon \cap F$. Thus for some ε_0, There exists some N such that for all $n > N$, $d(b_n, b) < \varepsilon_0$. If we choose $\varepsilon_0 = \varepsilon$ actually our definition for a point in S_ε, meaning that $b \in S_\varepsilon = \overline{S_\varepsilon}$. Thus S_ε is closed in F

S_ε is also open in F because for every $b \in S_\varepsilon$ there exists a z_k such that $d(z_k, b) < \varepsilon$ and thus we can construct a ball we can construct

an open ball $B(b, \varepsilon - d(z_k, b)) \subset F$. Since S_ε is closed and open, and F is a connected set, then $S_\varepsilon = \varnothing$ or $S_\varepsilon = F$. Trivially $a \in S_\varepsilon$. Therefore.

$$S_\varepsilon = F$$

$\square F$ does not need to be implied to be closed since closeness is later proven.

An example that proves that ε-chainable does not imply connectedness is the rationals. $\mathbb{Q} \subset \mathbb{R}$. It's disconnected because we can define disjoint, open, sets U, V such that:

$$U := (-\infty, \sqrt{2}) \cap \mathbb{Q} \qquad V := (\sqrt{2}, \infty) \cap \mathbb{Q} \qquad U \cup V = \mathbb{Q}$$

For any two points $a < b \in \mathbb{Q}$ and $\varepsilon > 0$. We can choose a finite $n \in \mathbb{N}$ such that:

$$n > \frac{b-a}{\varepsilon}$$

We can define the chain of rationals $z_0, z_1, ...z_n$ as the following

$$z_k = a + k\frac{b-a}{n}, \qquad k = 0, 1, ..., n$$

$$|z_{k-1} - z_k| = |a + (k-1)\frac{b-a}{n} - (a + k\frac{b-a}{n})| = \frac{b-a}{n} < \varepsilon$$

Therefore $\mathbb{Q}$ is ε-chainable but not connected. $\square$

2.3 Sequences and completeness

1. Prove proposition 3.4

3.4: (a) A set is closed if and only if it contains all it's limit points
(b) If $A \subset X$ then $\overline{A} = A \cup \{x : x \text{ is a limit point of } A\}$

(a) Assume for contradiction that $A \subset X$ is a closed set but does not contain all its limit points. That means there exists a sequence $\{x_n\}$ such that $\lim_{n\to\infty} x_n = x$ where $x \notin A$ and for some $\varepsilon > 0$ there exists N such that for all $n > N$ then $d(x_n, x) < \varepsilon$. Consider the compliment of the closed set, or the open set A^C. By definition of the compliment $x \in A^C$. Thus there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subset A^C$. However this is a contradiction because for every ε

there exist infinitely many $\{x_n\} \in A$ such that $d(x_n, x) < \varepsilon$. Therefore, X is not closed and thus, if a set contains all of its limit points, it's closed.

To prove the converse. Suppose $A \subset X$ contains all of its limit points. If we consider $x \in A^C$, x is not a limit point of A and there exists $B(x, \varepsilon) \subset A$, meaning A^C is open and thus A is closed. $\square$

(b) For simplicity, rewrite $A' := \{x : x \text{ is a limit point of } A\}$. Thus the proof becomes $\overline{A} = A' \cup A$.

Trivially $\overline{A} \supset A$. Since $\overline{A}$ contains A and is closed, it must contain all of its limit points ($\overline{A} \supset A'$) and thus $\overline{A} \supset A \cup A'$.

Next, we must show that $\overline{A} \subset A \cup A'$. Let $x \in \overline{A}$. Suppose $x \in A$, then we're done. Suppose $x \notin A$.Then we know that every open ball $B(x, \varepsilon)$ intersects A and by definition x is a limit point of A. Thus:

$$\overline{A} = A' \cup A$$

$\square$

2.Furnish the details of the proof of Corollary 3.8

Let (X, d) be a complete metric space and let $Y \subset X$. Suppose Y is closed. Show that (Y, d) is complete.

Let $\{y_n\}$ be a Cauchy sequence in Y. Since $Y \subset X$ then $\{y_n\}$ is a Cauchy sequence in the complete metric space (X, d). Thus, the sequence converges to some point in X.

$$\{y_n\} \to x \in X$$

Since Y is closed, it contains all of its limit points, and thus contains x. Thus $\{y_n\} \to x \in Y$. Therefore every Cauchy sequence in Y converges in Y. Thus Y is complete. $\square$

3. Show that diam A = diam $\overline{A}$

By definition:

$$\text{diam } A = \sup\{d(x, y) : x, y \in A\}$$

Since $A \subseteq \overline{A}$ then diam $A \leq$ diam $\overline{A}$

By definition of the closure, we can take $x, y \in \overline{A}$, and there exists

sequences $\{x_n\} \to x$ and $\{y_n\} \to y$ and thus:

$$\lim_{n\to\infty} d(x_n, y_n) = d(x, y)$$

For every n we get that $d(x, y) \leq \text{diam}A$ and taking the supremum for all $x, y \in \text{diam}A$ we get that

$$\text{diam } \overline{A} \leq \text{diam } A$$

Thus we can conclude that

$$\text{diam } A = \text{diam } \overline{A}$$

□

4. Let z_n, z be points in $\mathbb{C}$ and led d be the metric on $\mathbb{C}_\infty$. Show that $|z_n - z| \to 0$ if and only if $d(z_n, z) \to 0$. Also show that if $|z_n| \to \infty$ then $\{z_n\}$ is Cauchy in $\mathbb{C}_\infty$. (Must $\{z_n\}$ converge in $\mathbb{C}_\infty$?)

Assume that $|z_n - z| \to 0$, then

$$d(z_n, z) = \frac{2|z_n - z|}{\sqrt{(1 + |z_n|^2)(1 + |z|^2)}} \to 0$$

Since the denominator is strictly positive and ≥ 1 the convergence to zero holds.
To prove the converse: Let $d(z_n, z) \to 0$. Similarly, the denominator is ≥ 1 that means $2|z_n - z| \to 0$.

Let $|z_n| \to \infty$

For all ε, choose sufficiently large m, n such that $\frac{1}{|z_n|}, \frac{1}{|z_n|} < \frac{\varepsilon}{4}$ we have

$$d(z_n, z_m) = \frac{2|z_n - z_m|}{\sqrt{(1 + |z_n|^2)(1 + |z_m|^2)}} \leq \frac{2(|z_n| - |z_m|)}{|z_n||z_m|}$$

$$= 2(\frac{1}{|z_n|} + \frac{1}{|z_m|}) = \varepsilon$$

Therefore the sequence is Cauchy. Since $d(z_n, \infty) = \frac{2}{\sqrt{1+|z_n|^2}} \to 0, z_n \to \infty$ in $\mathbb{C}_\infty$. Then $\{z_n\}$ converges in $\mathbb{C}_\infty$

5. Show that every convergent sequence in (X, d) is a Cauchy sequence.

By definition, $\{x_n\}$ is a convergent sequence if for all $\varepsilon > 0$ there exists N such that for all $n > N$ then $d(x_n, x) < \varepsilon$.
Meaning there also exists M such that for all $m > M$ then $d(x_m, x) < \varepsilon/2$.

For any $\varepsilon > 0$ Choose N such that $n, m > N$ $d(x_m, x) < \varepsilon/2$ and $d(x_n, x) < \varepsilon/2$
By triangle inequality we can derive the following.

$$d(x_n, x_m) \leq d(x_n, x) + d(x_m, x) < \varepsilon/2 + \varepsilon/2 = \varepsilon$$

$$d(x_n, x_m) < \varepsilon$$

Thus for all $m, n > N$, then $d(x_n, x_m) < \varepsilon$ □

6. Give three examples of non complete metric spaces

1. $X = (0, 1)$ with metric $d(x, y) = |x - y|$. The sequence $\{\frac{1}{n}\}$ converges to 0 which is not in our space. Therefore (X, d) is a metric space that is incomplete
2. $X = \{(x, y) \in \mathbb{R}^2 : \sqrt{x^2 + y^2} < 1\}$ with metric $d(x_1, y_1, x_2, y_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. The sequence $\{(1 - \frac{1}{n}, 0)\}$ converges to $(1, 0)$ which is not in our space. Therefore (X, d) is a metric space that is incomplete.
3. $X = \mathbb{Q}$ with metric $d(x, y) = |x - y|$ The sequence

$$\{\sum_{k=1}^{n} \frac{1}{k^2}\}$$

converges to $\frac{\pi^2}{6}$ which is not in our space. Therefore (X, d) is a metric space that is incomplete.

7. Put a metric d on $\mathbb{R}$ such that $|x_n - x| \to 0$ if and only if $d(x_n, x) \to 0$ but that $\{x_n\}$ is a Cauchy sequence in $(\mathbb{R}, d)$ when $|x_n| \to \infty$ (Hint: Take inspiration from $\mathbb{C}_\infty$)

An incredibly boring answer would choose the metric to be

$$d(x_n, x_m) = \frac{|x_n - x_n|}{\sqrt{(1 + |x_m|^2)(1 + |x_n|^2)}}$$

The previous problems hold the justification □

8. Suppose $\{x_n\}$ is a Cauchy sequence and $\{x_{n_k}\}$ is a subsequence which is convergent. Show that $\{x_n\}$ is convergent.

To show that $\{x_n\}$ is convergent, then we have to show for every $\varepsilon > 0$, there is an N such that for all $n > N$, $d(x_n, x) < \varepsilon$
Since $\{x_{n_k}\}$ is convergent to some x, chose sufficiently large N such that for all $n > N$

$$d(x_{n_k}, x) < \frac{\varepsilon}{2}$$

Since $\{x_n\}$ is Cauchy, chose sufficiently large M such that for all $n, m > N$

$$d(x_n, x_m) < \frac{\varepsilon}{2}$$

Now for any $\varepsilon > 0$ choose $K = \max\{M, N\}$ such that for all $n, m > K$, By triangle inequality,

$$d(x_n, x,) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Thus $\{x_n\}$ Cauchy □

2.4 Compactness

1. Finish the proof of Proposition 4.4. (Prove the converse)
Suppose that every collection of closed sets $\mathcal{F}$ of K, with the f.i.p we have $\bigcap F_n \neq \varnothing$, then K is compact.

For contradiction, suppose K is non-compact. Thus, there exists an open cover $K \subset \mathcal{G} = \{G_\alpha : \alpha \in A\}$ such that there is no finite sub-cover cover $\mathcal{G}_i$. That being for any finite subset, of open sets in $\mathcal{G}$ then there exists $x \in K$ such that:

$$x \notin \bigcup_{i=1}^{n} G_i$$

Define an new family of open sets $\mathcal{F} := \{F_\alpha = (K - G_\alpha) : G \in \mathcal{G}\}$. Thus for our set of compliments, $x \in \bigcap F$. Thus $\mathcal{F}_\alpha \neq \varnothing$ has the f.i.p.

Our assumption says any closed-family with the f.i.p. has non-empty total intersection, so

$$\bigcap_{\alpha\in A} \mathcal{F}_\alpha \neq \varnothing$$

However

$$\bigcap_{\alpha\in A} \mathcal{F}_\alpha \ = \bigcap_{\alpha\in A} (K - G_\alpha) = K - \bigcup_{\alpha\in A} G_\alpha$$

But we established that $\bigcup G_\alpha$ is a cover of K:

$$K - \bigcup_{\alpha\in A} G_\alpha = K - K = \varnothing$$

A contradiction to $\bigcap_{\alpha\in A} \mathcal{F}_\alpha \neq \varnothing$. Thus, K is compact $\square$

2. Let $p = \{p_1, ..., p_n\}$ and $q = \{q_1, ..., q_n\}$ be points in $\mathbb{R}^n$ with $p_k < q_k$ for each k Let $R = [p_1, q_1] \times ... \times [p_n, q_n]$ and show that

$$\textbf{diam } R = d(p, q) = [\sum_{k=1}^{n} (q_k - p_k)^2]^{\frac{1}{2}}$$

In words, this is saying the "diagonal" of the box with sides orthogonal in $\mathbb{R}^n$ with p and q being opposite corners of the box, the longest distance between any two points in our box is also the distance between p and q.

By definition of the diameter and since R is compact, we have

$$\text{diam } R = \sup_{x\in R, y\in R} d(x, y) = \max_{x\in R, y\in R} d(x, y)$$

If we choose $x = \{x_1, ..., x_n\}$ and $y = \{y_1, ..., y_n\}$ to be points between p and q, we get for each k that:

$$p_k \leq x_k \leq q_k \qquad p_k \leq y_k \leq q_k$$

and

$$(y_k - x_k)^2 \leq (q_i - p_i)^2$$

by the above identity we can get the following

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (y_i - x_i)^2} \leq \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2} = d(p, q)$$

Equality holds is $x_i = p_i$ and $y_i = q_i$ for all i. Which is the maximum distance between x and y and thus

$$\text{diam } R = \max_{x \in R, y \in R} d(x, y) = d(p, q)$$

□

3. Let $F = [a_1, b_1] \times ... \times [a_n, b_n] \subset \mathbb{R}^n$ and let $\varepsilon > 0$; use exercise 2 to show that there are rectangle $R_1, ...R_m$ such that

$$F = \bigcup_{k=1}^{m} R_k$$

and diam $R_k < \varepsilon$ for each k. If $x_k \in F_k$ Then it follows that $R_k \subset B(x_k, \varepsilon)$

F is a big n-dimensional rectangle, then we can break it down into ε-small rectangles. For each side of the rectangle $[a_i, b_i]$ define $\ell_i := |a_i - b_i|$ and for every $\varepsilon > 0$ choose $N \in \mathbb{N}$ such that

$$N_i \geq \frac{\ell_i \cdot \sqrt{n}}{\varepsilon}$$

For each $i = 1, ...n$ divide the interval $[a_i, b_i]$ into N_i equal sub intervals each with length:

$$\delta_i = \frac{\ell_i}{N} \leq \frac{\ell_i}{\frac{\ell_i \cdot \sqrt{n}}{\varepsilon}} = \frac{\varepsilon}{\sqrt{n}}$$

Take every possible product of one sub interval from coordinate 1, coordinate 2, ... coordinate n.
Our m becomes:

$$m = N_1 ... N_n$$

For such products, label them each as $R_1 ... R_m$. And thus

$$F = \bigcup_{k=1}^{m} R_k$$

Since every $\ell_{R_k} < \frac{\varepsilon}{\sqrt{n}}$ we have the following distance between opposite corners

$$\text{diam } R < \sqrt{\sum_{i=1}^{n} (\frac{\varepsilon}{\sqrt{n}})^2} = \sqrt{\sum_{i=1}^{n} \frac{\varepsilon^2}{n}} = \sqrt{\frac{n\varepsilon^2}{n}} = \varepsilon$$

Since $\sup_{a,b \in R_k} d(a, b) < \varepsilon$ then for every $x_0 \in R_k$, we have $R_k \subset B(x_0, \varepsilon)$ □

4. Show that the union of a finite number of compact sets is compact Let $K = \bigcup_{i=1}^{n} K_i$ be the union of a finite number of compact sets. For each K_i that means for every open cover $\{G_i\}$, there exists a finite sub cover$\{G_{i,m}\}$. Thus:

$$K_i \subset \bigcup_{j=1}^{m} G_{i,j}$$

Thus, for each i, we have

$$K \subset \bigcup_{i=1}^{n} \bigcup_{j=1}^{m} G_{i,j} i$$

Which is a finite union of a finite union, and thus it's finite. Thus K is compact. □

5. Let X be the set of all bounded sequences of complex numbers. That is, $\{x_n\} \in X \iff \sup\{|x_n| : n \geq 1\} < \infty$. If $x = \{x_n\}$ and $y = \{y_n\}$, define $d(x, y) = \sup\{|x_n - y_n| : n \geq 1\}$ Show that for each $x \in X$ and $\varepsilon > 0$, $\overline{B}(x, \varepsilon)$ is not totally bounded although it is complete. By definition:

$$\overline{B}(x, \varepsilon) = \{y \in X : d(x, y) \leq \varepsilon\}$$

For every $x, y, z \in X$ we have $d(x, y) = d(x + z, y + z)$ meaning d is translation invariant. Thus for any x we have the translational mapping

$$T : X \to X,\ T_x(y) = y - x$$

And so we send $\overline{B}(x, \varepsilon) \to \overline{B}(0, \varepsilon)$

X is a Banach space, so it is complete. Since $\overline{B}(0, \varepsilon)$ is a closed subset of X, then it's also complete.

Next we construct an infinite set $\{k\}$ where $k_n = (0, 0, ..., 0, \varepsilon/2, ...)$ where $\varepsilon/2$ appears in every nth position and every remaining coordinate is 0. Every point lies in a ball since $|k_n| = \varepsilon/2 \leq \varepsilon$. However for two sequences in our set k_n, k_m where $n \neq m$ we get $d(k_n, k_m) = \sup |k_n - k_m| = \varepsilon/2$. Meaning for any $r < \varepsilon/4$ the metric can only contain at most one of the sequences. Thus it takes an infinite covering of balls of r-balls. Contradicting the "total boundedness" definition. Since our space is translation invariant. The

same applies to the ball

$$T_x^{-1}(\overline{B}(0,\varepsilon)) = \overline{B}(x,\varepsilon)$$

Therefore, the ball isn't totally bounded, but complete. □

6. Show that the closure of a totally bounded set is totally bounded.

If $A \subset X$ is totally bounded, there exists a finite set of points $\{x_1, ...x_n\} \subset X$ for every $\varepsilon > 0$ such that

$$A \subset \bigcup_{i=1}^{n} B(x_i, \frac{\varepsilon}{2})$$

Take any point $y \in \overline{A}$. By definition of the closure, for every $\varepsilon > 0$ then there is $a \in A$ such that $d(y,a) < \varepsilon/2$ Since a is also in A, it must exist in one of the balls $B(x_i, \varepsilon/2)$. Thus $d(a, x_i) < \varepsilon/2$. By triangle inequality

$$d(y, x_i) \leq d(y,a) + d(a, x_i) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

So $y \in B(x_i, \varepsilon)$ and therefore:

$$\overline{A} \subset \bigcup_{i=1}^{n} B(x_i, \varepsilon)$$

$\overline{A}$ is totally bounded. □

2.5 Continuity

1. Prove Proposition 5.2

5.2 Proposition Let $f : (X,d) \to (\Omega, \rho)$ be a function and $a \in X, \alpha = f(a)$ The following are equivalent statements:

1. f is continuous at a
2. For every $\varepsilon > 0, f^{-1}(B(\alpha, \varepsilon))$ contains a ball with center at a
3. $\alpha = \lim f(x_n)$ whenever $a = \lim x_n$

We have to show for each of the statements

$$(1) \iff (2) \iff (3)$$

$(1) \implies (2)$**:**
Statement 1, by definition, states:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } d(x,a) < \delta \implies \rho(f(x), f(a)) < \varepsilon$$

By definition $\rho(f(x), f(a)) < \varepsilon$ means $f(x) \in B(\alpha, \varepsilon)$. Thus:

$$B(a, \delta) \subset f^{-1}(B(\alpha, \varepsilon)$$

So, $f^{-1}(B(\alpha, \varepsilon)$ contains a ball centered at α

$(2) \implies (1)$**:**
Assume (2). Fix $\varepsilon > 0$. Then, $f^{-1}(B(\alpha, \varepsilon)$ contains the ball $B(a, \delta)$. Equivalently, for every x that satisfies $d(x,a) < \delta$ we have $\rho(f(x), \alpha) < \varepsilon$ Which is the ε-δ definition with continuity at a

$(1) \implies (3)$**:**
Let $\varepsilon > 0$, by continuity, there exists δ such that

$$d(x,a) < \delta \implies \rho(f(x), \alpha) < \varepsilon$$

Now suppose $x_n \to a$, meaning there exists N such that for all $n \geq N \implies d(x_n, a) < \delta$ Thus for for all $n \geq N \implies \rho(f(x), \alpha) < \varepsilon$ So, $\alpha = \lim f(x_n)$ whenever $a = \lim x_n$

$(3) \implies (1)$**:**
Prove the contrapositive. Suppose f is not continuous at a, then we can show that we can build a sequence $\{x_n\}$ such that

$$x_n \to a \text{ but } f(x_n) \not\to f(a) = \alpha$$

Non-continuity states that

$$\exists \varepsilon_0 > 0 \quad \forall \delta > 0 \quad \exists x \in X : d(x,a) < \delta \text{ and } \rho(f(x), \alpha) \geq \varepsilon_0$$

Choose $\delta_n := 1/n$. For every n, choose x_n such that

$$d(x_n, a) < \frac{1}{n} \quad \rho(f(x_n), \alpha) \geq \varepsilon_0$$

Thus

$$\lim_{n\to\infty} d(x_n, a) < \lim_{n\to\infty} \frac{1}{n} \implies x_n \to a$$

and f fails to converge to α
Therefore the statements are equivalent

$$(1) \iff (2) \iff (3)$$

□

2. Show that if f and g are uniformly continuous (Lipschitz) functions from X to $\mathbb{C}$ then so is $f + g$

let $h := f + g$ and we must show that h is uniformly continuous.
Since f is uniformly continuous, that means, for any $\varepsilon > 0$, there exists δ_1 such that $d(x, y) < \delta_1 \implies |f(x) - f(y)| < \varepsilon/2$
Similarly, since g is uniformly continuous, that means, for any $\varepsilon > 0$, there exists δ_2 such that $d(x, y) < \delta_2 \implies |g(x) - g(y)| < \varepsilon/2$
Choose δ to be

$$\delta := \min\{\delta_1, \delta_2\}$$

Thus, for any $\varepsilon > 0$ we have

$$|h(x) - h(y)| = |f(x) + g(x) - f(y) - g(y)|$$

By triangle inequality

$$\leq |f(x) - f(y)| + |g(x) - g(y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Therefore $f + g$ is also uniformly continuous □

3. We say that $f : X \to \mathbb{C}$ is bounded if there is a constant $M > 0$ with $|f(x)| \leq M$ for all x in X. Show that if f and g are bounded uniformly continuous (Lipschitz) functions from X into $\mathbb{C}$ then so it fg.

Since f and g are both bounded, there exist constants

$$M_f := \sup_{x\in X} |f(x)| < \infty \quad M_g := \sup_{x\in X} |g(x)| < \infty$$

Similar to the previous problem f is uniformly continuous, that means, for any $\varepsilon > 0$, there exists δ_g such that

$$d(x, y) < \delta_f \implies |f(x) - f(y)| < \frac{\varepsilon}{2M_g + 1}$$

Similarly, since g is uniformly continuous, that means, for any $\varepsilon > 0$, there exists δ_g such that

$$d(x,y) < \delta_g \implies |g(x) - g(y)| < \frac{\varepsilon}{2M_g + 1}$$

Choose δ to be

$$\delta := \min\{\delta_f, \delta_g\}$$

Thus, for any $\varepsilon > 0$ we have

$$\begin{aligned}
&|h(x) - h(y)| = |f(x)g(x) - f(y)g(y)| \\
&= |f(x)g(x) - f(x)g(y) + f(x)g(y) - f(y)g(y)| \\
&= |f(x)g(x) - f(x)g(y) + f(x)g(y) - f(y)g(y)| \\
&\leq |f(x)||g(x) - g(y)| + |g(y)||f(x) - f(y)| \\
&\leq M_f \frac{\varepsilon}{2M_f + 1} + M_g \frac{\varepsilon}{2M_g + 1} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon
\end{aligned}$$

Therefore fg is also uniformly continuous and bounded □

4. Is the composition of two uniformly continuous (Lipschitz) functions again uniformly continuous (Lipschitz)

Let us define two uniformly continuous functions where

$$f : (d_X, X) \to (d_Y, Y) \quad (d_Y, Y) \to (d_Z, Z)$$

The composition of two functions $h := g \circ f$ where

$$h : (d_X, X) \to (d_Z, Z)$$

Since f and g are both uniformly continuous, then there exist constants $M_1, M_2 > 0$ such that

$$d_Y(f(x), f(y)) \leq M_1 d_X(x,y) \qquad d_Z(g(x), g(y)) \leq M_2 d_Y(x,y)$$

Thus, we have

$$d_Z(g(f(x)), g(f(y))) \leq M_2 d_Y(f(x), f(y)) \leq M_1 M_2 d_X(x,y)$$

Choose $M_3 = M_1 \cdot M_2$

$$d_Z(h(x), h(y)) \leq M_3 d_X(x,y)$$

Thus, the composition of two uniformly continuous (Lipschitz) functions is uniformly continuous (Lipschitz). □

5. Suppose $f : X \to \Omega$ is uniformly continuous; show that if $\{x_n\}$ is a Cauchy sequence in X then $\{f(x_n)\}$ is a Cauchy sequence in Ω. Is this still true if we only assume that f is continuous? (Prove or give a counterexample)

Since $f : (X, d) \to (\Omega, \rho)$ is uniformly continuous, then for all $\varepsilon > 0$ there exists $\delta > 0$ such that

$$d(x, y) < \delta \implies \rho(f(x), f(y)) < \varepsilon$$

Since $\{x_n\}$ is a cauchy sequence, then for the same $\delta > 0$ there exists N such that for all $n, m > N$ we have

$$d(x_n, x_m) < \delta$$

Combining the inequalities, for all $n, m > N$ then

$$\rho(f(x), f(y)) < \varepsilon$$

Assuming only continuity is not strong enough. Let $X = (0, 1) \subset \mathbb{R}$ and $f : X \to \mathbb{R}$ where $f(x) = \frac{1}{x}$ which is a continuous function, but not uniformly continuous. Choose $\{x_n\} = \frac{1}{n}$ to be a Cauchy sequence.

$$d(f(x_n), f(x_m)) = d(n, m) = |m - n|$$

The distance can be made arbitrarily large and thus $f(x_n)$ is not Cauchy in $\mathbb{R}$ □

6. Recall the definition of a dense set (1.14). Suppose that Ω is a complete metric space and that $f : (D, d) \to (\Omega, \rho)$ is uniformly continuous, where D is dense in (X, d). Use Exercise 5 to show that there is a uniformly continuous function $g : X \to \Omega$ with $g(x) = f(x)$ for every x in D

(**1.14 Definition.** A subset A of a metric space X is *dense* if $\overline{A} = X$)

Exercise 5 states that uniform continuity carries Cauchy sequences to Cauchy sequences. Since D is dense, there is a sequence $\{x_n\} \subset D$ such that $\{x_n\} \to x$. $\{x_n\}$ is Cauchy in X and by the result in exercise 5, $\{f(x_n)\}$ is Cauchy in Ω. Since Ω is also complete, $(f(x_n))$ converges. Define a new function

$$g(x) := \lim_{n \to \infty} f(x_n) \in \Omega$$

To show that g is uniformly continuous, let $\varepsilon > 0$. Since f is uniformly continuous, f provides $\delta > 0$ such that

$$d(x, y) < \delta \implies \rho(f(x), f(y)) < \frac{\varepsilon}{2}$$

Choose sequences $\{x_n\}, \{y_n\} \subset D$ such that $x_n \to x, y_n \to y$ and since D is dense, $d(x_n, y_n) < \delta$ for every n. For each n

$$\rho(f(x_n), f(y_n)) < \frac{\varepsilon}{2}$$

Taking limits, we show that g is uniformly continuous.

$$\rho(g(x), g(y)) = \lim \rho(f(x_n), f(y_n)) \leq \limsup \rho(f(x_n), f(y_n)) < \frac{\varepsilon}{2} < \varepsilon$$

Confirming uniqueness, if $h : X \to \Omega$ is also uniformly continuous map with all the correct conditions then.

$$h(x) = \lim h(x_n) = \lim f(x_n) = g(x)$$

So $h = g$ and the equation is unique □

7. Let G be an open set of $\mathbb{C}$ and let P be a polygon in G from a to b Use theorems 5.15 and 5.17 to show that there is a polygon $Q \subset G$ from a to b which is composed of line segments which hare parallel to either the real or imaginary axes.

5.15. Theorem. Suppose $f : X \to \Omega$ is continuous and X is compact, then f is uniformly continuous

5.17 Theorem If A and B are disjoint sets in X with B closed and A compact, then $d(A, B) > 0$

Since $P \subset G$ that means that P and G^C are disjoint and thus we can define the following constant.

$$\varepsilon := d(P, G^C) > 0$$

P can be described as the image of a continuous path $f : [0, 1] \to \mathbb{C}$. Since $[0, 1]$ is compact, f is uniformly continuous and thus there exists $\delta > 0$ such that

$$|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon/2$$

We can partition $[0,1]$ into $0 = x_0 < x_1 < ... < x_N = 1$ such that $x_k - x_{k-1} < \delta$. Let $z_k := f(t_k)$. So every successive pair still satisfies $|z_k - z_{k-1}| < \varepsilon/2$.

For each k we make a path comprised of the horizontal and vertical paths. Define an intermediate point

$$h_k := (\mathrm{Re}(z_k), \mathrm{Im}(z_{k-1}))$$

h_k is horizontal from z_{k-1} and vertical from z_k and since the horizontal and vertical lengths are $\leq |z_k - z_{k-1}| < \varepsilon/2$ and thus every point in our L shape is within ε of P and the L lies inside of G. Choose Q to be the union of all the vertical and horizontal segments. $z_0 = f(0) = a$ and $z_N = f(1) = b$ so we still get a polygon from a to b in vertical and horizontal edges. □

8. Use Lebesgue's Covering Lemma (4.8) to give another proof of Theorem 5.15.

4.8 Lebesgue's Covering Lemma If (X, d) is sequential compact and $\mathscr{G}$ is an open cover of X then there is $\varepsilon > 0$ such that if x is in X, there is a set G in $\mathscr{G}$ with $B(x, \varepsilon) \subset G$

5.15. Theorem. Suppose $f : X \to \Omega$ is continuous and X is compact, then f is uniformly continuous

Fix $\varepsilon > 0$ For every $x \in X$ there exists δ_x such that

$$d(x, y) < \delta_x \implies \rho(f(x), f(y)) < \frac{\varepsilon}{2}$$

Define a ball $G_x := B(x, \delta_x)$ which is open in X. Thus for every $x \in X$ we construct a cover of X $\{G_x\}_{x \in X}$. Since X is compact and using the Lebesgue's Covering Lemma, there is a $\delta > 0$ such that for all $x \in X$, for some G_x

$$B(x, \delta) \subset G_x$$

For some $x, y \in X$ with $d(x, y) < \delta$ they both sit inside some ball with center x_0. By triangle inequality

$$\rho(f(x), f(y)) \leq \rho(f(x), f(x_0)) + \rho(f(x_0), f(y)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Thus

$$d(x, y) < \delta \implies \rho(f(x), f(y)) < \varepsilon$$

□

9. Prove the following converse to Exercise 2.5. Suppose (X, d) is a compact metric space having the property that for every $\varepsilon > 0$ and for any points a, b in X, there are points $z_0, z_1, ..., z_n$ in X with $z_0 = a$, and $z_n = b$ and $d(z_{k-1}, d_z) < \varepsilon$ for $1 \leq k \leq n$. Then (X, d) is connected. (Hint: Use Theorem 5.17.)

Prove the contrapositive. Assume X is not connected, then there exists two open sets A, B such that

$$A \cup B = X \quad A \cap B = \varnothing$$

Since B is open in X, then A is closed by $X \backslash B = A$. Similarly, since A is open, B is closed and thus A and B are also compact. By theorem 5.17 we get

$$d(A, B) = \inf\{d(a, b) : a \in A, b \in B\} > 0$$

Thus there exists $\varepsilon > 0$ such that $\varepsilon < d(A, B)$. Since $z_0, z_1, ..., z_n \subset X$, Set $z_0 \in A$ and $z_n \in B$.Then $d(z_{k-1}, z_k) < d(A, B)$ for all k. Thus there is no path. Meaning X must be connected. □

10. Let f and g continuous functions from (X, d) to (Ω, ρ) and let D be a dense subset of X. Prove that if $f(x) = g(x)$ for x in D then $f = g$ Use this to show that the function g obtained in Exercise 6 is unique.

Since D is dense in X, then for all $x \in X$ there exists $\{x_n\} \subset D$ such that

$$\lim_{n \to \infty} x_n = x$$

Since $f(x) = g(x)$ for $x \in D$ then $f(x_n) = g(x_n)$ for all n. Thus

$$\lim_{n \to \infty} f(x_n) = \lim_{n \to \infty} g(x_n)$$

Therefore $f(x) = g(x)$ for all $x \in X$ □

2.6 Uniform convergence

1. Let $\{f_n\}$ be a sequence of uniformly continuous functions from (X, d) to (Ω, ρ) and suppose that $f = u - \lim f_n$ exists. prove that f is uniformly continuous. If each f_n is a Lipschitz function with constant M_n and $\sup M_n < \infty$ show that f is a Lipschitz function. If $\sup M_n = \infty$ show that f may fail to be Lipschitz.

Fix $\varepsilon > 0$. Since $\{f_n\} \to f$ uniformly, choose N such that for all $n \geq N$ we have.

$$\sup_{\forall x \in X} \rho(f_N(x), f(x)) < \frac{\varepsilon}{3}$$

Thus there exists a $\delta > 0$ such that

$$d(x,y) < \delta \implies \rho(f_N(x), f_N(y) < \frac{\varepsilon}{3}$$

By triangle inequality we get

$$\rho(f(x), f(y) \leq \rho(f(x), f_N(x)) + \rho(f_N(x), f_N(y)) + \rho(f_N(y), f(y))$$
$$< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

Thus, f is uniformly continuous.

Now let f_n be Lipschitz for every n with a corresponding constant M_n. Meaning there exists $M := \sup\{M_n\}$ such that for all n

$$\rho(f_n(x), f_n(y)) < \varepsilon = M \cdot d(x,y)$$

Since f_n is also uniformly convergent, there exists N such that for all $n \geq N$ we have

$$\rho(f_n(x), f(x)) < \varepsilon = M \cdot d(x,y)$$

By triangle inequality

$$\rho(f(x), f(y)) \leq \rho(f(x), f_n(x)) + \rho(f_n(x), f_n(y)) + \rho(f_n(y), f(y))$$
$$< M \cdot d(x,y) + M \cdot d(x,y) + M \cdot d(x,y) = 3M \cdot d(x,y)$$

Thus, f is Lipschitz as well.

Assume that f is Lipschitz with some constant $M > 0$. We will derive a contradiction under the condition $\sup\{M_n\} = \infty$.
For any $\varepsilon > 0$ there exists N such that for all $n \geq N$ and for all $x \in X$

$$\rho(f_n(x), f(x)) < \varepsilon$$

Since f_n is Lipschitz with constant M_n, we have

$$\rho(f_n(x), f_n(y)) \leq M_n \cdot d(x,y)$$

Similarly, since f is Lipschits

$$\rho(f(x), f(y)) \leq M \cdot d(x,y)$$

By triangle inequality

$$\rho(f_n(x), f_n(y)) \leq \rho(f_n(x), f(x)) + \rho(f(x), f(y)) + \rho(f(y), f_n(y))$$

Thus

$$M_n \cdot d(x, y) \leq 2\varepsilon + M \cdot d(x, y)$$

Rehearing we get the following

$$(M_n - M) \cdot d(x, y) \leq \varepsilon$$

Since $\sup M_n = \infty$, for any M, there exists n such that $M_n > M + \frac{2\varepsilon}{d(x,y)}$ and thus

$$(M_n - M) \cdot d(x, y) > \varepsilon$$

which is a contradiction. Thus our assumption that f is Lipschitz fails.

3 Chapter 3: Elementary Properties and Examples of Analytic Functions

3.1 Power series

1. Prove proposition 1.5

1.5 Proposition Let $\sum a_n$ and $\sum b_n$ be two absolutely convergent series and put

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}$$

Then $\sum c_n$ is absolutely convergent with sum

$$(\sum a_n)(\sum b_n)$$

Define $A := \sum |a_n|$ and $B := \sum b_n$. Since $A, B < \infty$, then by triangle inequality

$$\sum_{n=0}^{\infty} c_n \leq \sum_{n=0}^{\infty} \sum_{k=0}^{n} |a_k||b_{n-k}|$$

By Fubini's theorem

$$= (\sum_{n=0}^{\infty} |a_n|)(\sum_{n=0}^{\infty} |b_n|) = AB < \infty$$

Since the sum is monotonically increasing and bounded above, it must be absolutely convergent.
To show $\sum c_n = (\sum a_n)(\sum b_n)$, rewrite

$$\sum_{n=0}^{\infty} \sum_{k=0}^{n} a_k b_{n-k} = (\sum_{n=0}^{\infty} a_n)(\sum_{n=0}^{\infty} b_n)$$

and show that

$$\sum_{n=0}^{\infty} \sum_{k=0}^{n} a_k b_{n-k} = \sum_{n=0}^{\infty} \sum_{m=0}^{n} a_k b_m$$

for some m. First, choose $m := n - k$. The map $(n, k) \mapsto (k, m)$ is a bijection from

$\{(n,k) : n \geq 0, 0 \leq k \leq n\}$ to $\mathbb{N} \times \mathbb{N}$ and hence the equality holds. Since each sum is absolutely convergent, by Fubini's theorem, since

$$\sum_{n=0}^{\infty}\sum_{m=0}^{n} a_k b_m = (\sum_{n=0}^{\infty} a_n)(\sum_{n=0}^{\infty} b_n)$$

□

2. Give the details of the proof of Proposition 1.6.

Recall $0 < s < r$, and thus for all $z \in \mathbb{C}$ with $|z - a| \leq s$

$$\sum_{n=0}^{\infty} |a_n + b_n||z-a|^n \leq \sum_{n=0}^{\infty} |a_n|s^n + |b_n|s^n < \infty$$

And thus the sums converge absolutely for s. Thus, by the previous supersize we can do the following

$$\sum_{n=0}^{\infty}(a_n + b_n)(z-a)^n = \sum_{n=0}^{\infty} a_n(z-a)^n + b_n(z-a)^n$$

$$= \sum_{n=0}^{\infty} a_n(z-a)^n + \sum_{n=0}^{\infty} b_n(z-a)^n$$

and

$$c_n = \sum_{n=0}^{\infty} a_k b_{n-k}$$

Thus

$$\sum_{n=0}^{\infty} c_n(z-a)^n = \sum_{n=0}^{\infty}(\sum_{k=0}^{n} a_k b_{n-k})(z-a)^n$$

$$= [\sum_{n=0}^{\infty} a_n(z-a)^n][\sum_{n=0}^{\infty} b_n(z-a)^n]$$

□

3. Prove that $\limsup(a_n + b_n) \leq \limsup(a_n) + \limsup(b_n)$ **and** $\liminf(a_n + b_n) \geq \liminf(a_n) + \liminf(b_n)$

By definition, if $A := \limsup(a_n)$ and $B := \limsup(b_n)$, then for all $\varepsilon > 0$ there exists N such that for all $n \geq N$ then,

$$a_n \leq \varepsilon + A \quad \text{and} \quad b_n \leq \varepsilon + B$$

Thus

$$a_n + b_n \leq A + B + 2\varepsilon$$

Taking the supremum for $n \geq N$

$$\sup(a_n + b_n) \leq A + B + 2\varepsilon$$

Taking the limit as $n \to \infty$

$$\limsup(a_n + b_n) = \lim_{n\to\infty}(\sup_{n\geq N}(a_n + b_n)) \leq A + B + 2\varepsilon$$

Since $\varepsilon > 0$ was arbitrarily small

$$\limsup(a_n + b_n) \leq A + B = \limsup(a_n) + \limsup(b_n)$$

Similarly, by definition, if $\alpha := \liminf(a_n)$ and $\beta := \liminf(b_n)$, then for all $\varepsilon > 0$ there exists N such that for all $n \geq N$ then,

$$a_n \geq \alpha - \varepsilon \quad \text{and} \quad b_n \geq \beta - \varepsilon$$

Thus

$$a_n + b_n \geq \alpha + \beta - 2\varepsilon$$

Taking the infimum for $n \geq N$

$$\inf(a_n + b_n) \geq \alpha + \beta - 2\varepsilon$$

Taking the limit as $n \to \infty$

$$\liminf(a_n + b_n) = \lim_{n\to\infty}(\inf_{n\geq N}(a_n + b_n)) \geq \alpha + \beta - 2\varepsilon$$

Since $\varepsilon > 0$ was arbitrarily small

$$\liminf(a_n + b_n) \geq \alpha + \beta = \liminf(a_n) + \liminf(b_n)$$

. □

4. Show that $\liminf a_n \leq \limsup a_n$ for any sequence in $\mathbb{R}$

Let $A := \limsup(a_n)$ and $\alpha := \liminf(a_n)$. By definition, for all $\varepsilon > 0$ there exists N_1 such that for all $n \geq N_1$ then,

$$a_n \leq \varepsilon + A$$

Similarly or all $\varepsilon > 0$ there exists N_2 such that for all $n \geq N_2$ then,

$$a_n \geq \alpha - \varepsilon$$

Choosing $N = \max\{N_1, N_2\}$, then for all $n \geq N$ and rearranging we get the following

$$\alpha - \varepsilon \leq a_n \leq A + \varepsilon$$

$$\alpha - \varepsilon \leq A + \varepsilon$$

Letting $\varepsilon \to 0$ we get

$$\liminf a_n \leq \limsup a_n$$

□

5. If $\{a_n\}$ is a convergent sequence in $\mathbb{R}$ and $a = \lim a_n$, show that $a = \liminf a_n = \limsup a_n$

Since $\{a_n\}$ is a convergent sequence, that means for every $\varepsilon > 0$ There exists N such that for all $n \geq N$ we have, $|a_n - a| < \varepsilon$ or

$$-\varepsilon < a_n - a < \varepsilon$$

And thus

$$a - \varepsilon < a_n < a + \varepsilon$$

Which is similar to the proof from before. Hence for every $n \geq N$

$$\liminf a_n \geq a - \varepsilon \quad \text{and} \quad \limsup \leq a + \varepsilon$$

Letting $\varepsilon \to 0$, and combining with the fact that $\liminf a_n \leq \limsup a_n$, equality holds an we get

$$a = \liminf a_n = \limsup a_n$$

□

Find the radius of convergence for each of the following power series: For all of the problems, we will use the following equations to determine radius of convergence.

$$\frac{1}{R} = \limsup |a_n|^{\frac{1}{n}} \quad \text{and} \quad R = \lim |\frac{a_n}{a_{n+1}}|$$

1.

$$\sum_{n=0}^{\infty} a^n z^n, a \in \mathbb{C}$$

$a_n = a^n$ for all n. Thus

$$R = \lim |\frac{a_n}{a_{n+1}}| = \lim |\frac{a^n}{a^{n+1}}| = \lim |\frac{1}{a}| = |\frac{1}{a}|$$

$$R = \begin{cases} \frac{1}{|a|} & a \neq 0 \\ \infty & a = 0 \end{cases}$$

2.

$$\sum_{n=0}^{\infty} a^{n^2} z^n, a \in \mathbb{C}$$

$a_n = a^{n^2}$ for all n. Thus

$$R = \lim |\frac{a_n}{a_{n+1}}| = \lim |\frac{a^{n^2}}{a^{(n+1)^2}}| = \lim |\frac{1}{a^{2n+1}}|$$

$$R = \begin{cases} 0 & |a| > 1 \\ 1 & |a| = 1 \\ \infty & |a| < 1 \end{cases}$$

3.

$$\sum_{n=0}^{\infty} k^n z^n, k \in \mathbb{Z}/\{0\}$$

$a_n = k^n$

$$\frac{1}{R} = \limsup |a_n|^{\frac{1}{n}} = \limsup |k^n|^{\frac{1}{n}} = \limsup |k|$$

Thus $R = \frac{1}{|k|}$

4.

$$\sum_{n=0}^{\infty} z^{n!}$$

Rewrite as $\sum_{m=1}^{\infty} a_m z^m$ where

$$a_m = \begin{cases} 1 & m = n! \text{ for some } n \in \mathbb{N} \\ 0 & m \neq n! \end{cases}$$

$$\frac{1}{R} = \limsup |a_m|^{\frac{1}{m}} = \limsup |1|^{\frac{1}{n!}} = 1$$

Thus $R = 1$

7. Show that the radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n} z^{n(n+1)}$$

is 1, and discuss convergence for $z = 1$, -1, and i.(Hint: the nth coefficient of this series is not $(-1)^n/n$)

Rewrite as $\sum_{m=1}^{\infty} a_m z^m$ where

$$a_n = \begin{cases} 1 & m = n(n+1) \text{ for some } n \in \mathbb{N} \\ 0 & m \neq n(n+1) \end{cases}$$

Thus we get (I will continue to rewrite m as n because consistently using m is a pain in the ass)

$$\frac{1}{R} = \limsup |a_n|^{\frac{1}{n}} = \limsup |\frac{(-1)^n}{n}|^{\frac{1}{n}} = \limsup |\frac{-1}{\sqrt[n]{n}}|$$

Using some calc 1 knowledge we know that $\sqrt[n]{n} \to 1$, is monotonically increasing, and always positive . Thus the reciprocal is monotonically decreasing for $n > 0$ and we get.

$$\frac{1}{R} = 1 \quad \text{and thus} \quad R = 1$$

Evaluating the sum at 1:
Trivially, letting $z = 1$ we get the sum

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

Which is the alternating harmonic series which converges to $-\ln 2$

Evaluating the sum at -1:
For all integers n, the product $n(n+1)$ is always even. Meaning that z is always raised to an even power and letting $z = -1$, we get the sum

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

Which is the same sum as before.

Evaluating the sum at i:
We check for every $n \mod 4$. If $n = 0 \mod 4, \quad (n = 4m)$
We get $n(n+1) = 4m(4m+1)$ so

$$i^{4m(4m+1)} = 1^{m(4m+1)} = 1$$

If $n = 1 \mod 4, \quad (n = 4m+1)$
We get $(4m+1)(4m+2) = 16m^2 + 12m + 2$ so

$$i^{16m^2+12m+2} = i^2 = -1$$

If $n = 2 \mod 4, \quad (n = 4m+2)$
We get $(4m+2)(4m+3) = 16m^2 + 20m + 6$ so

$$i^{16m^2+20m+6} = i^6 = -1$$

If $n = 3 \mod 4, \quad (n = 4m+3)$
We get $(4m+3)(4m+4) = 4(4m+3)(1m+1)$ so

$$i^{4(4m+3)(1m+1)} = 1^{(4m+3)(1m+1)} = 1$$

Thus our sum becomes

$$\sum_{n=1}^{\infty} \frac{1}{n} a_n$$

Where a_n is given as

$$a_n = \begin{cases} 1 & \text{for } n \mod 4 \in \{0, 3\} \\ -1 & \text{for } n \mod 4 \in \{1, 2\} \end{cases}$$

We can rewrite the sum as the following

$$\sum_{k=0}^{\infty} \left(\frac{1}{4k+1} - \frac{1}{4k+2} - \frac{1}{4k+3} + \frac{1}{4k+4}\right)$$

Which is actually the Taylor expansion of

$$\frac{\pi}{4} - \frac{\ln 2}{2}$$

□

3.2 Analytic functions

1. Show that $f(z) = |z|^2 = x^2 + y^2$ has a derivative only at the origin. By the Cauchy–Riemann equations let $u := x^2 + y^2$ and $v := 0$. Then we get

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial u}{\partial y} = 2y$$
$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0$$

Thus $2x = 0, \quad 2y = 0$ so f is only complex differentiable at the origin.

2. Prove that if b_n, a_n are real and positive and $0 < b = \lim b_n, a = \limsup a_n$ then ab=$\limsup(a_n b_n)$. Does this remain true if the requirement of positivity is dropped? Fix some $\varepsilon > 0$. By definition, there exists N such that for all $n > N$ we have

$$\sup(a_n b_n) \leq (b + \varepsilon) \sup ab$$

Since $a_n > 0$ then taking $n \to \infty$ we get that for every n

$$\limsup(a_n b_n) \leq (b + \varepsilon) \limsup a = (b + \varepsilon)a$$

Letting $\varepsilon \to 0$ we get

$$\limsup ab \leq ab$$

By definition of lim sup there exists subsequences $a_{n_j} \to a$ and $b_{n_j} \to b$. Thus

$$\limsup(a_n b_n) \geq \lim a_{n_j} b_{n_j} = ab$$

0 Thus $\limsup(a_n b_n) \geq ab$ and therefore $\limsup(a_n b_n) = ab$

Positivity is necessary because of the following counterexample. Let $a_n = (-1)^n$ and $b_n = -1$ $\limsup a_n = 1$ and $\lim b_n = -1$. But $\limsup a_n b_n = \limsup((-1)^n \cdot -1) = 1$ but $ab = -1$ Thus positivity is required.

Show that $\lim n^{(1/n)} = 1$.

Rewrite the limit as

$$\lim_{n \to \infty} e^{(1/n) \cdot \ln n}$$

Since e^x is continuous at $x = 0$, we get

$$e^{(1/n) \cdot \ln n} = e^{\lim_{n \to \infty} (1/n) \cdot \ln n} = e^0 = 1$$

Show that $(\cos z)' = -\sin z$ **and** $(\sin z)' = \cos z$ By the limit definition of the derivative, the derivative of $\cos z$

$$\lim_{h\to 0} \frac{\cos(z+h) - \cos(z)}{h} = \lim_{h\to 0} \frac{\cos(z)\cos(h) - \sin(z)\sin(h) - \cos(z)}{h}$$

$$\lim_{h\to 0} \frac{\cos(z)(\cos(h) - 1) - \sin(z)\sin(h)}{h}$$

Splitting the limits

$$\lim_{h\to 0} \frac{\cos(z)(\cos(h) - 1)}{h} - \lim_{h\to 0} \frac{\sin(z)\sin(h)}{h}$$

$$\cos(z) \lim_{h\to 0} \frac{(\cos(h) - 1)}{h} - \sin(z) \lim_{h\to 0} \frac{\sin(h)}{h}$$

Evaluating each limit

$$\cos(z) \cdot 0 - \sin(z) \cdot 1 = -\sin(z)$$

Similarly, by the limit definition of the derivative, the derivative of $\cos z$

$$\lim_{h\to 0} \frac{\sin(z+h) - \sin(z)}{h} = \lim_{h\to 0} \frac{\sin(z)\cos(h) + \cos(z)\sin(h) - \sin(z)}{h}$$

$$\lim_{h\to 0} \frac{\sin(z)(\cos(h) - 1) + \cos(z)\sin(h)}{h}$$

Splitting the limits

$$\lim_{h\to 0} \frac{\sin(z)(\cos(h) - 1)}{h} + \lim_{h\to 0} \frac{\cos(z)\sin(h)}{h}$$

$$\sin(z) \lim_{h\to 0} \frac{(\cos(h) - 1)}{h} + \cos(z) \lim_{h\to 0} \frac{\sin(h)}{h}$$

Evaluating each limit

$$\sin(z) \cdot 0 + \cos(z) \cdot 1 = \cos(z)$$

5. Derive formulas (2.14)

$$\textbf{2.14} \quad \cos z = \frac{e^{iz} + e^{-iz}}{2} \qquad \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Recall the series expansion for e^z

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

Evaluating the sum for e^{iz} and we get

$$e^{iz} = \sum_{n=0}^{\infty} \frac{(iz)^n}{n!}$$

Split the sum for each even and odd values of n

$$e^{iz} = \sum_{n \text{ is even}}^{\infty} \frac{(iz)^n}{n!} + \sum_{n \text{ is odd}}^{\infty} \frac{(iz)^n}{n!}$$

Fixing the indexes

$$e^{iz} = \sum_{m=0}^{\infty} \frac{(iz)^{2m}}{(2m)!} + \sum_{m=0}^{\infty} \frac{(iz)^{2m+1}}{(2m+1)!}$$

$$e^{iz} = \sum_{m=0}^{\infty} (-1)^m \frac{z^{2m}}{(2m)!} + i \sum_{m=0}^{\infty} (-1)^m \frac{z^{2m+1}}{(2m+1)!}$$

Which are the sums of $\sin z$ and $\cos z$ and we get

$$e^{iz} = \cos z + i \sin z$$

Similarly for the sum of e^{iz}

$$e^{-iz} = \sum_{m=0}^{\infty} (-1)^m \frac{(-z)^{2m}}{(2m)!} + i \sum_{m=0}^{\infty} (-1)^m \frac{(-z)^{2m+1}}{(2m+1)!}$$

$$e^{iz} = \sum_{m=0}^{\infty} (-1)^m \frac{z^{2m}}{(2m)!} - i \sum_{m=0}^{\infty} (-1)^m \frac{z^{2m+1}}{(2m+1)!}$$

And now we have the two following equations

$$e^{iz} = \cos z + i \sin z$$

$$e^{-iz} = \cos z - i \sin z$$

Adding and subtracting the equations we get

$$e^{iz} + e^{-iz} = 2 \cos z$$

$$e^{iz} - e^{-iz} = 2i \sin z$$

6. Describe the following sets

1. $S = \{z : e^z = i\}$
   Recall $i = e^{i\pi/2+2\pi k}, k \in \mathbb{Z}$. Thus
   $$S = \{z : e^z = e^{i(\pi/2+2\pi k)}\} = \{z : z = i(\pi/2 + 2\pi k), k \in \mathbb{Z}\}$$
2. $S = \{z : e^z = -1\}$
   $$S = \{z : e^z = e^{i(\pi+2\pi k)}\} = \{z : z = i(\pi + 2\pi k), k \in \mathbb{Z}\}$$
3. $S = \{z : e^z = -i\}$
   $$S = \{z : e^z = e^{i(3\pi/2+2\pi k)}\} = \{z : z = i(3\pi/2 + 2\pi k), k \in \mathbb{Z}\}$$
4. $S = \{z : \cos z = 0\}$
   $$S = \{z : \pi/2 + k\pi, k \in \mathbb{Z}\}$$
5. $S = \{z : \sin z = 0\}$
   $$S = \{z : k\pi, k \in \mathbb{Z}\}$$

7. Prove formulas for $\cos(z + w)$ and $\sin(z + w)$

Recall Euler's identity

$$e^{iz} = \cos z + i \sin z$$

Evaluating for $(z + w)$ we get two equations

$$e^{i(z+w)} = \cos(z + w) + i \sin(z + w)$$
$$e^{iz}e^{iw} = [\cos(z) + i \sin(z)][\cos(w) + i \sin(w)]$$

Thus

$$\cos(z + w) + i \sin(z + w) = [\cos(z) + i \sin(z)][\cos(w) + i \sin(w)]$$

Distributing and factoring out i on the right hand side

$$\cos(z + w) + i \sin(z + w) =$$
$$\cos z \cos w - \sin z \sin w + i(\sin z \cos w + \cos z \sin w)$$

Splitting each part into real and imaginary parts

$$\sin(z + w) = \sin z \cos w + \cos z \sin w$$
$$\cos(z + w) = \cos z \cos w - \sin z \sin$$

8. Define $\tan z = \frac{\sin z}{\cos z}$ Where is this function defined and analytic?

Since sin and cos are both complex analytic, then the only issues arise when $\cos z = 0$ Thus $\tan z$ is analytic in

$$\mathbb{C} - \{\frac{\pi}{2} + k\pi, k \in \mathbb{Z}\}$$

9. Suppose that $z_n, z \in G = \mathbb{C} - \{z : z \leq 0\}$ and $z_n = r_n e^{i\theta_n}, z = re^{i\theta}$ where $-\pi < \theta, \theta_n < \pi$. Prove that if $z_n \to z$, then $\theta_n \to \theta$ and $r_n \to r$

Fix some $\varepsilon > 0$. If $z_n \to z$ then there exists N such that for all $n \geq N$, $|z_n - z| < \varepsilon$. Recall that $r_n = |z_n|$. By inverse triangle inequality we get

$$||z_n| - |z|| \leq |z_n - z| < \varepsilon$$

Thus for any fixed ε we already chose N such that for all $n \geq N$ we have $|r_n - r| < \varepsilon$ and thus $r_n \to r$. Since $r_n, r > 0$, we can also conclude

$$\frac{z_n}{r_n} \to \frac{z}{r}$$

and thus

$$e^{i\theta_n} \to e^{i\theta}$$

Since $-\pi < \theta, \theta_n < \pi$, the map $\theta \to e^{i\theta}$ is a homomorphism into $\mathbb{S}^1 \backslash \{-1\}$ and thus

$$e^{i\theta_n} \to e^{i\theta} \implies \theta_n \to \theta$$

. □

10. Prove the following generalization of Proposition 2.20. Let G and Ω be open in $\mathbb{C}$, G connected, and suppose f and h are functions defined on G, $g : \Omega \to \mathbb{C}$ and Suppose that $f(G) \subset \Omega$. Suppose that g and h are analytic, that f is continuous, h is one-to-one, and that they satisfy $h(z) = g(f(z))$ for z in G. Show that f is analytic. Give the formula for $f'(z)$

Since h is one-to-one. We know that $h'(z) \neq 0$ for all $z \in G$. So, $g'(f(z)) \cdot f'(z) \neq 0$ and thus $g'(\omega) \neq 0$ for all $\omega \in f(G)$ meaning g is also one-to-one. Since g is analytic and one-to-one, this means

g^{-1} exists and is also analytic and we can create the following composition

$$f(z) = g^{-1}(h(z))$$

Since f is a composition of analytic functions, f must also be analytic.

Formula for $f'(z)$

Recall the chain rule states that if $h(z) = g(f(z))$ then

$$h'(z) = g'(f(z)) \cdot f'(z)$$

Thus

$$f'(z) = \frac{h'(z)}{g'(f(z))}$$

11. Suppose that $f : G \to \mathbb{C}$ is a branch of the logarithm and that n is an integer. Prove that $z^n = \exp(nf(z))$ for all z in G

Recall that by definition of the logarithm

$$\exp(f(z)) = z$$

and properties of the exponential

$$\exp(nz) = \exp(z)^n$$

Hence for all $z \in G$

$$\exp(nf(z))) = \exp(f(n))^n = z^n$$

for all $n \in \mathbb{Z}$

12. Show that the real part of the function $z^{\frac{1}{2}}$ is always positive

Take the principal branch of the logarithm $G = \mathbb{C}\backslash(\infty, 0]$ where $\arg(z) \in (-\pi, \pi)$. For all $z \in G$ there exists $r \in \mathbb{R}, r > 0$ and $\theta = \arg(z)$ such that

$$z = re^{i\theta}$$

The principal square root is defined as

$$z^{\frac{1}{2}} = \exp(\frac{1}{2}\text{Log}z)$$

Thus for $z^{\frac{1}{2}}$ we get

$$z^{\frac{1}{2}} = (re^{i\theta})^{\frac{1}{2}} = r^{\frac{1}{2}} e^{\frac{1}{2}i\theta}$$

Since $r > 0$ we just have to show that the real part of $e^{\frac{1}{2}i\theta}$ is also positive By the Cauchy Identity,

$$e^{\frac{1}{2}i\theta} = \cos(\theta/2) + i\sin(\theta/2)$$

Thus for $\theta \in (-\pi, \pi)$ we get

$$\mathrm{Re}(z^{\frac{1}{2}}) > 0$$

□

13. Let $G = \mathbb{C} - \{z : z \leq 0\}$ and let n be a positive integer. Find all analytic functions $f : G \to \mathbb{C}$ such that $z = (f(z))^n$ for all $z \in G$

Let $\mathrm{Log}(z)$ be the principle branch. Then $\log(z) = \mathrm{Log}(z) + 2\pi ki$ for some $k \in \mathbb{Z}$. Thus

$$f(z) = z^{\frac{1}{n}} = e^{\frac{1}{n}\log z} = e^{\frac{1}{n}(\mathrm{Log}(z)+2\pi ki))} = e^{\frac{1}{n}\mathrm{Log}(z)} e^{\frac{2\pi ki}{n}}$$

Notice that $e^{\frac{2\pi ki}{n}}$ are the nth roots of unity for some value of $k = 0, 1, ..., n-1$. So

$$f(z) = e^{\frac{1}{n}\mathrm{Log}(z)} e^{\frac{2\pi ki}{n}}$$

14. Suppose $f : G \to \mathbb{C}$ is analytic and that G is connected. Show that if $f(z)$ is real for all z in G then f is constant

Since f is analytic and G is connected, f is differentiable , and thus it satisfies the Cauchy–Riemann equations, thus if $f(z) = u(x, y) + iv(x, y)$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

But since f is real, then $v(z) \equiv 0$ So we have

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 0 \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = 0$$

So $f'(z) = 0$ meaning $f(z)$ is constant for all $z \in G$

15. For $r > 0$ let

$$A = \{\omega : \omega = \exp(\frac{1}{z}) \text{ where } 0 < |z| < r\}$$

determine the set A

Trivially $\omega = e^{\frac{1}{z}} \neq 0$. Thus the image of $A \subseteq \mathbb{C} - \{0\}$. Next fix $\omega \in \mathbb{C} - \{0\}$. For $z = x + iy$, choose $x = \log w$ and $y = \arg(w) + 2k\pi$. Choose large enough k such that $x^2 + y^2 > \frac{1}{r}$ And we found $z \in A$ thus $\mathbb{C} - \{0\} \subseteq A$. Therefore $\mathbb{C} - \{0\} = A$

16. Find an open connected set $G \subset \mathbb{C}$ and two continuous functions f and g defined on G such that $f(z)^2 = g(z)^2 = 1 - z^2$ for all $z \in G$. Can you make G maximal? Are f and g analytic

Let $g = -f$. If $F(z) = 1 - z^2 = 0$ then $z = -1, 1$. Thus we choose

$$G = \mathbb{C} - ((-\infty, -1] \cup [1, \infty))$$

Removing two principle branches of the log and G is simply connected. We also know $F = (1 - z)(1 + z)$ and thus

$$\text{Log } F(z) = \log(1 + z) + \log(1 - z)$$

Which is analytic on G. So now we can define $f(z)$ as

$$f(z) = \exp(\frac{1}{2}\text{Log } (1 - z^2))$$

Which is a composition of analytic functions and is thus analytic in G.
Suppose $G' \supsetneq G$ is open and connected. Then G' contains some point $p \in (-\infty, -1] \cup [1, \infty)$. Draw a loop $\gamma \subset G'$ that crosses one of the rays through p that winds once around either -1 or 1. Which contradicts means that p makes G not maximal and thus it is.

17. Give the principal branch of $\sqrt{1 - z}$

The principal branch is obtained by composing the principal log with $1 - z$

$$\sqrt{1 - z} = \exp(\frac{1}{2}\text{Log } (1 - z))$$

Thus the preimage is $\mathbb{C} - [1, \infty)$ and we can also derive

$$\sqrt{|1 - z|}\exp(\frac{1}{2}i\arg(1 - z)) = \sqrt{|1 - z|} \cdot e^{\frac{1}{2}i\theta}$$

18. Let $f : G \to \mathbb{C}$ and $g : G \to \mathbb{C}$ be branches of z^a and z^a respectively. Show that fg is a brach of z^{a+b} and f/g is a brach of z^{a-b}. Suppose that $f(G) \subset G$ and $g(G) \subset G$ and prove that both $f \circ g$ and $g \circ f$ are branches of z^{ab}

If f is a branch of z^a then there exists a branch of the logarithm on G such that

$$f(z) = \exp(a \text{ Log } z)$$

Thus we can derive the following properties

$$f(z) \cdot g(z) = \exp(a \text{ Log } z) \exp(b \text{ Log } z)$$

By exponent properties

$$\exp(a \text{ Log } z) \exp(b \text{ Log } z) = \exp((a + b) \text{ Log } z)$$

Meaning that fg is a brach of z^{a+b}. Similarly we can show that

$$f(z)/g(z) = \frac{\exp(a \text{ Log } z)}{\exp(b \text{ Log } z)} = \exp(a \text{ Log } z) \exp(b \text{ Log } z)^{-1}$$

Again, by exponent properties

$$\exp(a \text{ Log } z) \exp(b \text{ Log } z)^{-1} = \exp(a \text{ Log } z) \exp(-b \text{ Log } z)$$

$$= \exp((a - b) \text{ Log } z)$$

Meaning that f/g is a brach of z^{a-b}.
Now to show that $f \circ g$ is a branch of z^{ab}, there exists a branch such that

$$f(g(z)) = \exp(a \text{ Log } (\exp(b \text{ Log } z)))$$

Properties of logarithms tells us that $\text{Log } (\exp(z)) = z$. Thus we get

$$\exp(ab \text{ Log } z)$$

Thus $f \circ g$ is a brach of z^{ab}. Since the order was arbitrary and $z^{ab} = z^{ba}$, then $f \circ g = g \circ f$ and the proof is similar.

19. Let G be a region and define $G^* = \{z : \overline{z} \in G\}$. If $f : G \to \mathbb{C}$ is analytic prove that $f^* : G^* \to \mathbb{C}$, defined by $f^*(z) = \overline{f(\overline{z})}$ is also analytic.

If f is analytic in G then satisfies the Cauchy–Riemann equations, thus if $f(z) = u(x,y) + iv(x,y)$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Write $\overline{f(\overline{z})}$ as $u(x,-y) - iv(x,-y)$ and check if it follows the Cauchy–Riemann equations.

$$\frac{\partial u}{\partial x} = \frac{(-\partial v)}{(-\partial y)} \quad \frac{\partial u}{(-\partial y)} = -\frac{(-\partial v)}{\partial x}$$

Which is in line with f's Cauchy–Riemann equations. Thus $f^*(z)$ is analytic

20. Let $z_1, z_2, ..., z_n$ be complex numbers such that $\Re(z_k) > 0$ and $\Re(z_l ... z_k) > 0$ for all $1 \leq k \leq n$. Show that $\log(z_1 ... z_n) = \log z_1 + ... + \log z_n$ where $\log z$ is the principal branch of the logarithm. If the restrictions on the z_k are removed, does the formula remain valid?

Assuming Conway means the complex logarithm, we'll denote it as Log z. If $\Re(z) > 0$ then $-\pi < \arg(z) < \pi$ and thus if $\Re(z_l ... z_k) > 0$ then $-\pi < \arg(z_l ... z_k) < \pi$. By definition of the "principle" complex logarithm.

$$\text{Log } z_k = \log|z_k| + i \arg(z_k)$$

Recall that $\log|z_k| \cdot \log|z_l| = \log|z_k z_l|$ and that $\arg(z_k) + \arg(z_l) = \arg(z_k \cdot z_l)$. Thus

$$\text{Log } (z_1 z_2) = \log|z_1 z_2| + i \arg(z_1 z_2)$$

$$= \log|z_1| + \log|z_2| + i \arg(z_1) + i \arg(z_2) = \text{Log } (z_1) + \text{Log } (z_2)$$

Now suppose the formula is true for $n = k - 1$, then

$$\text{Log}(z_1 ... z_n) = \text{Log}((z_1 ... z_{n-1}) z_n) = \text{Log}(z_1 ... z_{n-1}) + \text{Log}(z_n)$$

$$\text{Log}(z_1) + \text{Log}(z_2) + ... + \text{Log}(z_n)$$

If the restrictions on the z_k are removed, does the formula remain valid?

Choose $z_1 = z_2 = -1/2 - \sqrt{3}/2 = e^{-2\pi i/3}$. We get

$$\text{Log}(z_1) + \text{Log}(z_1) = \log 1 + -2\pi i/3 + \log 1 + -2\pi i/3 = -4\pi i/3$$

But

$$\text{Log}(z_1 z_2) = 2\pi i/3$$

Thus the formula fails since $\text{Log}(z_1 z_2) \neq \text{Log}(z_1) + \text{Log}(z_1)$ $\square$

21. Prove that there is no branch of the logarithm defined on $G = \mathbb{C} - \{0\}$ (Hint: Suppose such a branch exists and compare this with the principal branch.)

Assume for contradiction such a branch L exists such that $L : G \to \mathbb{C}$ and $\exp(Lz) = z$ for all $z \in G$. Restricting G to the preimage of the principle logarithm we get a branch of a log on G and thus L differs by the principle logarithm by $2\pi ik$ (for $k \in \mathbb{Z}$) and we get

$$L(z) = \log|z| + i(\arg z + 2\pi k)$$

However $L(z)$ is analytic and continuous at -1. Thus by contradiction is not a branch of the logarithm.

3.3 Analytic functions as mappings. Möbius transformations

(I actually initially struggled a lot with these when I first learned them)

1. Find the image of $\{z : \Re z < 0, |\Im z| < \pi\}$ under the exponential function.

Rewriting the set, we get

$$S := \{z = x + iy : x < 0, -\pi < y < \pi\}$$

Applying the exponential function, we get $\exp(x+iy) = \exp(x)\exp(iy)$. Meaning at as x ranges from $(-\infty, 0)$ the radius r ranges from $(0, 1)$ and the argument ranges from $(-\pi, \pi)$. Thus we get the open disc centered at 0 and radius 1 minus the negative real line. $\exp(S) = \{z : 0 < |z| < 1\} - \{z \in \mathbb{R} : z < 0\}$

2. Do exercise 1 for the set $\{z : |\Im z| < \pi/2\}$

Rewriting the set, we get

$$S := \{z = x + iy : x \in \mathbb{R}, -\pi/2 < y < \pi/2\}$$

Applying the exponential function, we get $\exp(x+iy) = \exp(x)\exp(iy)$. Meaning at as x ranges from $(-\infty, \infty)$ the radius r ranges from $(0, \infty)$ and the argument ranges from $(-\pi/2, \pi/2)$. Thus we get the positive, open, half plane. $\exp(S) = \{z : |\Re z| > 0\}$

3. Discuss the mapping properties of $\cos z$ **and** $\sin z$

(I'm not sure what Conway means by "Discuss", so I'll just derive some things)

Recall the following identities

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i}$$

Using $z = x + iy$ and angle sum identities we can derive the following

$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

$$\cos z = \cos x \cosh y - i \sin x \sinh y$$

Discuss the mapping properties of z^n **and** $z^{1/n}$ **for** $n > 2$ (I'll do the same thing here)

Rewriting in polar coordinates we get the following

$$z^n = |z|^n \quad z^{\frac{1}{n}} = |z|^{\frac{1}{n}} e^{\frac{i\theta}{n}}$$

For z^n if $|z| > 1$ then $z \to \infty$ as $n \to \infty$ If $|z| < 1$ then $z \to 0$ as $n \to \infty$. And the inverse is true for $z^{1/n}$.

5. Find the fixed points of a dilation, a translation and the inversion on $\mathbb{C}_\infty$

A dilation is defined as

$$D(z) = az \quad (a \in \mathbb{C})$$

Thus a fixed point is for when $az = z$ or when $z = 0$ and $z = \infty$.
A translation is defined as

$$T(z) = b + z \quad (b \in \mathbb{C})$$

Thus a fixed point is for when $b + z = z$ or when $z = \infty$.
An inversion is inversion as

$$I(z) = \frac{1}{z}$$

Thus a fixed point is for when $z = \frac{1}{z}$ or when $z = 1$ and $z = -1$.

6. Evaluate the following cross ratios

(a) $(7+i, 1, 0, \infty)$
Recall that

$$S(z) = \left(\frac{z - z_3}{z - z_4}\right) / \left(\frac{z_2 - z_3}{z_2 - z_4}\right) \quad (\text{if } z_2, z_3, z_4 \in \mathbb{C})$$

then

$$S(z) = \frac{z - z_3}{z_2 - z_3} \quad (\text{if } z_4 = \infty)$$

Thus

$$S(7+i) = \frac{7 + i - 0}{1 - 0} = 7 + i$$

(b) $(2, 1-i, 1, 1+i)$
Plugging into the above formula we get

$$\left(\frac{2-1}{2-(1+i)}\right) / \left(\frac{1-i-1}{1-i-(1+i)}\right) = \left(\frac{1}{1-i}\right) / \left(\frac{-i}{-2i}\right) = \frac{2}{1-i} \cdot \frac{1+i}{1+i}$$

$$= \frac{2+2i}{2} = 1 + i$$

(c) $(0, 1, i, -1)$

$$\left(\frac{0-i}{0-(-1)}\right) / \left(\frac{1-i}{1-(-1)}\right) = \left(\frac{-i}{1}\right) / \left(\frac{1-i}{2}\right)$$

$$= \frac{-2i}{1-i}\frac{1+i}{1+i} = \frac{2-2i}{2} = 1 - i$$

(d) $(i-1, \infty, 1+i, 0)$

$$S(z) = \frac{z - z_3}{z - z_4} = \frac{i - 1 - (1+i)}{i - 1 - (0)} = \frac{-2}{i-1}\frac{i+1}{i+1}$$

$$\frac{-2i-2}{-2} = 1 + i$$

7. If $T(z) = \frac{az+b}{cz+d}$ find z_2, z_3, z_4 (in terms of a, b, c, d) such that $T(z) = (z, z_2, z_3, z_4)$

Step 1. Make a really lucky guess

$$z_4 = -\frac{d}{c} \quad z_3 = -\frac{b}{a} \quad z_2 = \frac{d-b}{a-c}$$

(I'm joking)

For $T(z) = 1 \implies az + b = cz + d \implies (a - c)z = d - b$ and thus

$$z_2 = \frac{d - b}{a - c}$$

For $T(z) = 0 \implies az + b = 0$ and thus

$$z_3 = -\frac{b}{a}$$

For $T(z) = \infty \implies cz + d = 0$ and thus

$$z_4 = -\frac{d}{c}$$

8. If $T(z) = \frac{az+b}{cz+d}$ show that $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$ iff we can choose a, b, c, d to be real numbers

For $z \in \mathbb{R}_\infty$, then if a, b, c, d are real, then $T(z)$ is also real and we preserve the mappings from above for $T(z) = 1$, $T(z) = 0$, and $T(z) = \infty$. Thus $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$. To prove the converse, suppose that $T(\mathbb{R}_\infty) = \mathbb{R}_\infty$. Then

$$\frac{az + b}{cz + d} = \overline{\frac{az + b}{cz + d}} = \frac{\overline{a}z + \overline{b}}{\overline{c}z + \overline{d}}$$

Using the properties from problem 7. we get that

$$z_4 = -\frac{d}{c} = -\frac{\overline{d}}{\overline{c}}$$

Meaning that d and c have or real, or have no imaginary part. we can do the same for z_3 to find that a and b also have or real, or have no imaginary part. And the following identity

$$z_2 = \frac{d - b}{a - c}$$

confirms that a, b, c, d must either all be real or all be imaginary. □

9. If $T(z) = \frac{az+b}{cz+d}$, find necessary and sufficient conditions that $T(\Gamma) = \Gamma$ where Γ is the unit circle $\{z : |z| = 1\}$.

If $|z| = 1$ then $|T(z)| = 1$ and thus

$$|az + b| = |cz + d| \quad (\text{for all } |z| = 1)$$

Squaring both sides and following the algebra we get

$$|az + b|^2 = |cz + d|^2$$

$$|az + b|^2 - |cz + d|^2 = 0$$

$$(az+b)(\overline{az+b})-(cz+d)(\overline{cz+d})=0$$
$$az\overline{az}+b\overline{az}+az\overline{b}+b\overline{b}-(cz\overline{cz}+d\overline{cz}+cz\overline{d}+d\overline{d})=0$$
$$az\overline{az}+b\overline{az}+az\overline{b}+b\overline{b}-(cz\overline{cz}+d\overline{cz}+cz\overline{d}+d\overline{d})=0$$
$$z\overline{z}(|a|^2-|c|^2)+z(a\overline{b}-c\overline{d})+\overline{z}(b\overline{a}-d\overline{c})+|b|^2-|d|^2=0$$

Using $z\overline{z}-1=0$ we get

$$|a|^2-|c|^2=1$$
$$|b|^2-|d|^2=-1$$
$$a\overline{b}-c\overline{d}=0$$
$$b\overline{a}-d\overline{c}=0$$

Those conditions are equivalent to

$$(c,d)=(\lambda\overline{b},\lambda\overline{a})\quad(\text{for }|\lambda|=1)$$

Thus we get

$$T(z)=\frac{az-b}{\lambda(\overline{b}z+a)}=e^{i\theta}\frac{az+b}{\overline{b}z+a}$$

□

10. Let $D=\{z:|z|<1\}$ and find all Möbius transformations such that $T(D)=D$

Since the transformation takes circles to circles, then we get a problem similar to the above problem. Where $T(\Gamma)=\Gamma$ and Γ is the unit circle $\{z:|z|=1\}$. Because $0\in D$, there exists $\alpha\in D$ such that $T(\alpha)=0$. The point symmetric to α with respect to Γ is

$$z^*-a=\frac{R^2}{\overline{z-a}}\implies\alpha^*=\frac{1}{\overline{\alpha}}$$

Thus, we have $T(\alpha^*)=\infty$. Writing in general form we get

$$T(z)=\frac{z-\alpha}{\overline{\alpha}z-1}$$

We can also rotate as the circle has rotational symmetry and we get

$$T(z)=e^{i\theta}\frac{z-\alpha}{\overline{\alpha}z-1},\quad\alpha\in D,\theta\in\mathbb{R}$$

11. Show that the definition of symmetry (3.17) does not depend on the choice of points z_2, z_3, z_4. That is, show that if $\omega_2, \omega_3, \omega_4$ are also in Γ then equation (3.18) is satisfied iff $(z^*, \omega_2, \omega_3, \omega_4) = \overline{(z, \omega_2, \omega_3, \omega_4)}$. (Hint: Use Exercise 8.)

Let Γ be the circle and pick points $z_2, z_3, z_4 \in \Gamma$.If $T(z)$ is unique then $T(z_2) = 1, T(z_3) = 0, T(z_4) = \infty$ and thus

$$T(\Gamma) = \mathbb{R}_\infty$$

We are given that

$$T(z^*) = \overline{T(z)}$$

Pick more points $\omega_2, \omega_3, \omega_4 \in \Gamma$. Define another map $S(z)$ such that $S(\omega_2) = 1, S(\omega_3) = 0, S(\omega_4) = \infty$ and thus

$$S(\Gamma) = \mathbb{R}_\infty$$

Now, define a new function such that $M := S \circ T^{-1}$. Thus $M : \mathbb{R}_\infty \to \mathbb{R}_\infty$ and thus

$$M(\overline{\omega}) = \overline{M(\omega)}$$

Finally, we can do the following

$$S(z^*) = M(T(z^*)) = M(\overline{T(z)}) = \overline{M(T(z))} = \overline{S(z)}$$

Thus

$$(z^*, \omega_2, \omega_3, \omega_4) = \overline{(z, \omega_2, \omega_3, \omega_4)}$$

To prove the converse, suppose $S(z^*) = \overline{S(z)}$ and define $N(\omega) = M^{-1}(\omega)$. The rest of the proof is similar.

12. Prove Theorem 3.4

(I'm actually very fond of this problem) 3.4 Theorem If $f : G \to \mathbb{C}$ is analytic then f preserves angles at each point z_0 of G where $f'(z_0) \neq 0$

Choose two curves $\gamma_1, \gamma_2 \subset G$ that intersect at z_0. To find the angle of the two curves at z_0, Choose two more points $z_1 \in \gamma_1, z_2 \in \gamma_2$ such that $|z_1 - z_0| = |z_0 - z_2| = \varepsilon$. We get the following

$$z_1 - z_0 = \varepsilon e^{i\theta_1} \quad z_2 - z_0 = \varepsilon e^{i\theta_2}$$

$$\frac{z_2 - z_0}{z_1 - z_0} = e^{i(\theta_1 - \theta_2)}$$

Thus the angle between these curves is $\theta_1 - \theta_2$ or

$$\lim_{\varepsilon \to 0} \arg\left(\frac{z_2 - z_0}{z_1 - z_0}\right)$$

Similarly, considering the image of f. the angle between $f(\gamma_1)$ and $f(\gamma_2)$ at $f(z_0)$ is

$$\lim_{\varepsilon\to 0} \arg\left(\frac{f(z_2)-f(z_0)}{f(z_1)-f(z_0)}\right)$$

Rewriting we can get

$$\lim_{\varepsilon\to 0} \arg\left(\frac{\frac{f(z_2)-f(z_0)}{z_2-z_0}}{\frac{f(z_1)-f(z_0)}{z_1-z_0}} \cdot \frac{z_2-z_0}{z_1-z_0}\right)$$

Since $|z_1 - z_0| = |z_2 - z_0| = \varepsilon$ this means that the numerator and denominator are both the derivative of f with respect to z at z_0. Thus

$$\lim_{\varepsilon\to 0} \arg\left(\frac{\frac{f(z_2)-f(z_0)}{z_2-z_0}}{\frac{f(z_1)-f(z_0)}{z_1-z_0}} \cdot \frac{z_2-z_0}{z_1-z_0}\right) = \lim_{\varepsilon\to 0} \arg\left(\frac{f'(z_0)}{f'(z_0)} \cdot \frac{z_2-z_0}{z_1-z_0}\right)$$

$$= \lim_{\varepsilon\to 0} \arg\left(\frac{z_2-z_0}{z_1-z_0}\right)$$

□

13. Give a discussion of the mapping $f(z) = \frac{1}{2}(z+\frac{1}{z})$ (I don't like these "discussion" questions.)

Since $f(z) = f(1/z)$ and $f(\overline{z}) = \overline{f(z)}$ then it's symmetric. For the unit circle $D := \{z : |z| = 1\}$ we get that $f(D) = [-1, 1]$ because $f(e^{i\theta}) = \cos\theta$. $f(i) = f(-i) = 0$.

14. Let G be the open connected set between two circles which are tangent at the point a. Map G conform ally onto the open unit disk. (Hint: first try $(z-a)^{-1}$)

Suppose that one circles is inside another. Consider the Möbius transformation $T := \frac{1}{z-a}$ Since both circles are tangent at a their image under T is two parallel lines that intersect at infinity. Thus the region between the two circles is an infinite strip of the complex plane. Generally the mapping of the strip

$$\{z : |\Re(z)| < \pi/2\}$$

to the half plane is just $\exp z$. Thus in order to normalize our strip, we need the map

$$z \mapsto \alpha z + \beta$$

Where we choose α and β to normalize the strip. Finally, the mapping of the half plane to the unit circle

$$z \mapsto \frac{z-i}{z+i}$$

Composing all of these functions we get

$$z \mapsto z - a \mapsto \frac{\alpha}{z-a} + \beta \mapsto e^{\left(\frac{\alpha}{z-a}+\beta\right)} \mapsto$$

$$\frac{e^{\left(\frac{\alpha}{z-a}+\beta\right)} - i}{e^{\left(\frac{\alpha}{z-a}+\beta\right)} + i}$$

Next suppose that the circles are next to each other. Choose one circle with center z_0 and radius r. Map one circle to the unit circle with a translation and dilation, and then perform an inversion and we get one circles is inside another.

$$z \mapsto \frac{1}{r(z - z_0)}$$

And compose this mapping with this mapping to get the unit circle. □

15. Let $G = \{z : 0 < |z| < 1\}$. Can you map G conformally onto the open unit disk?

If a bijective map f exists, any path γ that surrounds 0 is homo topic to a point in the image of f but not in G. Thus no map exists. However if we square a disc automorphism, we get an injective, conformal, map. Choosing to send $f : 0 \mapsto 1/2$. We get

$$f(z) = \frac{z + \frac{1}{2}}{1 + \frac{1}{2}z}$$

Squaring, we get an injective map.

$$\left(\frac{z + \frac{1}{2}}{1 + \frac{1}{2}z}\right)^2$$

16. Map $G = \mathbb{C} - \{z : -1 \leq z \leq 1\}$ onto the open unit disk by an analytic function f. Can f be one-to-one?

Send the slit to the negative real axis

$$g_1 := \frac{z-1}{z+1}$$

Now we have the plane minus the negative real axis. Next take the principal square root to get the positive real half plane.

$$g_2 := \sqrt{z}$$

We take the principal root to keep the function analytic and one to one. Finally map the half plane to the unit circle

$$g_3 := \frac{z-1}{z+1}$$

Taking the composition of all the functions

$$f = g_3 \circ g_2 \circ g_1 = \frac{\sqrt{\frac{z-1}{z+1}} - 1}{\sqrt{\frac{z-1}{z+1}} + 1}$$

Each part of the composition is one-to-one, thus f is one to one.

17. Let G be a region and suppose that $f : G \to \mathbb{C}$ is analytic such that $f(G)$ is a subset of a circle. Show that f is constant.

Since $f(G)$ is a subset of the circle, choose $z_2, z_3, z_4 \in f(G)$ and a transform T such that

$$T(z_2) \mapsto 1 \quad T(z_3) \mapsto 0 \quad T(z_4) \mapsto \infty$$

Let $g := T \circ f$. Thus $g : G \to \mathbb{R}_\infty$. Since g is also analytic, we get that it follows the Cauchy–Riemann equations. Meaning that if $g = u + iv$ we get $v \equiv 0$ and thus $u_x = u_y = 0$ and by extension $f' = 0$ and thus f is constant. □

18. Let $-\infty < a < b < \infty$ and put $Mz = \frac{z-ia}{z-ib}$. Define lines $L_1 = \{z : \Im z = b\}$, $L_2 = \{z : \Im z = a\}$, and $L_3 = \{z : \Re z = 0\}$. Determine which of the regions A, B, C, D, E, F in Figure 1 are mapped by M onto the regions U, V, W, X, Y, Z in Figure 2.

Since $M(ia) = 0$, the neighborhood of ia gets mapped to the neighborhood 0. So B, E, C, F gets mapped to U, V, X, Y and A, D gets mapped to W, Z. Also $M(ib) = \infty$ Thus the neighborhood of ib gets mapped to the regions that reach ∞ So A, D, B, E gets mapped to X, U, W, Z and C, F gets mapped to V, Y. Thus,

$$A, D \to W, Z \quad C, F \to V, Y \quad B, E \to U, X$$

Also L_3 gets mapped to the Real axis. Thus, all we need to do is determine orientation and we get the mappings since adjacent regions should map to adjacent regions. For $z = x + iy$ then

$$M(x + iy) = \frac{x + i(y-a)}{x + i(y-b)} = \frac{x^2 - ix(y-b) + ix(y-a) + (y-a)(y-b)}{x^2 + (y-b)^2}$$

The imaginary part

$$\Im M(x + iy) = \frac{x(b-a)}{x^2 + (y-b)^2}$$

If $x > 0$ then $\Im M(x + iy) > 0$ and if $x < 0$ then $\Im M(x + iy) < 0$. Thus the right half plane gets mapped to the upper half plane and we get.

$$M(A) = Z \quad M(B) = X \quad M(C) = Y$$

$$M(D) = W \quad M(E) = U \quad M(F) = V$$

19. Let a. b, and M be the same as in Exercise 18 and let $\log$ be the principal branch of the logarithm.

(a) Show that $\log Mz$ is defined for all z except $z = ic$, $a \leq c \leq b$ and if $h(z) = \Im[\log Mz]$ then $0 < h < \pi$ For $\Re z > 0$.

Recall from Exercise 18 that since B, E was mapped to U, X, the imaginary axis between ai and bi gets mapped to the negative imaginary axis, where the logarithm isn't defined. Also from Exercise 18, we know that the right half plane gets mapped to the vertical half plane, or all points $re^{i\theta}$ such that $\theta \in (0, \pi)$ Thus $\log M(z) = \ln r + i\theta$ thus $0 < \Im(\log Mz) < \pi$

(b) Show that $\log(z - ic)$ is defined for $\Re z > 0$ and any real number c; also prove that $|\Im \log(z - ic)| < \pi/2$ if $\Re z > 0$

$\log z$ is already defined for $\Re z > 0$ so $\log(z - ic)$ is also defined. Also if $\Re(z) > 0$ then z is the right half plane and it's all points $re^{i\theta}$ such that $|\theta| < \pi/2$. And $\Re(z - ic) = \Re(z - ic)$ and thus it's also all points in the right half plane. Thus

$$|\Im \log(z - ic)| = |\arg(z - ic)| < \pi/2$$

(c) Let h be as in (a) and prove that $h(z) = \Im[\log(z - ia) - \log(z - ib)] = \Im[\log \frac{z-ia}{z-ib}]$

Recall that

$$h(z) = \Im\left[\log\left\{\frac{z - ai}{z - bi}\right\}\right]$$

$$M(x+iy) = \frac{x^2 - ix(y-b) + ix(y-a) + (y-a)(y-b)}{x^2 + (y-b)^2} = \frac{x^2 + (y-a)(y-b) + ix(b-a)}{x^2 + (y-b)^2}$$

Thus

$$\Im\left[\log\left\{M(x+iy\right\}\right] = \arg\left(\frac{x^2 + (y-a)(y-b) + ix(b-a)}{x^2 + (y-b)^2}\right)$$

And thus

$$h(x+iy) = \arctan\left(\frac{x(b-a)}{x^2 + (y-a)(y-b)}\right)$$

Evaluating the other expression

$$\Im\left[\log(z - ia)\right] - \Im\left[\log(z - ib)\right]$$

$$= \arctan\left(\frac{y-a}{x}\right) - \arctan\left(\frac{y-b}{x}\right)$$

Recall that $\arctan u + \arctan v = \arctan \frac{u+v}{1-uv}$ and we get

$$\arctan\left(\frac{x(b-a)}{x^2+(y-a)(y-b)}\right) = h(x+iy)$$

(d) Show that

$$\int_a^b \frac{dt}{z-it} = i[\log(z-ib) - \log(z-ia)]$$

(Hint: Use the Fundamental Theorem of Calculus.)

Define a function

$$F(t) := i\log(z-it)$$

Then we get

$$F'(t) = i\frac{d}{dt}\log(z-it) = i\frac{-i}{z-it} = \frac{1}{z-it}$$

By FTC we get

$$\int_a^b \frac{dt}{z-it} = F(b) - F(a) = i[\log(z-ib) - \log(z-ia)]$$

(e) Combine (c) and (d) to get that

$$h(x+iy) = \int_a^b \frac{x}{x^2+(y-t)^2}dt = \arctan\left(\frac{y-a}{x}\right) - \arctan\left(\frac{y-b}{x}\right)$$

We already know that

$$h(x+iy) = \arctan\left(\frac{y-a}{x}\right) - \arctan\left(\frac{y-b}{x}\right)$$

So we just need to show that

$$\int_a^b \frac{x}{x^2+(y-t)^2}dt = \arctan\left(\frac{y-a}{x}\right) - \arctan\left(\frac{y-b}{x}\right)$$

Substitute $u = (y-t)/x$ so that $dt = -xdu$

$$\int_a^b \frac{x}{x^2+(y-t)^2}dt = \int_{(y-a)/x}^{(y-b)/x} \frac{x}{x^2+(xu)^2}(-xdu)$$

$$\int_{(y-b)/x}^{(y-a)/x} \frac{du}{1+u^2} = [\arctan(u)]_{(y-b)/x}^{(y-a)/x}$$
$$= \arctan\left(\frac{y-a}{x}\right) - \arctan\left(\frac{y-b}{x}\right)$$

(f) Interpret part (e) geometrically and show that for $\Re z > 0$ $h(z)$ is the angle depicted in the figure.

The angle of the line between ai and $z = x + iy$ is just arctan $\frac{y-a}{x}$. The negative angle is between bi and z is arctan $\frac{y-b}{x}$ so the difference of the angles is the angle made at z. So

$$h(x+iy) = \arctan\left(\frac{y-a}{x}\right) - \arctan\left(\frac{y-b}{x}\right)$$

20. Let

$$Sz = \frac{az+b}{cz+d} \quad Tz = \frac{\alpha z+\beta}{\gamma z+\delta};$$

Show that $S = T$ iff there is a non-zero complex number λ such that $\alpha = \lambda a$, $\beta = \lambda b$, $\gamma = \lambda c$, $\delta = \lambda d$

Trivially, if there exists a λ, then factor out from Tz and we get $Tz = Sz$. To prove the converse, suppose $Tz = Sz$ where $ad - bc \neq 0$. Thus $T(z_2) = S(z_2) = 0$, $T(z_3) = S(z_3) = 0$, and thus $T(z_4) = S(z_4) = \infty$. By a previous problem we get

$$z_4 = -\frac{d}{c} = -\frac{\delta}{\gamma} \quad z_3 = -\frac{b}{a} = -\frac{\beta}{\alpha} \quad z_2 = \frac{d-b}{a-c} = \frac{\delta-\beta}{\alpha-\gamma}$$

Thus we can derive complex numbers λ_1 and λ_2 such that

$$\delta = \lambda_1 d \quad \gamma = \lambda_1 c \quad \alpha = \lambda_2 a \quad \beta = \lambda_2 b$$

We can also derive that

$$\frac{d-b}{a-c} = \frac{\delta-\beta}{\alpha-\gamma} = \frac{d\lambda_1 - b\lambda_2}{a\lambda_2 - c\lambda_1}$$
$$\frac{d-b}{a-c} = \frac{d\lambda_1 - b\lambda_2}{a\lambda_2 - c\lambda_1}$$
$$(d-b)(a\lambda_2 - c\lambda_1) = (a-c)(d\lambda_1 - b\lambda_2)$$
$$da\lambda_2 - ba\lambda_2 - dc\lambda_1 + bc\lambda_1 = ad\lambda_1 - cd\lambda_1 - ab\lambda_2 + cb\lambda_2$$
$$\lambda_2(da - ba + ab - cb) = \lambda_1(ad - cd + dc - bc)$$
$$\lambda_2(ad - bc) = \lambda_1(ad - bc)$$
$$(\lambda_1 - \lambda_2)(ad - bc) = 0$$

Thus, either $\lambda_1 = \lambda_2$ or $ad - bc = 0$ which is a contradiction. Thus there exists λ such that $\alpha = \lambda a$, $\beta = \lambda b$, $\gamma = \lambda c$, $\delta = \lambda d$ □

Let T be a Möbius transformation with fixed points z_1 and z_2. If S is a Möbius transformation show that $S^{-1}TS$ has fixed points $S^{-1}z_1$ and $S^{-1}z_2$

(This seemed a bit too trivial so I'm definitely missing something)
Given that $T(z_1) = z_1$ Evaluate $S^{-1}TS(S^{-1}z_1)$

$$S^{-1}TSS^{-1}(z_1) = S^{-1}T(z_1) = S^{-1}(z_1)$$

Thus $S^{-1}(z_1)$ is a fixed point of $S^{-1}TS$. Similarly Given that $T(z_2) = z_2$ Evaluate $S^{-1}TS(S^{-1}z_2)$

$$S^{-1}TSS^{-1}(z_2) = S^{-1}T(z_2) = S^{-1}(z_2)$$

22. (a) Show that a Möbius transformation has 0 and ∞ as its only fixed points iff it is a dilation, but not the identity. (b) Show that a Möbius transformation has ∞ as its only fixed point iff it's a translation.

Let $Dz = \frac{az+b}{cz+d}$ be a transformation. Since $D(0) = 0$, thus $\frac{b}{d} = 0$, meaning $b = 0$. And if $D(\infty) = \infty$ then $\frac{a}{c} = \infty$ and $c = 0$. Thus we get

$$D(z) = \frac{a}{d}z$$

Which is a dilation. Now suppose $D(z) = \alpha z$ $(\alpha \neq 0, 1)$ which is dilatation. To find the fixed points, let $z = \alpha z$. Clearly $z = 0$ and $z = \infty$

Now let $Tz = \frac{az+b}{cz+d}$ be a transformation with fixed points $z = \infty$. Since $T(\infty) = \infty$ then $\frac{a}{c} = \infty$ and $c = 0$. Thus we get

$$T(z) = \frac{az+b}{d} = \frac{a}{d}z + \frac{b}{d}$$

Solving for fixed points we get

$$z = \frac{a}{d}z + \frac{b}{d}$$

With fixed points $z = \infty$ and

$$z = \frac{b}{d\left(1 - \frac{a}{d}\right)}$$

However since $z = \infty$ is the only fixed point, that means

$$\frac{b}{d\left(1 - \frac{a}{d}\right)} = \infty \implies \frac{a}{d} = 1$$

Which gives us

$$D(z) = z + \frac{b}{d}$$

Which is a translation Now suppose $T(z) = z + \beta$ ($\beta \neq 0$) which is translation. To find the fixed points, let $z = z + \beta$. Clearly $z = \infty$

23. Show that a Möbius transformation T satisfies $T(0) = \infty$ and $T(\infty) = 0$ iff $Tz = az^{-1}$ for some $a \in \mathbb{C}$

Let $Tz = \frac{az+b}{cz+d}$. We get

$$T(0) = \frac{b}{d} = \infty \implies d = 0 \quad T(\infty) = \frac{a}{c} = 0 \implies a = 0$$

Thus we get

$$T(z) = \frac{b}{c}\frac{1}{z}$$

Letting $\alpha = \frac{b}{c}$ we get

$$Tz = az^{-1}$$

Trivially, if $Tz = az^{-1}$, then $T(0) = \infty$ and $T(\infty) = 0$

24. Let T be a Möbius transformation, $T \neq$ the identity. Show that a Möbius transformation S satisfies $ST = TS$ iff S and T have the same fixed points. (Hint: Use Exercises 21 and 22. Also note that for a given T the set $\{S : TS = ST\}$ is an abelian group by Exercises 21 and 22.)

Suppose S and T have the same fixed pointsz_1, z_2.

Case 1: $z_1 \neq z_2$
Choose a new Möbius transformation M such that $M(z_1) = 0$ and $M(z_2) = \infty$. Then the transformation MTM^{-1} has fixed points 0 and ∞. Similarly, the following compositions are also dilations

$$MSM^{-1} \quad \text{and its inverse} \quad MS^{-1}M^{-1}$$

Since dilations commute we get

$$MTM^{-1} = (MSM^{-1})(MTM^{-1})(MS^{-1}M^{-1})$$

$$MTM^{-1} = MSTS^{-1}M^{-1}$$

$$T = STS^{-1}$$

$$TS = ST$$

Case 2: $z_1 = z_2$
We still choose another transformation M such that $M(z_1) = \infty$. Similarly

MTM^{-1} has ∞ as a fixed point and thus is a translation. Since translations commute we perform the same algebra from **Case 1** and thus.

$$TS = ST$$

□

25. Find all the abelian subgroups of the group of Möbius transformations.

Recall from the previous problem that S and T commute iff they have the same fixed points. If we look at the group of transformations that share the same fixed point $G_1 = \{T : T(z_1) = z_1\}$, choose another transformation M such that $M(z_1) = \infty$. Thus the group of transformations $MTM^{-}1$ has a fixed point at ∞ and is a group of translations, thus the group G_1 is isomorphic to the additive group $(\mathbb{C}, +)$
Similarly, the group that shares exactly two fixed points $G_2 = \{T : T(z_1) = z_1, T(z_2) = z_2\}$ is isomorphic to the multiplicative group $(\mathbb{C}^\times, \times)$ because we can choose a transform M such that $M(z_1) = \infty$ and $M(z_2) = 0$.
Finally we construct a finite set composed of inversions and their inverses

$$G_3 = \{e, -z, 1/z, -1/z\}$$

26 (a). Let $GL_2(\mathbb{C})$ = all invertible 2×2 matrices with entries in $\mathbb{C}$ and let $\mathcal{M}$ be the group of Möbius transformations. Define $\varphi : GL_2(\mathbb{C}) \to \mathcal{M}$ by

$$\phi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{az + b}{cz + d}$$

Show that φ is a group homomorphism of $GL_2(\mathbb{C})$ onto $\mathcal{M}$

We need to show that for two matrices $A, B \in GL_2(\mathbb{C})$ we get

$$\varphi(A) \circ \varphi(B) = \varphi(AB)$$

First just evaluate each side with the following matrices

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad \text{where} \quad ad - bc \neq 0, \quad \alpha\delta - \beta\gamma \neq 0$$

For the LHS we get

$$\varphi(A) \circ \varphi(B) = \frac{a\left(\frac{\alpha z - \beta}{\gamma z - \delta}\right) - b}{c\left(\frac{\alpha z - \beta}{\gamma z - \delta}\right) - d} = \frac{(a\alpha + b\gamma)z + (a\beta + b\delta)}{(c\alpha + d\gamma)z + (c\beta + d\delta)}$$

Evaluating the RHS we get

$$AB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} a\alpha + b\gamma & a\beta + b\delta \\ c\alpha + d\gamma & c\beta + d\delta \end{pmatrix}$$

$$\varphi(AB) = \frac{(a\alpha + b\gamma)z + (a\beta + b\delta)}{(c\alpha + d\gamma)z + (c\beta + d\delta)}$$

26 (b). Let $SL_2(\mathbb{C})$ be the subgroup of $GL_2\mathbb{C}$ consisting of all matrices of determinant 1. Show that the image of $SL_2(\mathbb{C})$ under φ is all of $\mathcal{M}$. What part of the kernel of φ is in $SL_2(\mathbb{C})$?

For any transformation $\varphi(A) \in \mathcal{M}$ where

$$(A) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

We can modify the transformation where $\lambda = \frac{1}{\sqrt{ad-bc}}$ to get

$$\frac{\frac{a}{\sqrt{ad-bc}}z + \frac{b}{\sqrt{ad-bc}}}{\frac{c}{\sqrt{ad-bc}}z + \frac{d}{\sqrt{ad-bc}}}$$

which creates a surjective map $\varphi : SL_2(\mathbb{C}) = \mathcal{M}$
Recall the kernel of a group homomorphism is

$$\ker \varphi = \{A \in SL_2(\mathbb{C}) : \varphi(A) = e_{\mathcal{M}} = z\}$$

So to start start by solving for the identity

$$\frac{az+b}{cz+d} = z \quad az + b = cz^2 + dz \iff a = d, b = 0, c = 0$$

Which forces $\det(A) = 1$ iff $a = \pm 1$ meaning

$$\ker(\varphi | SL_2\mathbb{C}) = \{I, -I\}$$

27. If $\mathcal{G}$ is a group and $\mathcal{N}$ is a subgroup then $\mathcal{N}$ is said to be a *normal subgroup* of $\mathcal{G}$ if $STS^{-1} \in \mathcal{N}$ whenever $T \in N$ and $S \in \mathcal{G}$. $\mathcal{G}$ is a simple group is the only normal subgroups of $\mathcal{G}$ are $\{I\}$ (I the identity of $\mathcal{G}$) and $\mathcal{G}$ itself. Prove that the group $\mathcal{M}$ of Möbius transformations is a simple group.

Recall that any transformation M can be expressed as a matrix $A \in SL_2(\mathbb{C})$. Suppose that we have a non-identity element N in our normal subgroup $\mathcal{N} \subset \mathcal{M}$. Since the subgroup is normal, it's closed under conjugations and thus we can conjugate N and we can conjugate to a convenient form. If N has one fixed point, it can be conjugated to a translation (WLOG say $f(z) = z + 1$) and can be represented as the following matrix.

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

If it has two fixed points it can be conjugated to a dilation and can be represented as the following matrix

$$A = \begin{pmatrix} \alpha & 0 \\ 0 & \alpha^{-1} \end{pmatrix}$$

Computing the commutator $ABA^{-1}B^{-1}$ we get

$$[AB] = \begin{pmatrix} 1 & \alpha^2 - 1 \\ 0 & 1 \end{pmatrix}$$

Which gives us $(AB)(z) = z + \alpha^2 - 1$, a non-trivial translation and hence $\mathcal{N}$ contains all non-trivial translations. For simplicity we'll rewrite it as

$$[AB] = C = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$$

Next choose another matrix in $\mathcal{M}$, being the inversion

$$D = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad D^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

We get that $DCD^{-1} \in \mathcal{N}$ which gives us

$$DCD^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}$$

Thus $\mathcal{N}$ also contains $\begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}$ AND $\begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$ which is sufficient to generate all of $\mathcal{M}$ and thus $\mathcal{N} = \mathcal{M}$ meaning $\mathcal{M}$ only has two subgroups (the identity and itself). thus $\mathcal{M}$ is normal.

4 Chapter 4: Complex Integration

4.1 Riemann-Stieltjes integrals

1. Let $\gamma : [a, b] \to \mathbb{R}$ be non decreasing. Show that γ is of bounded variation and $V(\gamma) = \gamma(b) - \gamma(a)$

Define a partition of $[a, b]$ to be

$$P = \{a = t_0 < t_1 ... t_n = b\}$$

Since γ is non-decreasing, that means for all $a \leq t_n < t_m \leq b$ we have $\gamma(t_n) \geq \gamma(t_m)$. Thus the variation gives us

$$V(\gamma) = \sum_{k=1}^{n} \gamma(t_k) - \gamma(t_{k-1}) = \gamma(b) - \gamma(a)$$

Thus

$$V(\gamma) = \gamma(b) - \gamma(a)$$

2. Prove Proposition 1.2.

1.2 Proposition Let $\gamma : [a, b] \to \mathbb{C}$ be of bounded variation. Then:

(a) If P and Q are partitions of $[a, b]$ and $P \subset Q$, then $v(\gamma; P) \leq v(\gamma; Q)$

(b) If $\sigma : [a, b] \to \mathbb{C}$ is also of bounded variation and $\alpha, \beta \in \mathbb{C}$ then $\alpha\gamma + \beta\sigma$ is of bounded variation and $V(\alpha\gamma + \beta\sigma) \leq |\alpha|V(\gamma) + |\beta|V(\sigma)$

Recall that

$$v(\gamma; P) = \sum_{k=1}^{n} |\gamma(t_k) - \gamma(t_{k-1})|$$

If Q is a refinement of P then fix k and define points of Q that lie between $[t_{k-1}, t_k]$ be

$$t_{k-1} = s_0 < s_1 ... < s_l < t_k$$

Thus we get

$$\gamma(t_k) - \gamma(t_{k-1}) = \sum_{m=1}^{l} \gamma(s_m) - \gamma(s_{m-1})$$

By triangle inequality

$$|\gamma(t_k) - \gamma(t_{k-1})| \leq \sum_{m=1}^{l} |\gamma(s_m) - \gamma(s_{m-1})|$$

$$v(\gamma; P) = \sum_{k=1}^{n} |\gamma(t_k) - \gamma(t_{k-1})| \leq \sum_{k=1}^{n} \sum_{m=1}^{l} |\gamma(s_m) - \gamma(s_{m-1})| = v(\gamma; Q)$$

And thus

$$v(\gamma; P) \leq v(\gamma; Q)$$

For any partition P we have

$$v(\alpha\gamma + \beta\sigma; P) = \sum_{k=1}^{n} |\alpha(\gamma(t_k) - \gamma(t_{k-1})) + \beta(\sigma(t_k) - \sigma(t_{k-1}))|$$

by triangle inequality

$$\leq \sum_{k=1}^{n} |\alpha||(\gamma(t_k) - \gamma(t_{k-1}))| + |\beta||(\sigma(t_k) - \sigma(t_{k-1}))| = |\alpha| v(\gamma; P) + |\beta| v(\sigma; P)$$

Taking the supremum over all partitions gives us

$$V(\alpha\gamma + \beta\sigma) \leq |\alpha| V(\gamma) + |\beta| V(\sigma)$$

3. Prove Proposition 1.7.

1.2 Proposition Let f and g be continuous functions on $[a, b]$ and let γ and σ be functions of bounded variation on $[a, b]$. Then for any scalars α and β

(a)
$$\int_a^b (\alpha f + \beta g)\, d\gamma = \alpha \int_a^b f\, d\gamma + \beta \int_a^b g\, d\gamma$$

(b)
$$\int_a^b f d(\alpha\gamma + \beta\sigma) = \alpha \int_a^b f\, d\gamma + \beta \int_a^b f\, d\sigma$$

Recall that we have $I \in \mathbb{C}$ such that for every $\varepsilon > 0$ there is $\delta > 0$ such that when $P = \{t_0 < t_1 < ... < t_m\}$ is a partition of $[a, b]$ with $||P|| = \max\{(t_k - t_{k-1} : 1 \leq k \leq m\} < \delta$ then

$$\left| I - \sum_{k=1}^{m} f(\tau_k)[\gamma(t_{k-1})] \right| < \varepsilon$$

where I is the integral

$$I = \int_a^b f\, d\gamma$$

Fix $\varepsilon > 0$. Let $I_f = \int f\, d\gamma$, $I_g = \int g\, d\gamma$ and, $I_h = \int h\, d\gamma = \int (\alpha f + \beta g)\, d\gamma$. By the above property we can choose delta meshes δ_f, δ_g, $\delta_h > 0$ with $||P|| < \delta := \min\{\delta_f, \delta_g, \delta_h\}$ and τ_k such that

$$\left| I_h - \sum_{k=1}^{m} h(\tau_k)[\gamma(t_{k-1})] \right| < \frac{\varepsilon}{1 + |\alpha| + |\beta|}$$

and similarly for g and f. Using linearity of sums

$$\sum_{k=1}^{m} h(\tau_k)[\gamma(t_{k-1})] = \alpha \sum_{k=1}^{m} f(\tau_k)[\gamma(t_{k-1})] + \beta \sum_{k=1}^{m} g(\tau_k)[\gamma(t_{k-1})]$$

Thus, by triangle inequality

$$|I_h - \alpha I_f - \beta I_g| < \frac{\varepsilon}{1 + |\alpha| + |\beta|} + \frac{|\alpha|\varepsilon}{1 + |\alpha| + |\beta|} + \frac{|\beta|\varepsilon}{1 + |\alpha| + |\beta|} = \frac{\varepsilon(1 + |\alpha| + |\beta|)}{1 + |\alpha| + |\beta|} = \varepsilon$$

□

Let $J := \int f d(\alpha\gamma + \beta\sigma)$, $I_\gamma = \int f\, d\gamma$, and $I_\sigma = \int f\, d\sigma$. If we fix $\varepsilon > 0$ there exist δ_γ, δ_σ, $\delta_h > 0$ with $||P|| < \delta := \min\{\delta_f, \delta_g, \delta_h\}$. Perform something similar to the previous example

$$\left| J - \sum f(\tau_k)[\alpha\gamma(t_{k-1}) + \beta\sigma(t_{k-1})] \right| < \frac{\varepsilon}{1 + |\alpha| + |\beta|}$$

$$\left| I_\gamma - \sum f(\tau_k)[\gamma(t_{k-1})] \right| < \frac{\varepsilon}{1 + |\alpha| + |\beta|}$$

$$\left| I_\sigma - \sum f(\tau_k)[\sigma(t_{k-1})] \right| < \frac{\varepsilon}{1 + |\alpha| + |\beta|}$$

Using linearity of sums

$$|I_\gamma - \alpha I_\gamma - \beta I_\sigma| < \frac{\varepsilon}{1 + |\alpha| + |\beta|} + \frac{|\alpha|\varepsilon}{1 + |\alpha| + |\beta|} + \frac{|\beta|\varepsilon}{1 + |\alpha| + |\beta|} = \frac{\varepsilon(1 + |\alpha| + |\beta|)}{1 + |\alpha| + |\beta|} = \varepsilon$$

4. Prove Proposition 1.8 (Use induction)

1.8 Proposition. Let $\gamma : [a, b] \to \mathbb{C}$ be of bounded variation and let $f : [a, b] \to \mathbb{C}$ be continuous. If $a = t_0 < t_1 < ... < t_n = b$ then

$$\int_a^b f\, d\gamma = \sum_{k=1}^{n} \int_{t_{k-1}}^{t_k} f\, d\gamma$$

Proof of the base case
Show that if $c \in (a, b)$ then

$$\int_a^b f\, d\gamma = \int_a^c f\, d\gamma + \int_c^b f\, d\gamma$$

Write the integrals as the following

$$I = \int_a^b f\, d\gamma \quad I_1 = \int_a^c f\, d\gamma \quad I_2 = \int_c^b f\, d\gamma$$

Fixing $\varepsilon > 0$, by definition there exists $\delta, \delta_1, \delta_2 > 0$ such that for any partition P we have

$$\left| I - \sum f(\tau_k)[\gamma(t_{k-1})] \right| < \varepsilon/3$$
$$\left| I_1 - \sum f(\tau_k)[\gamma(t_{k-1})] \right| < \varepsilon/3$$
$$\left| I_2 - \sum f(\tau_k)[\gamma(t_{k-1})] \right| < \varepsilon/3$$

Doing the same as previous problems, by triangle inequality we get

$$|I - (I_1 + I_2)|$$
$$\leq \left| I - \sum f(\tau_k)[\gamma(t_{k-1})] \right| + \left| I_1 - \sum f(\tau_k)[\gamma(t_{k-1})] \right| + \left| I_2 - \sum f(\tau_k)[\gamma(t_{k-1})] \right|$$
$$< \varepsilon/3 + \varepsilon/3 + \varepsilon = \varepsilon$$

Induction on n

Suppose the statement holds for $n-1$ points in $[a, b]$. Apply the lemma with $c = t_{n-1}$

$$\int_a^b f\, d\gamma = \int_a^{t_{n-1}} f\, d\gamma + \int_{t_{n-1}}^{t_n} f\, d\gamma$$

By the induction hypothesis on the first term

$$\int_a^{t_{n-1}} f\, d\gamma = \sum_{k=1}^{n-1} \int_{t_{n-1}}^{t_n} f\, d\gamma$$

Thus we get

$$\int_a^b f\, d\gamma = \sum_{k=1}^{n} \int_{t_{k-1}}^{t_k} f\, d\gamma$$

5. Let $\gamma(t) = \exp((-1+i)t^{-1})$ for $0 < t \leq 1$ and $\gamma(0) = 0$. Show that γ is a rectifiable path and find $V(\gamma)$. Give a rough sketch of the trace of γ

Rewriting $\gamma(t)$ we get

$$\gamma(t) = r(t)e^{i\theta(t)} = e^{-1/t}e^{i/t}$$

Thus, extending continuity into $t = 0$. For $t > 0$ we have

$$\gamma'(t) = \exp\left(\frac{-1+i}{t}\right)\frac{-1+i}{t^2}, \qquad |\gamma'(t)| = \frac{\sqrt{2}e^{-1/t}}{t^2}$$

By Prop 1.3.

$$V(\gamma) = \int_a^b |\gamma'(t)|\, dt$$

$$V(\gamma_\varepsilon) = \int_\varepsilon^1 \frac{\sqrt{2}e^{-1/t}}{t^2}\, dt$$

$$V(\gamma_\varepsilon) = \sqrt{2}\int_\varepsilon^1 \frac{e^{-1/t}}{t^2}\, dt$$

$$V(\gamma_\varepsilon) = \sqrt{2}e^{-1/t}\Big|_\varepsilon^1$$

$$V(\gamma_\varepsilon) = \sqrt{2}(e^{-1} - e^{-1/\varepsilon})$$

Letting $\varepsilon \to 0$

$$V(\gamma_\varepsilon) = \sqrt{2}e^{-1} < \infty$$

and thus γ is rectifiable.

Since $r(t) = e^{-1/t}$, and $\theta(t) = \frac{1}{t}$ then $\theta = [1, \infty)$. Meaning the path of γ is a tight spiral in the circle $|z| \leq e^{-1}$

6. Show that if $\gamma; [a, b] \to \mathbb{C}$ us a Lipschitz function then γ is of bounded variation.

Recall if γ is Lipschitz, then there is a constant M such that for any δ Partition P_δ then

$$\sum_{k=1}^m |\gamma(t_k) - \gamma(t_{k-1})| \leq \sum_{k=1}^m M\,|t_k - t_{k-1}| = M(b-a)$$

Taking the supremum over P gives $V(\gamma) \leq M(b-a) < \infty$. Thus γ is of bounded variation.

7. Show that $\gamma : [0, 1] \to \mathbb{C}$ defined by $\gamma(t) = t + it\sin(1/t)$ for $t \neq 0$ and $\gamma(0) = 0$ is a path but is not rectifiable. Sketch this path

Consider $|\Im(\gamma(t))| \leq 1$. Meaning that it's the topologists sine curve confined to the right cone and it oscillates back and forth like a mf (idk how to draw shit in LaTeX yet).

8. Let γ and σ be the two polygons $[1, i]$ and $[1, 1 + i, i]$. Express γ and σ as paths and calculate $\int_\gamma f$ and $\int_\sigma f$ where $f(z) = |z|^2$

We can write γ as $\gamma(t) = (1 - t) + it$. for $t \in [0, 1]$. We can also evaluate $\gamma'(t) = -1 + i$. Writing $|z|^2 = z\overline{z}$ we get

$$|\gamma(t)|^2 = \gamma(t)\overline{\gamma(t)} = ((1-t) + it)((1-t) - it) = 2t^2 - 2t + 1$$

Thus our integral becomes

$$(-1+i)\int_0^1 (2t^2-2t+1) = (-1+i)\left[\frac{2}{3}t^3 - t^2 + t\right]_0^1 = \frac{2}{3}(-1+i)$$

Next, writing σ as the sum of two lines $\sigma(t) = \sigma_1(t) + \sigma_2(t)$ where

$$\sigma_1(t) = 1 + ti \quad \sigma_2(t) = (1-t) + i$$

$$\sigma_1'(t) = i \quad \sigma_2'(t) = -1$$

$$|\sigma_1(t)|^2 = 1 + t^2 \quad |\sigma_2(t)|^2 = t^2 - 2t + 2$$

Computing the integrals

$$\int_\sigma f = \int_{\sigma_1} f + \int_{\sigma_1} f$$

$$= i\int_0^1 1 + t^2 - 1\int_0^1 t^2 - 2t + 2$$

$$= i\left[t + t^3\frac{1}{3}\right]_0^1 - 1\left[t^3\frac{1}{3} - t^2 + 2t\right]_0^1$$

$$= i\frac{4}{3} - \frac{4}{3}$$

9. Define $\gamma : [0, 2\pi] \to \mathbb{C}$ by $\gamma(t) = \exp(int)$ where n is some integer ((positive, negative, or zero). Show that $\int_\gamma 1/z\, dz = 2\pi in$

Given $\gamma(t) = e^{int}$ we have $\gamma'(t) = e^{int} in\, dt$, evaluating the integral

$$\int_\gamma z^{-1}\, dz = \int_0^{2\pi} e^{-int}e^{int} in\, dt = in\int_0^{2\pi} dt = 2\pi in$$

10. Define $\gamma(t) = e^{it}$ for $0 \le t \le 2\pi$ and find $\int_\gamma z^n\, dz$ for every integer n

Doing something similar to the previous problem we get

$$\int_\gamma z^n\, dz = i\int_0^{2\pi} e^{int}e^{it}\, dt = i\int_0^{2\pi} e^{it(n+1)}\, dt = i\left[\frac{e^{it(n+1)}}{i(n+1)}\right]$$

$$= \frac{1}{(n+1)}\left[e^{it(n+1)}\right]_0^{2\pi} = \frac{1}{(n+1)}(e^{i2\pi(n+1)} - e^0)$$

$$= \frac{1}{(n+1)}(1-1) = 0$$

That is only if $n \neq -1$ if $n = -1$ then by the previous problem $\int_\gamma z^{-1}\, dz = 2\pi i$

11. Let γ be a polygon $[1-i, 1+i, -1+i, -1-i, 1-i]$. Find $\int_\gamma \frac{1}{z}\,dz$

Define the following paths

$$\gamma_1 = 1+it \quad \gamma_2 = -t+i \quad \gamma_3 = -1-it \quad \gamma_4 = t-i \quad (\text{for } t \in [-1,1])$$

We get

$$\int_\gamma \frac{1}{z}\,dz = \int_{\gamma_1} \frac{1}{z}\,dz + \int_{\gamma_2} \frac{1}{z}\,dz + \int_{\gamma_3} \frac{1}{z}\,dz + \int_{\gamma_4} \frac{1}{z}\,dz$$

Recall by definition

$$\int_{\gamma_n} \frac{1}{z}\,dz = \int_a^b \frac{\gamma_n'(t)}{\gamma_n(t)}\,dt = \log \gamma_n(t)\Big|_a^b$$

Thus for each of the following γ_n we get

$$\log \gamma_1(t)\Big|_{-1}^1 = \log \frac{1+i}{1-i} = \log \frac{\sqrt{2}e^{i\pi/4}}{\sqrt{2}e^{-i\pi/4}} = \log e^{i\pi/2} = i\frac{\pi}{2}$$

$$\log \gamma_2(t)\Big|_{-1}^1 = \log \frac{-1+i}{1+i} = \log \frac{\sqrt{2}e^{3i\pi/4}}{\sqrt{2}e^{i\pi/4}} = \log e^{i\pi/2} = i\frac{\pi}{2}$$

$$\log \gamma_3(t)\Big|_{-1}^1 = \log \frac{-1-i}{-1+i} = \log \frac{\sqrt{2}e^{5i\pi/4}}{\sqrt{2}e^{3i\pi/4}} = \log e^{i\pi/2} = i\frac{\pi}{2}$$

$$\log \gamma_4(t)\Big|_{-1}^1 = \log \frac{1-i}{-1-i} = \log \frac{\sqrt{2}e^{7i\pi/4}}{\sqrt{2}e^{5i\pi/4}} = \log e^{i\pi/2} = i\frac{\pi}{2}$$

Thus, we get

$$\int_\gamma \frac{1}{z}\,dz = i\frac{\pi}{2} + i\frac{\pi}{2} + i\frac{\pi}{2} + i\frac{\pi}{2} = 2i\pi$$

12. Let $I(r) = \int_\gamma \frac{e^{iz}}{z}\,dz$ where $\gamma : [0,\pi] \to \mathbb{C}$ is defined by $\gamma(t) = re^{it}$. Show that $\lim_{r\to\infty} I(r) = 0$

We have the following paths $\gamma(t) = re^{it}$ and $\gamma'(t) = ire^{it}\,dt$. Thus we get the following integral

$$\int_\gamma \frac{e^{iz}}{z}\,dz = \int_0^\pi \frac{e^{i(re^{it})}}{re^{it}} \cdot ire^{it}\,dt$$

Simplifying, we get

$$= \int_0^\pi e^{i(re^{it})}\,dt$$

By triangle inequality

$$= \int_0^\pi e^{i(re^{it})} \le \int_0^\pi \left|e^{i(re^{it})}\right| |dt| = \int_0^\pi e^{-r\sin t}$$

In the interval $[0, \frac{\pi}{2}]$ we have that $\sin t \geq \frac{2}{\pi}t$ and in the interval $[\frac{\pi}{2}, \pi]$ we have that $\sin t \geq 2 - \frac{2x}{\pi}$. Thus we get

$$e^{-r\sin t} \leq e^{-r\left(\frac{2}{\pi}t\right)} \quad \text{and} \quad e^{-r\sin t} \leq e^{-r\left(2-\frac{2}{\pi}t\right)}$$

Splitting the integral we have

$$\int_0^\pi e^{-r\sin t} = \int_0^{\pi/2} e^{-r\sin t} + \int_{\pi/2}^\pi e^{-r\sin t} \leq \int_0^{\pi/2} e^{-r\left(\frac{2}{\pi}t\right)} + \int_{\pi/2}^\pi e^{-r\left(2-\frac{2}{\pi}t\right)}$$

Evaluating the integral we get

$$= -\frac{\pi e^{-\frac{2rt}{\pi}}}{2r}\Bigg|_0^{\pi/2} + \frac{\pi e^{\frac{2r(t-\pi)}{\pi}}}{2r}\Bigg|_{\pi/2}^{\pi}$$

$$= \frac{\pi - \pi e^{-r}}{2r} + \frac{\pi - \pi e^{-r}}{2r} = \frac{\pi - \pi e^{-r}}{r}$$

Letting $r \to \infty$ we get $I(r) = 0$

13. Find $\int_\gamma z^{-1/2}$ where γ is (a) γ is the upper half of the unit circle from +1 to -1: (b) γ is the lower half of the unit circle from +1 to -1.

(a) Parameterizing $\gamma(t)$ we get $\gamma : [0, \pi] \to \mathbb{C}$ defined by $\gamma(t) = e^{it}$ with $\gamma'(t) = ie^{it}\,dt$. Thus our integral is

$$\int_0^\pi e^{-it/2} ie^{it} = \int_0^\pi ie^{it-it/2} = \int_0^\pi ie^{it/2}$$

Evaluating the integral we get

$$2\ e^{it/2}\Big|_0^\pi = 2(e^{i\pi/2} - 1) = -2 + 2i$$

(b) Parameterizing $\gamma(t)$ we get $\gamma : [0, \pi] \to \mathbb{C}$ defined by $\gamma(t) = e^{-it}$ with $\gamma'(t) = -ie^{-it}\,dt$ Thus our integral is

$$\int_0^\pi -e^{it/2} ie^{-it} = \int_0^\pi -ie^{-it+it/2} = \int_0^\pi -ie^{-it/2}$$

Evaluating the integral we get

$$2\ e^{-it/2}\Big|_0^\pi = 2(e^{-i\pi/2} - 1) = -2 - 2i$$

14. Prove that if $\varphi : [a, b] \to [c, d]$ is continuous and $\varphi(a) = c,\ \varphi(b) = d$ then φ is one-to-one iff φ is strictly increasing

Since $[a, d]$ is an interval and φ is continuous, assume for contradiction that φ is not strictly increasing but still one-to-one. There exists $x < y < z$ such that

$$\varphi(x) < \varphi(y) > \varphi(z) \quad \text{or} \quad \varphi(x) > \varphi(y) < \varphi(z)$$

WLOG take the former case. By the intermediate value theorem, there exists $u \in (x, y)$ and $v \in (y, z)$ such that $\varphi(u) = \varphi(v)$, a contradiction.

For the converse, suppose that φ is strictly increasing. Thus for any $x < y$ we have $\varphi(x) < \varphi(y)$. Meaning φ is an injective map and thus is one-to-one.

15. Show that the relation in Definition 1.16 is an equivalence relation.

Definition 1.16. Let $\sigma : [c, d] \to \mathbb{C}$ and $\gamma : [a, b] \to \mathbb{C}$ be rectifiable paths. The path σ is *equivalent* to γ if there is a function $\varphi : [c, d] \to [a, b]$ which is continuous and strictly increasing, and with $\varphi(a) = c,\ \varphi(b) = d$ such that $\sigma = \gamma \circ \varphi$. We call the function φ a change of parameter.

Reflexivity:
Choose φ to be the identity. We get $\gamma \circ \varphi = \gamma$ and thus $\gamma \sim \gamma$

Symmetry:
If we have $\sigma = \gamma \circ \varphi$ and the function $\varphi : [a, b] \to [c, d]$ which is continuous and strictly increasing, then choose a new map $\psi := \varphi^{-1} : [c, d] \to [a, b]$. Thus, composing ψ on both sides we get

$$\sigma \circ \psi = \gamma \circ \varphi \circ \psi$$

$$\gamma = \sigma \circ \psi$$

And thus $\sigma \sim \gamma \implies \gamma \sim \sigma$

Transitivity: $(\tau \sim \sigma,\ \sigma \sim \gamma \implies \tau \sim \gamma)$
Suppose that $\tau = \sigma \circ \varphi_1$ where $\varphi_1 : [a, b] \to [c, d]$ is a change of parameter and $\sigma = \gamma \circ \varphi_2$ where $\varphi_1 : [c, d] \to [e, f]$ is also a change of parameter. Then

$$\tau = \gamma \circ (\varphi_2 \circ \varphi_1)$$

Since φ_1 and φ_2 are both continuous and strictly increasing, then $\varphi_2 \circ \varphi_1$ is also continuous and strictly increasing.

16. Show that if γ and σ are equivalent rectifiable paths then $V(\gamma) = V(\sigma)$

Let $\gamma : [a,b] \to \mathbb{C}$ and $\sigma : [c,d] \to \mathbb{C}$ be equivalent paths. Thus, there exists $\varphi : [c,d] \to [a,b]$ and $\sigma = \gamma \circ \sigma$. For any partition $P = \{c = s_0 < ... < s_m = d\}$, the set $t_k = \varphi(s_k)$. Then $\{a = t_0 < ... < t_m = b\}$ is also a partition since φ is strictly increasing and we get

$$v(\sigma, P) = \sum_{k=1}^{m} |\sigma(s_k) - \sigma(s_{k-1})| = \sum_{k=1}^{m} |\gamma(t_k) - \gamma(t_{k-1})| = v(\gamma, \varphi(P)) \leq V(\gamma)$$

Taking the supremum over all partitions of P we get $V(\sigma) \leq V(\gamma)$

Since $\psi = \varphi^{-1}$ is also continuous and strictly increasing, we get $\gamma = \sigma \circ \psi$ and thus $V(\gamma) \leq V(\sigma)$, therefore.

$$V(\sigma) = V(\gamma)$$

17. Show that if $\gamma : [a,b] \to \mathbb{C}$ is a path then there is an equivalent path $\sigma : [0,1] \to \mathbb{C}$.

Just give the following reparameterization

$$\varphi : [0,1] \to [a,b], \quad \varphi(t) = a + (b-a)t$$

Which is strictly increasing, linear (and thus continuous) and $\varphi(0) = a$, $\varphi(1) = b$.

18. Prove Proposition 1.17.

1.17 Proposition. Let γ be a rectifiable curve and suppose that f is a function continuous on γ. Then:

(a) $\int_\gamma f = -\int_{-\gamma} f$

(b) $\left|\int_\gamma f\right| \leq \int_\gamma |f|\,|dz| \leq V(\gamma) \sup[|f(z)| : z \in \gamma]$

(c) If $c \in \mathbb{C}$ then $\int_\gamma f(z)\,dz = \int_{\gamma+c} f(z-c)\,dz$

(a) Recall the definition of a line integral, for some path $\gamma(t) : [a,b] \to \mathbb{C}$ (and defining $s := -t$) we get

$$-\int_{-\gamma} f = \int_{-b}^{-a} f(\gamma(-t))(-\gamma'(t))\,dt = -\left(-\int_a^b f(\gamma(s))(-\gamma'(s))\,(-ds)\right) = \int_\gamma f$$

(b) Take any fine δ partition $P = \{t_k\}$ of γ such that our integral is

$$\int_\gamma f = \sum_{k=1}^{m} f(\tau_k)[\gamma(t_k) - \gamma(t_k - 1)]$$

By triangle inequality

$$\left|\int_\gamma f\right| = \left|\sum_{k=1}^{m} f(\tau_k)[\gamma(t_k) - \gamma(t_k - 1)]\right| \leq \sum_{k=1}^{m} |f(\tau_k)||\gamma(t_k) - \gamma(t_k - 1)| = \int_\gamma |f||dz|$$

Passing the limit we get

$$\int_\gamma |f||dz| = \int_a^b |f(\gamma(t))||\gamma'(t)||dt| \leq (\sup|f(z)|)(|\gamma(a)| - |\gamma(b)|) = V(\gamma) \sup_{z\in\gamma} |f(z)|$$

(c) Expressing the integral $\int_{\gamma+c} f(z-c)\,dz$ as a sum, we get

$$\sum_{k=1}^{m} f((\gamma+c)(\tau_k) - c)((\gamma+c)(t_k) - (\gamma+c)(t_{k-1})) = \sum_{k=1}^{m} f(\tau_k)[\gamma(t_k) - \gamma(t_k - 1)] = \int_\gamma f$$

19. In the proof of Case II of Theorem 1.18, where was the assumption that γ lies in a disk used.

20. Let $\gamma(t) = 1 + e^{it}$ for $0 \leq t \leq 2\pi$ and find $\int_\gamma (z^2 - 1)^{-1} dz$

We have $\gamma'(t) = ie^{it}\,dt$, you know the rest

$$\int_\gamma \frac{1}{z^2 - 1}\,dz = \int_0^{2\pi} \frac{ie^{it}}{(e^{it} + 1)^2 - 1}\,dt =$$

$$= \int_0^{2\pi} \frac{ie^{it}}{e^{2it} + 2e^{it}}\,dt = \int_0^{2\pi} \frac{i}{e^{it} + 2}\,dt$$

Perform a the following substitution

$$u = -it \quad du = -i\,dt$$

$$\int_0^{2\pi} \frac{i}{e^{it} + 2}\,dt = -\int_{t=0}^{t=2\pi} \frac{1}{e^{-u} + 2}\,du = -\int_{t=0}^{t=2\pi} \frac{e^u}{2e^u + 1}\,du$$

Perform another substitution

$$v = 2e^u + 1 \quad dv = 2e^u du$$

$$= -\int_{t=0}^{t=2\pi} \frac{1}{e^{-u} + 2}\,du = -\frac{1}{2}\int_{t=0}^{t=2\pi} \frac{1}{v}\,dv$$

$$= -\frac{1}{2}\int_{t=0}^{t=2\pi} d(\ln|v| + i\arg(v))\,dv$$

Undo the substitution $v = 2e^u + 1$

$$= -\frac{1}{2}\left[(\ln|2e^u+1| + i\arg(v))\right]_{t=0}^{t=2\pi} = -\frac{1}{2}(-2\pi i) = \pi i$$

21. Let $\gamma(t) = 2e^{it}$ for $-\pi \le t \le \pi$ and find $\int_\gamma (z^2-1)^{-1}dz$

We have $\gamma'(t) = 2ie^{it}\,dt$, here we go again

$$\int_\gamma \frac{1}{z^2-1}\,dz = 2i\int_{-\pi}^{\pi} \frac{e^{it}}{4e^{2it}-1}\,dt =$$

Perform a the following substitution

$$u = -it \quad du = -i\,dt$$

$$2i\int_{t=-\pi}^{t=\pi} \frac{e^{-u}}{4e^{-2u}-1}\,dt = -2\int_{t=-\pi}^{t=\pi} \frac{e^u}{e^{2u}-4}\,dt =$$

Perform another substitution

$$v = e^u \quad dv = e^u du$$

$$-2\int_{t=-\pi}^{t=\pi} \frac{1}{v^2-4}\,dt$$

$$= -2\left(\frac{\ln(v-2) + i\arg(v-2)}{4} - \frac{\ln(v+2)+i\arg(v+2)}{4}\right)$$

Undo the substitution $v = e^u$

$$= \left(\frac{\ln(e^u+2) + i\arg(e^u+2)}{2} - \frac{\ln(e^u-2)+i\arg(e^u-2)}{2}\right)$$

Undo the substitution $u = -it$

$$= \left.\frac{\ln(e^{-it}+2) + i\arg(e^{-it}+2)}{2} - \frac{\ln(e^{-it}-2)+i\arg(e^{-it}-2)}{2}\right|_{-\pi}^{\pi}$$

$$= 0$$

(just trust me bro I just did my best to copy the steps from an integral solver)

22. Show that if F_1 and F_2 are primitives for $f : G \to \mathbb{C}$ and G is connected, then there is a constant c such that $F_1(z) = c + F_2(z)$ for each z in G. Define a new function $H := F_1 - F_2$. Thus H is analytic in G and thus for all $z \in G$ we have

$$H'(z) = F_1'(z) - F_2'(z) = f(z) - f(z) = 0$$

Since the primitive gives the endpoint difference, then for some rectifiable path $\gamma : [\alpha, \beta] \to G$ we have

$$\int_\gamma H' = H'(\alpha) - H(\beta)$$

But the integrand is identically 0, so the integral is 0 and thus $H(\alpha) = H(\beta)$. Since α and β are arbitrary, we get that H is constant on G. Meaning $c := F_1(\alpha) - F_2(\alpha)$

$$F_1(z) = F_2(z) + c$$

23. Let γ be a closed rectifiable curve in G and $a \notin G$. Show that for $n \geq 2$ $\int_\gamma (z-a)^{-n}\, dz = 0$

Since $a \notin G$, the function has the following primitive

$$F(z) = \frac{1}{1-n}(z-a)^{1-n}$$

Which is analytic on all of G and this primitivity implies that for some path $\gamma \subset G$ with endpoints α and β we get

$$\int_\gamma (z-a)^{-n}\, dz = F(\beta) - F(\alpha)$$

Since γ is closed $\alpha = \beta$ and thus

$$\int_\gamma (z-a)^{-n}\, dz = 0 \quad (n \geq 2)$$

24. Prove the following integration by parts formula. Let f and g be analytic in G and let γ be a rectifiable curve from a to b in G. Then

$$\int_\gamma fg' = f(b)g(b) - f(a)g(a) - \int_\gamma f'g$$

Be cause f and g are analytic, the product fg is also analytic and by the product rule

$$(fg)' = f'g + fg'$$

Taking the integral of both sides along γ we get

$$\int_\gamma (fg)'\, dz = \int_\gamma f'g\, dz + \int_\gamma fg'\, dz$$

Since fg is the primitive of $(fg)'$ on G we get

$$\int_\gamma (fg)'\,dz = f(b)g(b) - f(a)g(a)$$

Rearranging the terms we get

$$\int_\gamma fg'\,dz = f(b)g(b) - f(a)g(a) - \int_\gamma f'g\,dz$$

4.2 Power series representation of analytic functions

1. Show that the function defined by (2.2) is continuous

For a compact set $K := [a,b] \times [c,d]$ and $\varphi : K \to \mathbb{C}$ be a continuous function and and define $g : [c,d] \to \mathbb{C}$ by

$$g(t) = \int_a^b \varphi(s,t)\,ds$$

Fix $\varepsilon > 0$ and arbitrary $t_0 \in [c,d]$. Since φ is continuous on K, it's also uniformly continuous and thus there exists $\delta > 0$ such that for all $s \in [a,b]$ and $t \in [c,d]$ with $|t - t_0| < \delta$ we have

$$|\varphi(s,t) - \varphi(s,t_0)| < \frac{\varepsilon}{b-a}$$

Then, we have

$$|g(t) - g(t_0)| = \left| \int_a^b \varphi(s,t) - \varphi(s,t_0)\,ds \right| \leq \int_a^b |\varphi(s,t) - \varphi(s,t_0)|\,|ds|$$

$$\leq \int_a^b \frac{\varepsilon}{b-a}\,ds = \frac{\varepsilon}{b-a}(b-a) = \varepsilon$$

Since t_0 was arbitrary, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|g(t) - g(t_0)| < \varepsilon$ and thus g is continuous. □

2. Prove the following analogue of Leibniz's rule. Let G be an open set and let γ be a rectifiable curve in $\mathbb{C}$. Suppose that $\varphi : \gamma \times G \to \mathbb{C}$ is a continuous function and define $g : G \to \mathbb{C}$ by

$$g(z) = \int_\gamma \varphi(w,z)\,dw$$

Then g is continuous. If $\frac{\partial \varphi}{\partial z}$ exists, for each (w,z) in $\{\gamma\} \times G$ and is continuous then g is analytic and

$$g'(z) = \int_\gamma \frac{\partial \varphi}{\partial z}(w,z)\, dw$$

Suppose $\gamma : [a,b] \to \mathbb{C}$ Choose a compact set $K \subset\subset G$ that contains γ. Thus we know φ is uniformly continuous on γ. Fix, $z_0 \subset K$ and $\varepsilon > 0$. Since φ is uniformly continuous, choose $\delta > 0$ such that

$$|z - z_0| < \delta \implies \sup_{t\in[a,b]} |\varphi(\gamma(t), z) - \varphi(\gamma(t), z_0)| < \frac{\varepsilon}{L}$$

Where $L := \int_a^b |\gamma'(t)|\, dt$ Thus we get

$$|g(z) - g(z_0)| = \left| \int_\gamma (\varphi(w,z) - \varphi(w,z_0))\, dw \right|$$

$$\left| \int_a^b (\varphi(\gamma(t),z) - \varphi(\gamma(t),z_0))\, \gamma'(t)\, dt \right| \le \int_a^b |\varphi(\gamma(t),z) - \varphi(\gamma(t),z_0)||\gamma'(t)| dt$$

$$\le \sup |\varphi(w,z) - \varphi(w,z_0)| \int_a^b |\gamma'(t)| dt$$

$$< L\frac{\varepsilon}{L} = \varepsilon$$

Thus φ is continuous.

To show g is analytic, suppose $\frac{\partial \varphi}{\partial z}$ exists and is continuous on $\gamma \times G$. This implies that the difference of quotients converge to $\frac{\partial \varphi}{\partial z}$ uniformly in $w \in \gamma$. Fix z_0 and $\varepsilon > 0$. Consider the difference quotient

$$\frac{g(z) - g(z_0)}{z - z_0} = \int_\gamma \frac{\varphi(w,z) - \varphi(w,z_0)}{(z - z_0)}$$

Since φ is differentiable in z for each $w \in \gamma$ we can define a function

$$f(t) = \varphi(w, z_0 + t(z - z_0)) \quad \text{for } t \in [0,1]$$

Then $f(0) = \varphi(w, z_0)$ and $f(1) = \varphi(w,z)$. By FTC

$$f(1) - f(0) = \int_0^1 f'(t)\, dt$$

By the chain rule

$$f'(t) = \frac{\partial \varphi}{\partial z}(w, z_0 + t(z - z_0))(z - z_0)$$

Thus,

$$\varphi(w,z)-\varphi(w,z_0)=(z-z_0)\int_0^1 \frac{\partial\varphi}{\partial z}(w,z_0+t(z-z_0))\,dt$$

$$\frac{\varphi(w,z)-\varphi(w,z_0)}{(z-z_0)}=\int_0^1 \frac{\partial\varphi}{\partial z}(w,z_0+t(z-z_0))\,dt$$

Substituting the difference quotient

$$\frac{g(z)-g(z_0)}{(z-z_0)}=\int_\gamma\left[\int_0^1 \frac{\partial\varphi}{\partial z}(w,z_0+t(z-z_0))\,dt\right]$$

Since the partial is continuous in the compact set, we can interchange the order of the integrals by Fubini's theorem

$$\frac{g(z)-g(z_0)}{(z-z_0)}=\int_0^1\left[\int_\gamma \frac{\partial\varphi}{\partial z}(w,z_0+t(z-z_0))\,dt\right]$$

For simplicity, define a new function

$$F(z)=\int_0^1\left[\int_\gamma \frac{\partial\varphi}{\partial z}(w,z_0+t(z-z_0))\,dt\right]$$

For every $\varepsilon>0$ there exists $\delta>0$ such that $w\in\gamma$, $t\in[0,1]$, and z with $|z-z_0|<\delta$

$$\left|\frac{\partial\varphi}{\partial z}(w,z_0+t(z-z_0))-\frac{\partial\varphi}{\partial z}(w,z_0)\right|<\frac{\varepsilon}{L}$$

Thus

$$\left|F(z)-\int_\gamma \frac{\partial\varphi}{\partial z}(w,z_0)\,dw\right|\le\int_0^1\int_\gamma\left|\frac{\partial\varphi}{\partial z}(w,z_0+t(z-z_0))-\frac{\partial\varphi}{\partial z}(w,z_0)\right||dw|\,dt$$

$$<L\frac{\varepsilon}{L}=\varepsilon$$

Therefore

$$\lim_{z\to z_0}\frac{g(z)-g(z_0)}{(z-z_0)}=\int_\gamma \frac{\partial\varphi}{\partial z}(w,z_0)\,dw$$

Thus g is analytic and sending $z\to z_0$ we get

$$g'(z_0)=\int_\gamma \frac{\partial\varphi}{\partial z}(w,z_0)\,dw$$

Since z_0 was arbitrary

$$g'(z)=\int_\gamma \frac{\partial\varphi}{\partial z}(w,z)\,dw$$

3. Suppose that γ is a rectifiable curve in $\mathbb{C}$ and φ is defined and continuous on $\{\gamma\}$. Use Exercise 2 to show that

$$g(z) = \int_\gamma \frac{\varphi(w)}{w - z}$$

is analytic on $\mathbb{C} - \gamma$ and

$$g^{(n)}(z) = n! \int_\gamma \frac{\varphi(w)}{(w - z)^{n-1}}$$

For simplicity, define a function

$$\Phi(w, z) := \frac{\varphi(w)}{w - z}$$

The map $(w, z) \mapsto 1/(w - z)$ is continuous on $\mathbb{C} - \{(w, z) : w = 0\}$ and hence, since $w \neq z$ and $w \in \{\gamma\}$, Φ is continuous on $\gamma \times (\mathbb{C} - \{(w, z) : w = 0\})$. Thus $\partial\Phi/\partial z$ exists. For some fixed w we have

$$\frac{\partial}{\partial z}\frac{1}{w - z} = \frac{1}{(w - z)^2}$$

$$\frac{\partial \Phi}{\partial z}(w, z) = \frac{\varphi(w)}{(w - z)^2}$$

By the previous exercise we get the function g

$$g(z) := \int_\gamma \Phi \, dw = \int_\gamma \frac{\varphi(w)}{w - z}$$

Is analytic on G and we get

$$g'(z) = \int_\gamma \frac{\partial \Phi}{\partial z}(w, z) \, dw = \int_\gamma \frac{\varphi(w)}{(w - z)^2} \, dw$$

For the inductive step define $\Phi_n(w, z) := \varphi/(w - z)^{n+1}$ Thus we get

$$\frac{\partial^n}{\partial z^n}\frac{1}{w - z} = n!\frac{1}{(w - z)^{n+1}}$$

Assuming Φ_n is contentious then

$$\frac{\partial \Phi_{n+1}}{\partial z} = n\Phi_n$$

Thus we get

$$g^{(n)}(z) = n! \int_\gamma \frac{\varphi(w)}{(w - z)^{n+1}}$$

4. (a) Prove Abel's Theorem: Let $\sum a_n(z-a)^n$ have radius of convergence to 1 and suppose that $\sum a_n$ converges to A. Prove that

$$\lim_{r\to 1-} \sum a_n r^n = A$$

(Hint: Find a summation formula which is the analogue of integration by parts.)
(b) Use Abel's Theorem to prove that $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - ...$

Let

$$s_n := a_0 + a_1 + ...a_n$$

Since the series converges to A, we have $s_n \to A$ as $n \to \infty$. For any integer N and $0 \leq r < 1$

$$\sum_{n=0}^{N} a_n r^n = \sum_{n=0}^{N} s_n r^n - \sum_{n=0}^{N-1} s_n r^{n+1}$$

$$s_N r^N + \sum_{n=0}^{N-1} s_n r^n - \sum_{n=0}^{N-1} s_n r^{n+1} = s_N r^N + (1-r)\sum_{n=1}^{N-1} s_n r^n$$

Sending $N \to \infty$, for $r < 1$ we get that $s_n r^n \to A \cdot 0 = 0$ we get

$$\sum_{n=0}^{\infty} a_n r^n = (1-r)\sum_{n=1}^{\infty} s_n r^n$$

Since $r < 1$ we also get

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

So we get

$$\sum_{n=0}^{\infty} a_n r^n - A = (1-r)\sum_{n=0}^{\infty}(s_n - A)r^n$$

Thus for any $\varepsilon > 0$ we can choose N such that $|s_n - A| < \varepsilon/2$ for all $n > N$. Thus we get

$$\left|(1-r)\sum_{n=0}^{\infty}(s_n - A)r^n\right| \leq (1-r)\sum_{n=0}^{N-1}|s_n - A|r^n + (1-r)\sum_{n=N}^{\infty}\frac{\varepsilon}{2}r^n$$

The first sum is $(1-r)$ times some finite constant (because it's a finite sum), so it tends to 0 as $r \to 1^-$. The second sum converges to the following

$$(1-r)\frac{\varepsilon}{2}\frac{1}{1-r} = \frac{\varepsilon}{2}$$

Thus for some r sufficiently close to 1, $LHS \leq \varepsilon/2 < \varepsilon$ and thus

$$\lim_{r\to 1-}\sum_{n=0}^{\infty} a_n r^n = A$$

Consider the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = \log(1+x)$$

Which converges for $|x| < 1$ and such that for all $-1 < x < 1$

$$\lim_{x \to 1-} \log(1+x) = \log 2$$

Thus by Abel's theorem

$$\lim_{x \to 1-} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = \lim_{x \to 1-} \log(1+x) = \log 2$$

5. Give the power series expansion of $\log z$ about $z = i$ and find its radius of convergence

Write $z = i + w$. Then

$$\log z = \log(i+w) = \log i + \log\left(1 + \frac{w}{i}\right) = \log i + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \left(\frac{w}{i}\right)^n$$

$$\log i - \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \left(\frac{z}{i} - 1\right)^n = \log i - \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(-i)^n}{n} (z-i)^n$$

$$\log i - \sum_{n=1}^{\infty} \frac{i^n}{n} (z-i)^n$$

Also

$$\log i = \log e^{i\pi/2} = i\frac{\pi}{2}$$

So we get

$$i\frac{\pi}{2} - \sum_{n=1}^{\infty} \frac{i^n}{n} (z-i)^n$$

Recall that the radius of convergence r can be expressed as

$$\frac{1}{r} = \lim_{n \to \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

Where $a_n = i^n/n$. So

$$\frac{1}{r} = \lim_{n \to \infty} \left| \frac{i^{n+1}}{n+1} \frac{n}{i^n} \right| = \lim_{n \to \infty} \left| i \frac{n}{n+1} \right| = 1 \quad r = 1$$

6. Give the power series expansion of $\sqrt{z}$ about $z = 1$ and find its radius of convergence

Suppose we're taking the principal branch. Write $z = 1 + w$. Then

$$\sqrt{z} = \sqrt{1+w} = \sum_{n=0}^{\infty} \binom{1/2}{n} w^n$$

Where

$$a_n := \binom{1/2}{n} = \frac{1}{n!2^n} \prod_{k=0}^{n-1} (1-2k)$$

So our sum is

$$\sum_{n=0}^{\infty} a_n (z-1)^n$$

Recall that the radius of convergence r can be expressed as

$$\frac{1}{r} = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

Where $a_n = \binom{1/2}{n}$. So

$$\frac{1}{r} = \lim_{n\to\infty} \left| \frac{\binom{1/2}{n+1}}{\binom{1/2}{n}} \right| = \lim_{n\to\infty} \left| \frac{\frac{1}{2} - n}{n+1} \right| = 1 \quad r = 1$$

7. Use the results of this section to evaluate the following integrals

For a few of these, we'll use **Corollary 2.13:**

$$f^{(n)}(a) = \frac{n!}{2\pi i} \int_\gamma \frac{f(w)}{(w-a)^{n+1}}\, dw$$

Where $\gamma(t) = a + re^{it},\ 0 \le t \le 2\pi$

(a)

$$\int_\gamma \frac{e^{iz}}{z^2}\, dz, \quad \gamma(t) = e^{it}, \quad 0 \le t \le 2\pi$$

Let $a = 0$, $r = 1$, $n = 1$, and $f(z) = e^{iz}$ and we get

$$ie^{i0} = \frac{1}{2\pi i} \int_\gamma \frac{e^{iz}}{z^2}\, dz$$

$$-2\pi i = \int_\gamma \frac{e^{iz}}{z^2}\, dz$$

(b)

$$\int_\gamma \frac{dz}{z-a}, \quad \gamma(t) = a + re^{it}, \quad 0 \le t \le 2\pi$$

Let $a = a$, $r = r$, $n = 0$, and $f(z) = 1$ and we get

$$1 = \frac{0!}{2\pi i} \int_\gamma \frac{1}{z-a}\, dz$$

$$2\pi i = \int_\gamma \frac{1}{z-a}\, dz$$

(c)

$$\int_\gamma \frac{\sin z}{z^3}\, dz, \quad \gamma(t) = e^{it}, \quad 0 \le t \le 2\pi$$

Let $a = 0$, $r = 1$, $n = 2$, and $f(z) = \sin z$. So $f''(a) = 0$ and we get

$$0 = \frac{2!}{2\pi i} \int_\gamma \frac{\sin z}{z^3}\, dz$$

$$0 = \int_\gamma \frac{\sin z}{z^3}\, dz$$

(d)

$$\int_\gamma \frac{\log z}{z^n}\, dz, \quad \gamma(t) = 1 + \frac{1}{2}e^{it}, \quad 0 \le t \le 2\pi \text{ and } n \ge 0$$

Give the power expansion of the logarithm about $z = 1$

$$\log z = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}(z-1)^k \quad |z-1| < 1$$

So by our integral

$$\int_\gamma \frac{\log z}{z^3} = \int_\gamma \frac{1}{z^3} \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}(z-1)^k$$

$$= \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \int_\gamma \frac{(z-1)^k}{z^3}$$

The integrand $(z-1)^k/z^3$ is analytic in a neighborhood in the γ disk and thus every term is 0. Thus

$$\int_\gamma \frac{\log z}{z^n}\, dz = 0$$

8. Use a Möbius transformation to show that Proposition 2.15 holds if the disk $B(a,R)$ us a replaced by the half plane

2.15 Proposition. Let f be analytic in the disc $B(a,R)$ and suppose that γ is a closed rectifiable curve in $B(a,R)$. Then

$$\int_\gamma f = 0$$

Define the right half plane H and recall Möbius maps are analytic bijections. Choose a Möbius transformation $T(z)$ that maps the half plane to the unit disk.

$$T(z) = \frac{w-1}{w+1}$$

Let f be analytic on H. Define $g(z)$ such that

$$g := (f \circ T^{-1}) \cdot T^{-1}$$

Which is analytic by composition of analytic functions and let $\Gamma = T \circ \gamma$ be a closed rectify able curve in the unit disk. By the chain rule

$$\int_\gamma f(z)\,dz = \int (f\circ\gamma)(t)\,\gamma' t\,dt = \int (f\circ T^{-1}\circ\Gamma)(t)\cdot (T^{-1})'(\Gamma(t))(\Gamma'(t))\,dt$$

$$= \int g(\Gamma(t))\Gamma'(t) = \int_\Gamma g$$

By proposition 2.15

$$\int_\Gamma g = 0 \int_\gamma f$$

Use Corollary 2.13 to evaluate the following integrals:

Corollary 2.13:

$$f^{(n)}(a) = \frac{n!}{2\pi i}\int_\gamma \frac{f(w)}{(w-a)^{n+1}}\,dw$$

(a)

$$\int_\gamma \frac{e^z - e^{-z}}{z^n}\,dz \quad n\in\mathbb{N}, \quad \gamma(t) = e^{it},\ 0\le t\le 2\pi$$

Let $a=0$, $r=1$, $n=n-1$, $f(z)=e^z-e^{-z}$ and thus $f^{(n-1)}(z) = e^z + (-1)^n e^{-z}$

$$f^{(n-1)}(0) = e^0 + (-1)^n e^{-0} = 1 + (-1)^n$$

thus we get

$$1 + (-1)^n = \frac{(n-1)!}{2\pi i}\int_\gamma \frac{e^z - e^{-z}}{z^n}$$

$$\int_\gamma \frac{e^z - e^{-z}}{z^n} = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \frac{4\pi i}{(n-1)!} & \text{if } n \text{ is even} \end{cases}$$

(b)

$$\int_\gamma \frac{dz}{(z-1/2)^n} \quad n \in \mathbb{N}, \quad \gamma(t) = e^{it}, \ 0 \le t \le 2\pi$$

Let $a = 1/2$, $r = 1$, $n = n-1$, $f(z) = 1$ and thus $f^{(n-1)}(a) = 0$ and $f(a) = 1$ Thus we get for $n = 1$

$$1 = \frac{1}{2\pi i} \int_\gamma \frac{dz}{z-1/2}$$

$$\int_\gamma \frac{dz}{z-1/2} = 2\pi i$$

Thus we get for $n > 1$

$$0 = \frac{(n-1)!}{2\pi i} \int_\gamma \frac{dz}{(z-1/2)^n}$$

$$\int_\gamma \frac{dz}{(z-1/2)^n} = 0$$

(c)

$$\int_\gamma \frac{dz}{z^2+1} \quad \gamma(t) = 2e^{it}, \ 0 \le t \le 2\pi$$

Split the integral using partial fractions

$$\int_\gamma \frac{dz}{z^2+1} = \frac{1}{2i} \int_\gamma \frac{dz}{z-i} - \frac{1}{2i} \int_\gamma \frac{dz}{z+i}$$

Evaluate the first integral

$$\frac{1}{2i} \int_\gamma \frac{dz}{z-i}$$

Let $a = i$, $r = 2$, $n = 0$, $f(z) = 1$ and thus

$$1 = \frac{1}{2\pi i} \int_\gamma \frac{dz}{z-i}$$

$$\frac{1}{2i} \int_\gamma \frac{dz}{z-i} = \pi i$$

Evaluate the 2nd integral

$$\frac{1}{2i} \int_\gamma \frac{dz}{z+i}$$

Let $a = -i$, $r = 2$, $n = 0$, $f(z) = 1$ and thus

$$1 = \frac{1}{2\pi i} \int_\gamma \frac{dz}{z+i}$$

$$\frac{1}{2i}\int_\gamma \frac{dz}{z+i} = \pi$$

Summing the integrals we get

$$\int_\gamma \frac{dz}{z^2+1} = \pi - \pi = 0$$

(d)

$$\int_\gamma \frac{\sin z}{z}\,dz \quad \gamma(t) = e^{it},\ 0 \le t \le 2\pi$$

Let $a = 0$, $r = 1$, $n = 0$, $f(z) = \sin z$, $f(0) = 0$ and thus

$$0 = \frac{1}{2\pi i}\int_\gamma \frac{\sin z}{z}\,dz$$

$$\int_\gamma \frac{\sin z}{z}\,dz = 0$$

(e)

$$\int_\gamma \frac{z^{1/m}}{(z-1)^m}\,dz \quad \gamma(t) = 1 + \frac{1}{2}e^{it},\ 0 \le t \le 2\pi$$

Let $a = 1$, $r = 1$, $n = m-1$, $f(z) = z^{1/m}$ and with this mess we get

$$f^{(m-1)}(1) = \prod_{k=1}^{m-2}\left(\frac{1}{m} - k\right)$$

So

$$\int_\gamma \frac{z^{1/m}}{(z-1)^m}\,dz = 2\pi i \prod_{k=1}^{m-2}\left(\frac{1}{m} - k\right)$$

10. Evaluate $\int_\gamma \frac{z^2+1}{z(z^2+4)}\,dz$ Where $\gamma(t) = re^{it}$, $d0 \le t \le 2\pi$ for all possible values of r, $0 < r < 2$ and $2 < r < \infty$

First do partial fraction decomposition.

$$\frac{z^2+1}{z(z^2+4)} = \frac{A}{z} + \frac{B}{z+2i} + \frac{C}{z-2i} = \frac{A(z^2+4) + B(z)(z-2i) + C(z)(z+2i)}{z(z^2+4)}$$

$$A + B + C = 1 \quad -2iB + 2iC = 0 \quad 4A = 1$$

$$A = \frac{1}{4} \quad B = \frac{3}{8} \quad C = \frac{3}{8}$$

$$\int_\gamma \frac{z^2+1}{z(z^2+4)}\,dz = \frac{1}{4}\int_\gamma \frac{1}{z}\,dz + \frac{3}{8}\int_\gamma \frac{1}{z+2i}\,dz + \frac{3}{8}\int_\gamma \frac{1}{z-2i}\,dz$$

For $|r| < 2$ we get that the only singularities in the ball $B(0, r)$ are $z = 0$. Thus we only need to evaluate the first integral

$$1 = \frac{1}{2\pi i} \int_\gamma \frac{1}{z}\, dz$$

$$\frac{1}{4} \int_\gamma \frac{1}{z}\, dz = \frac{\pi i}{2}$$

For $|r| > 2$ we evaluate all the integrals

$$1 = \frac{1}{2\pi i} \int_\gamma \frac{1}{z + 2i}\, dz$$

$$\frac{3}{8} \int_\gamma \frac{1}{z + 2i}\, dz = \frac{3\pi i}{4}$$

And similarly

$$\frac{3}{8} \int_\gamma \frac{1}{z - 2i}\, dz = \frac{3\pi i}{4}$$

$$\int_\gamma \frac{z^2 + 1}{z(z^2 + 4)}\, dz = \frac{\pi i}{2} + \frac{3\pi i}{4} + \frac{3\pi i}{4} = 2\pi i$$

11. Find the domain of analyticity of

$$f(z) = \frac{1}{2i} \log\left(\frac{1 + iz}{1 - iz}\right)$$

also, show that $\tan f(z) = z$ **(i.e.,** f **is a branch of** $\arctan z$**). Show that**

$$f(z) = \sum_{k=0}^{\infty} (-1)^k \frac{z^{2k+1}}{2k + 1} \text{ for } |z| < 1$$

(Hint: see Exercise III. 3.19.)

Suppose we're working with the principle branch of the logarithm. Which is defined on $\mathbb{C} - (\infty, 0]$ So we require the internal Möbius transformation $\notin (\infty, 0]$. So set

$$\frac{1 + iz}{1 - iz} = t, \quad t \leq 0$$

$$1 + iz = t - itz \quad 1 - t = -iz - itz \quad 1 - t = -iz(1 + t)$$

$$z = -i\frac{t - 1}{t + 1}$$

Sending $t \to -t$ we get that $t \in [0, \infty)$

$$z = i\frac{t + 1}{1 - t}$$

Hence as t varies, the imaginary axis where $|z| \geq 1$ and thus f is analytic where

$$\{z : |\Im z| < 1\}$$

To show that $\tan fz = z$ then for simplicity let $w = f(z)$.

$$2wi = \log\left(\frac{1+iz}{1-iz}\right)$$

$$e^{2wi} = \frac{1+iz}{1-iz}$$

Stealing algebraic manipulation from the previous part of the exercise we get

$$z = \frac{e^{2wi}-1}{i(1+e^{2wi})}$$

Multiply the numerator and denominator by e^{-iw}

$$z = \frac{e^{2wi}-1}{i(1+e^{2wi})} \cdot \frac{e^{-iw}}{e^{-iw}} = \frac{e^{wi}-e^{-iw}}{i(e^{-iw}+e^{wi})} = \frac{\sin w}{\cos w} = \tan w$$

Thus

$$z = \tan w = \tan f(z)$$

To show the Power series of $f(z)$, rewrite as

$$f(z) = \frac{1}{2i}\left(\log(1+iz) - \log(1-iz)\right)$$

For $|z| < 1$, we expand the Taylor series for both terms

$$\log(1+iz) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(iz)^n}{n} \qquad \log(1-iz) = -\sum_{n=1}^{\infty} \frac{(iz)^n}{n}$$

Subtracting we get

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}(iz)^n}{n} + \frac{(iz)^n}{n} = \sum_{n=1}^{\infty}(1+(-1)^{n+1})\frac{(iz)^n}{n}$$

If n is even then the term is zero. Thus n is odd and we write $n = 2k+1$ and we get

$$2\sum_{k=0}^{\infty} i^{2k+1}\frac{z^{2k+1}}{2k+1}$$

$$\log\left(\frac{1+iz}{1-iz}\right) = 2i\sum_{k=0}^{\infty}(-1)^k\frac{z^{2k+1}}{2k+1}$$

Thus

$$f(z) = \sum_{k=0}^{\infty}(-1)^k\frac{z^{2k+1}}{2k+1} \text{ for } |z| < 1$$

12. Show that

$$\sec z = 1 + \sum_{k=1}^{\infty} \frac{E_{2k}}{(2k)!} z^{2k}$$

for some constants E_2, E_4.... These numbers are called Euler's constants. What is the radius of convergence of this series? Use the fact that $1 = \cos z \cdot \sec z$ to show that

$$E_{2n} - \binom{2n}{2n-2} E_{2n-2} + \binom{2n}{2n-4} E_{2n-4} ... (-1)^n \binom{2n}{2} E_2 + (-1)^n = 0$$

Evaluate E_2, E_4, E_6, E_8.($E_{10} = 50521$ and $E_{12} = 2702765$)

Similar to $\cos z$, $\sec z$ is also an even function so we get

$$\sum_{k=0}^{\infty} \frac{E_{2k}}{(2k)!} z^{2k}$$

Since for some constant E_{2k} (which we have no clue yet). Since $\sec 0 = 1$ then $E_0 = 1$ and we get

$$\sec z = 1 + \sum_{k=1}^{\infty} \frac{E_{2k}}{(2k)!} z^{2k}$$

To define E_{2k}, multiply the secant and cosine power series (both starting at k=0 for simplicity)

$$\cos z \cdot \sec z = \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} z^{2k} \right) \left(\sum_{k=0}^{\infty} \frac{E_{2k}}{(2k)!} z^{2k} \right) = 1 + 0$$

Multiplying these series and equating coefficients of z^{2N} for $N \geq 1$ we get

$$\sum_{k=0}^{N} \frac{E_{2k}}{(2k)!} \cdot \frac{(-1)^{N-k}}{(2(N-k))!} = 0$$

Multiplying through by $(2N)!$ we get

$$\sum_{k=0}^{N} E_{2k} (-1)^{N-k} \cdot \frac{(2N)!}{(2k)!(2(N-k))!} = \sum_{k=0}^{N} E_{2k} (-1)^{N-k} \cdot \binom{2N}{2k} = 0$$

Multiplying by $(-1)^N$

$$\sum_{k=0}^{N} E_{2k} (-1)^k \cdot \binom{2N}{2k} = 0$$

For $n = 1 : E_2 - 1 = 0$ so $E_2 = 1$
For $n = 2 : E_4 - \binom{4}{2} E_2 + 1 = 0$, $E_4 - 6 + 1 = 0$ so $E_4 = 5$
For $n = 3 : E_6 - \binom{6}{4} E_4 + \binom{6}{2} E_2 - 1 = 0$, $E_6 - 75 + 15 - 1 = 0$ so, $E_6 = 61$
For $n = 4 : E_8 - \binom{8}{6} E_6 + \binom{8}{4} E_4 - \binom{8}{2} E_2 + 1$, $E_8 - 1708 + 350 - 28 + 1 = 0$ so, $E_8 = 1385$

13. Find the series expansion of $\dfrac{e^z-1}{z}$ about zero and determine its radius of convergence. Consider $f(z)=\dfrac{z}{e^z-1}$ and let

$$f(z)=\sum_{k=0}^{\infty}\frac{a_k}{k!}z^k$$

be its power series expansion about zero. What is the radius of convergence? Show that

$$0=a_0+\binom{n+1}{1}a_1+...+\binom{n+1}{n}a_n$$

Using the fact that $f(z)+\frac{1}{2}z$ z is an even function show that $a_k=0$ for k odd and $k>1$. The numbers $B_{2n}=(-1)^{n-1}a_{2n}$ are called Bernoulli numbers for $n\geq 1$. Calculate $B_2, B_4...B_n$

Using $e^z=\sum z^n/n!$ we get

$$\frac{e^z-1}{z}=\sum_{n=0}^{\infty}\frac{z^n}{(n+1)!}$$

The power series converges everywhere since $z=0$ is a removable singularity. Thus the radius of convergence is ∞.

For $f(z)=\dfrac{z}{e^z-1}$, it's analytic everywhere except where $e^z-1=0$. Thus where $z=2\pi ik$ but the closet singularity to the origin (besides the origin) is $k=1$ so the radius of convergence is $R=2\pi$

Note that

$$\left(\frac{e^z-1}{z}\right)\cdot f(z)=1$$

Inserting the series we get

$$\left(\sum_{n=0}^{\infty}\frac{z^n}{(n+1)!}\right)\left(\sum_{k=0}^{\infty}\frac{a_k}{k!}z^k\right)=1$$

The Cauchy product for these series and equating coefficients of z^N for $N\geq 1$ we get

$$\sum_{k=0}^{N}\frac{a_k}{k!}\cdot\frac{1}{(N-k+1)!}=0$$

Multiplying through by $(N+1)!$

$$\sum_{k=0}^{N}a_k\cdot\frac{(N+1)}{k!(N-k+1)!}=\sum_{k=0}^{N}a_k\binom{N+1}{k}=0$$

Since $f(z) + \frac{1}{2}z$ is even that means all odd power terms, other than the z^1 term, must vanish. So $a_3 = a_5 = a_7 = ... = 0$. We also can say that $a_1 + \frac{1}{2} = 0$ and so $a_1 = -\frac{1}{2}$

(I actually can't be bothered to show the work for the Bernoulli numbers so here's the results)

$$B_2 = \frac{1}{6} \qquad B_4 = -\frac{1}{30} \qquad B_6 = \frac{1}{42} \qquad B_8 = -\frac{1}{30} \qquad B_8 = \frac{5}{66}$$

14. Find the power series expansion of $\tan z$ about $z = 0$, expressing the coefficients in terms of Bernoulli numbers. (Hint: use Exercise 13 and the formula $\cot 2z = \frac{1}{2}\cot z - \frac{1}{2}\tan z$.)

Recall that the following expansion

$$\frac{z}{e^z - 1} = \sum_{k=0}^{\infty} \frac{B_k}{k!} z^k$$

Since $B_1 = -\frac{1}{2}$ and $B_{2k+1} = 0$ for $k > 1$ we get for

$$\frac{z}{e^z - 1} = 1 - \frac{z}{2} + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!} z^{2k}$$

Using another identity

$$\tan z = \frac{e^{2zi} - 1}{i(1 + e^{2zi})}$$

Which gives us

$$\cot z = i\frac{e^{2zi} + 1}{e^{2zi} - 1} = i\frac{e^{2zi} - 1 + 2}{e^{2zi} - 1} = i + \frac{2i}{e^{2zi} - 1}$$

Multiplying by z we get our series

$$z\cot z = zi + \frac{2zi}{e^{2zi} - 1} = zi + 1 - \frac{2zi}{2} + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)!}(2zi)^{2k}$$

Simplifying and dividing by z we get (written oddly for convenience)

$$\cot z = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{(-1)^k B_{2k}}{(2k)!} \frac{(2z)^{2k}}{z}$$

Modifying the sum slightly we get

$$2\cot 2z = \frac{1}{z} + 2\sum_{k=1}^{\infty} \frac{(-1)^k B_{2k}}{(2k)!} \frac{(4z)^{2k}}{2z}$$

We then use $\tan z = \cot z - 2\cot 2z$

$$\cot z - 2\cot 2z = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{(-1)^k B_{2k}}{(2k)!} \frac{(2z)^{2k}}{z} - \left(\frac{1}{z} + 2\sum_{k=1}^{\infty} \frac{(-1)^k B_{2k}}{(2k)!} \frac{(4z)^{2k}}{2z} \right)$$

After some magic algebraic manipulation that you just have to trust me on

$$\tan z = \sum_{k=1}^{\infty} \frac{(-1)^k 2^{2k}(1-2^{2k}) B_{2k}}{(2k)!} z^{2k-1}$$

4.3 Zeros of an analytic function

1. Let f be an entire function and suppose that there is a constant M, $R > 0$, and an integer $n \geq 1$ such that $|f(z)| \leq M|z|^n$ for $|z| > R$. Show that f is a polynomial of degree $\leq n$.

Fix $r > R$ and choose a point on the circle $|z_0| = r$. Thus we have $|f(z_0)| \leq Mr^n$ by the Cauchy estimate for $k \geq 0$ we get

$$\left| f^{(k)}(0) \right| \leq \frac{k!}{r^k} \max_{|z|=r} |f(z_0)| \leq k! M^{r-k}$$

Since f is entire we have the following power series expansion

$$f(z) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} z^k$$

If $k > n$ then the right hand side goes to 0 and $r \to \infty$, thus $f^{(k)}(0) = 0$ and every term $k > n$ is zero and we get

$$f(z) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} z^k$$

Which is a polynomial of degree $\leq n$

2. Give an example to show that G must be assumed to be connected in Theorem 3.7.

3.7 Theorem
Let G be a connected open set and let $f : G \to \mathbb{C}$ be an analytic function. Then the following are equivalent statements:

(a) $f \equiv 0$

(b) there is a point a in G such that $f^{(n)}(a) = 0$ for each $n \geq 0$

(c) $\{z \in G : f(z) = 0\}$ has a limit point in G

Let G be a disconnected set such that

$$G = \{z : \Re(z) < 0\} \cup \{z : \Re(z) > 0\}$$

And a function f such that

$$f(z) = \begin{cases} 0, & \Re(z) < 0 \\ 1, & \Re(z) > 0 \end{cases}$$

Clearly (b) is satisfied because we choose a point (say $z = -1$) and thus $f^{(n)}(-1) = 0$ for each $n \geq 0$ but (a) is clearly not satisfied. Thus G must be connected.

3. Find all entire functions f such that $f(x) = e^x$ for x in $\mathbb{R}$

Define a new function

$$g(z) := f(z) - e^z$$

Which is 0 for all $z \in \mathbb{R}$. Since g is also entire and $\mathbb{C}$ is a connected and open set. the $g(z) \equiv 0$ and thus $f(z) - e^z \equiv 0$ and thus the only function that agrees with $f(x)$ is

$$f(z) = e^z$$

4. Prove that $e^{z+a} = e^z e^a$ by applying Corollary 3.8

3.8 Corollary. If f and g are analytic on a region G then $f \equiv g$ iff $\{z \in G : f(z) = g(z)\}$ has a limit point in G.

Define two functions

$$f(z) := e^{z+a} \quad g(z) := e^z e^a$$

We know that $f(x) = g(x)$ for all $x \in \mathbb{R}$. Since f and g are entire functions and since the real numbers form a set with limit points in $\mathbb{C}$, the set $\{z \in \mathbb{C} : f(z) = g(z)\}$ has a limit point in $\mathbb{C}$. By Corollary 3.8, since f and g agree on $\mathbb{R}$ we get $f(z) = g(z)$ for all $z \in \mathbb{C}$

5. Prove that $\cos(a+b) = \cos a \cos b - \sin a \sin b$ by applying Corollary 3.8.

(I feel like I'm missing something but isn't this just the exact same problem as the one before?)
Fix an arbitrary $b \in \mathbb{C}$ and define two functions of a

$$f(a) := \cos(a + b) \quad g(a) := \cos a \cos b - \sin a \sin b$$

We know that $f(x) = g(x)$ for all $x \in \mathbb{R}$. Since f and g are entire functions and since the real numbers form a set with limit points in $\mathbb{C}$, the set $\{z \in \mathbb{C} : f(z) = g(z)\}$ has a limit point in $\mathbb{C}$. By Corollary 3.8, since f and g agree on $\mathbb{R}$ we get $f(z) = g(z)$ for all $z \in \mathbb{C}$

6. Let G be a region and suppose that $f : G \to \mathbb{C}$ is analytic and $a \in G$ such that $|f(a)| \leq |f(z)|$ for all z in G. Show that either $f(a) = 0$ or f is constant.

Suppose for the first case $f(a) \neq 0$. Meaning there is no $z_0 \in G$ such that $f(z_0) = 0$ because

$$0 \leq |f(a)| \leq |f(z_0)| = 0$$

So, the function

$$g(z) = \frac{1}{f(z)}$$

Is analytic in G. Thus there exists $a \in G$ such that $|g(a)| \geq |g(z)|$ for all z in G. By the maximum modulus theorem, $g(z)$ must be constant and thus $f(z)$ is constant.

7. Give an elementary proof of the Maximum Modulus Theorem for polynomials.

Let $p : G \to \mathbb{C}$ be a polynomial with an interior point $z_0 \in G$ such that for all $z \in G$ $|p(z_0)| \geq |p(z)|$. For contradiction, assume that p is non-constant. We can rewrite the polynomial around z_0

$$p(z) = p(z - z_0 + z_0) = p(z_0) + (z - z_0)^n q(z)$$

Where n is the degree of the polynomial and $q(z)$ is another polynomial where $q(z_0) \neq 0$. To describe all points near z_0 choose a small $r > 0$ such that $z = z_0 + re^{it}$ and we get.

$$p(z) = p(z_0) + r^n e^{itn} q(z)$$

Performing a similar expansion of $q(z)$ around z_0 in terms of a new polynomial $q_1(z)$ we get that $q_1(z)$ is of order r^{n+1} and since we chose points very close to z we can rewrite as

$$p(z) = p(z_0) + r^n e^{itn} q(z_0) + O(r^{n+1})$$

Since $q(z) \neq 0$ we can choose t such that for some constant $c > 0$

$$e^{itn} q(z_0) = c\, p(z_0)$$

Thus

$$p(z) = p(z_0)(1 + cr^n) + O(r^{n+1})$$

For sufficiently small r, the higher-order terms are negligible, and thus

$$|p(z)| \approx |p(z_0)(1 + cr^n)| = |p(z_0)|(1 + cr^n) > |p(z_0)|$$

Which contradicts $|p(z_0)| \geq |p(z)|$ meaning that p is either constant, or z_0 is not an interior point.

8. Let G be a region and let f and g be analytic functions on G such that $f(z)g(z) = 0$ for all a in G. Show that either $f(z) \equiv 0$ or $g(z) \equiv 0$.

Assume that $f \not\equiv 0$. Then there exists $z_0 \in G$ such that $f(z_0) \neq 0$. Since the zeros of an analytic function are isolated, there exists some small r such that $f(z) \neq 0$ for all $z \in B(z_0, r) \subset G$. Thus $g(z) = 0$ for all $z \in B(z_0, r)$. The zero set of g thus has a limit point in G and thus $g \equiv 0$.
Similarly, if we assume $g \not\equiv 0$ then $f \equiv 0$

9. Let $U : \mathbb{C} \to \mathbb{R}$ be a harmonic function such that $U(z) \geq 0$ for all z in $\mathbb{C}$. Prove U is constant.

Since U is on the whole plane, it must have a harmonic conjugate and thus there exits a function

$$f := U + iV$$

Where U is the real part of f which is an entire function. To show that next consider

$$g(z) = e^{-f(z)}$$

Which is also an entire function and $|g(z)| \leq 1$. Since g is bounded and entire, it must be constant. Thus

$$|e^{-f(z)}| = e^{-U(z)}$$

is also constant

10. Show that if f and g are analytic functions on a region G such that $\overline{f}g$ is analytic then either f is constant or $g \equiv 0$

Suppose $g \not\equiv 0$. Let G_0 be the set such that

$$G_0 = \{z \in G : g(z) \neq 0\}$$

Then since f and $\overline{f}g$ are both analytic in G_0, $\overline{f}$ must also be analytic. By the Cauchy–Riemann equations, if f and $\overline{f}$ are both analytic, then f is constant in G_0.
Now suppose f is non-constant. Thus $\overline{f}$ can't be analytic. Then we can't assume that $g \not\equiv 0$ and thus. g must be the zero function.

4.4 The index of a closed curve

1. Prove Proposition 4.3.

4.3 Proposition. If γ and σ are closed rectifiable curves having the same initial points then

(a) $n(\gamma, a) = -n(-\gamma, a)$ for every $a \notin \{\gamma\}$

(b) $n(\gamma + \sigma, a) = n(\gamma, a) + n(\sigma, a)$

Since γ and σ have the same initial points. Then we can say that

$$\gamma(t) : [0,1] \to \mathbb{C} \quad \sigma(t) : [0,1] \to \mathbb{C} \quad (\gamma(0) = \gamma(1) = \sigma(0) = \sigma(1))$$

Recall the definition of the winding number of γ around a

$$n(\gamma, a) = \frac{1}{2\pi i} \int_\gamma (z-a)^{-1} dz$$

(a) Thus the definition for $-\gamma$ is

$$n(-\gamma, a) = \frac{1}{2\pi i} \int_{-\gamma} (z-a)^{-1} dz$$

Recall that $\int_{-\gamma} f = -\int_\gamma f$, and thus

$$n(-\gamma, a) = -n(\gamma, a)$$

(b) Define the sum of the paths as

$$(\gamma + \sigma)(t) = \begin{cases} \gamma(2t) & 0 \le t \le 1/2 \\ \sigma(2t-1) & 1/2 \le t \le 1 \end{cases}$$

Thus we get

$$\begin{aligned} n(\gamma + \sigma, a) &= \frac{1}{2\pi i} \int_{(\gamma+\sigma)} \frac{1}{z-a}\, dz \\ &= \frac{1}{2\pi i} \int_0^1 \frac{(\gamma+\sigma)'(t)}{(\gamma+\sigma)(t) - a}\, dt \\ &= \frac{1}{2\pi i} \int_0^{1/2} \frac{(\gamma+\sigma)'(t)}{(\gamma+\sigma)(t) - a}\, dt + \frac{1}{2\pi i} \int_{1/2}^0 \frac{(\gamma+\sigma)'(t)}{(\gamma+\sigma)(t) - a}\, dt \\ &= \frac{1}{2\pi i} \int_0^{1/2} \frac{\gamma'(2t)}{\gamma(2t) - a}\, dt + \frac{1}{2\pi i} \int_{1/2}^1 \frac{\sigma'(2t+1)}{\sigma(2t+1) - a}\, dt \\ &= \frac{1}{2\pi i} \int_0^1 \frac{\gamma'(t)}{\gamma(t) - a}\, dt + \frac{1}{2\pi i} \int_0^1 \frac{\sigma'(t)}{\sigma(t) - a}\, dt \end{aligned}$$

And thus

$$n(\gamma + \sigma, a) = n(\gamma, a) + n(\sigma, a)$$

2. Give and example of a closed rectifiable curve γ in $\mathbb{C}$ such that for any integer k there exists a point $a \notin \{\gamma\}$ with $n(\gamma, a) = k$

(We're assuming k is finite because if k was infinite then γ would not be rectifiable)
Consider the curve

$$\gamma_n : [n-1, n] \to \mathbb{C} \quad \gamma_n(t) = n - ne^{2\pi i(t-n+1)}$$

In other words, for some integer n the path is a circle that starts and ends at 0 with radius n and center at n. Thus for $k > 0$ we have the path $\gamma : [0 \to k]$ such that

$$\gamma = \sum_{n=1}^{k} \gamma_n$$

Which is made of k nested circles all tangent at $z = 0$. Thus it contains points with winding numbers 1 through k.
For a winding number of 0 choose $a = -1$.

3. Let $p(z)$ be a polynomial of degree n and let $R > 0$ be sufficiently large so that p never vanishes in $\{z : |z| \geq R\}$. If $\gamma(t) = Re^{it},\ 0 \leq t \leq 2\pi$ show that

$$\int_\gamma \frac{p'(z)}{p(z)}\, dz = 2\pi i n$$

Write $p(z)$ as

$$p(z) = c \prod_{k=1}^{n} (z - a_k)^{m_k}$$

Where m_k is the multiplicity of each zero. So we get $m_1 + m_2 ... m_n = n$, the degree of our polynomial. Recall that $\frac{p'}{p} = (\log(p))'$. Thus we get for some constant c.

$$\frac{p'}{p} = \left(\log(c) + \sum_{k=1}^{n} m_k \log(z - a_k) \right)'$$

Evaluating the derivative we get

$$\frac{p'}{p} = 0 + \sum_{k=1}^{n} \frac{m_k}{z - a_k}$$

So

$$\int_\gamma \frac{p'(z)}{p(z)} = \int_\gamma \sum_{k=1}^{n} \frac{m_k}{z - a_k} = 2\pi i \sum_{k=1}^{n} m_k = 2\pi i n$$

4. Fix $w = re^{i\theta} \neq 0$ and let γ be a rectifiable path in $\mathbb{C} - \{0\}$ from 1 to w. Show that there is an integer k such that $\int_\gamma z^{-1}\,dz = \log r + i\theta + 2\pi ik$

Let $\gamma : [0,1] \to [1,w]$. Our integral becomes

$$\int_0^1 \frac{\gamma'(t)}{\gamma(t)}\,dt$$

Which is $\log(\gamma(t))$. Thus we get

$$\int_0^1 \frac{\gamma'(t)}{\gamma(dt)}\,dt = \log(\gamma(t))|_0^1 = \log(\gamma(1)) - \log(\gamma(0))$$

Next, choose a branch of the log such that for some integer k, $1 = e^{2\pi ik}$. We get

$$\begin{aligned}\log(\gamma(1)) - \log(\gamma(0)) &= \log(w) - \log(1) = \log(re^{i\theta}) - \log(e^{2\pi ik}) \\ &= \log r + i\theta + 2\pi i\end{aligned}$$

4.5 Cauchy's Theorem and Integral Formula

1. Suppose $f : G \to \mathbb{C}$ is analytic and define $\varphi : G \times G \to \mathbb{C}$ by $\varphi(z,w) = [f(z) - f(w)](z-w)^{-1}$ if $z \neq 0$ and $\varphi(z,z) = f'(z)$. Prove that φ is continuous and for each fixed $w, z \to \varphi(z,w)$ is analytic.

Since f is analytic, then f' is also analytic so for $z = w$ φ must be continuous.
For $z \neq w$ we have $z - w \neq 0$ and thus, φ is composed of analytic functions meaning it must be continuous.

Proving analyticity
Fixing w, since f is analytic, we expand $f(z)$ around w

$$f(z) = \sum_{n=0}^{\infty} a_n(z-w)^n \quad f(w) = a_0$$

Where

$$f'(z) = \sum_{n=0}^{\infty} a_n(z-w)^{n-1}\,n$$

Thus we get

$$\varphi = \frac{f(z) - f(w)}{z-w} = \sum_{n=0}^{\infty} a_n(z-w)^{n-1}\,n$$

2. Give the details of the proof of Theorem 5.6.

Define a function $\varphi(w,z)$ such that

$$\varphi(z,w)\begin{cases}\frac{f(z)-f(w)}{z-w}, & z\neq w\\ f'(z) & z=w\end{cases}$$

For some fixed a we can do the following.

$$\frac{f(z)}{z-a}=\frac{f(z)-f(a)}{z-a}+\frac{f(a)}{z-a}=\varphi(z,a)+\frac{f(a)}{z-a}$$

Integrating both sides over some curve γ_k

$$\int_{\gamma_k}\frac{f(z)}{z-a}\,dz=\int_{\gamma_k}\varphi(z,a)\,dz+f(a)\int_{\gamma_k}\frac{1}{z-a}\,dz$$

Since φ is analytic everywhere on G, then for all $a\in G$

$$\int_{\gamma_k}\varphi(z,a)\,dz=0$$

Next taking the sum overall curves γ_k

$$f(a)\sum_{k=1}^{m}\int_{\gamma_k}\frac{1}{z-a}\,dz=\sum_{k=1}^{m}\int_{\gamma_k}\frac{f(z)}{z-a}\,dz$$

Taking the first integral

$$f(a)2\pi i\sum_{k=1}^{m}n(\gamma_k,a)=\sum_{k=1}^{m}\int_{\gamma_k}\frac{f(z)}{z-a}\,dz$$

Rearranging

$$f(a)\sum_{k=1}^{m}n(\gamma_k,a)=\sum_{k=1}^{m}\frac{1}{2\pi i}\int_{\gamma_k}\frac{f(z)}{z-a}\,dz$$

3. Let $B_{\pm}=\overline{B}(\pm1,\frac{1}{2})$, $G=B(0,3)-(B_+\cup B_-)$. Let γ_1, $\gamma_2\,\gamma_3$ be curves whose traces are $|z-1|=1$, $|z+1|=1$, $|z|=2$ respectively. Give γ_1, γ_2 and γ_3 orientations such that $n(\gamma_1,w)+n(\gamma_2,w)+n(\gamma_3,w)=0$ for all w in $\mathbb{C}-G$.

Choose $w\in\mathbb{C}-G$ such that, $|w>3|$. Then

$$n(\gamma_1,w)+n(\gamma_2,w)+n(\gamma_3,w)=0+0+0=0$$

Choose $w\in B_+$ such that, $|w>3|$. Then

$$n(\gamma_1,w)+n(\gamma_2,w)+n(\gamma_3,w)=0\implies n(\gamma_1,w)+0+n(\gamma_3,w)=0$$

So

$$n(\gamma_1, w) = -n(\gamma_3, w)$$

Thus γ_1 and γ_3 have opposite orientations. Similarly if we choose $w \in B_-$, then we know γ_2 and γ_3 have opposite orientations. Thus if we choose an orientation for γ_3 then γ_1 and γ_2 should both have the opposite orientation.

4. Show that the Integral Formula follows from Cauchy's Theorem.

Recall Cauchy's Theorem (5.7). We get

$$\sum_{k=1}^{m} \int_{\gamma_k} f\, dz = 0$$

From problem 2, define φ similarly. Then we get that for one of our curves γ we have.

$$\int_\gamma \varphi(z, a)\, dz = 0$$

$$\int_\gamma \frac{f(z) - f(a)}{z - a}\, dz = 0$$

$$\int_\gamma \frac{f(z)}{z - a}\, dz = f(a) \int_\gamma \frac{1}{z - a}\, dz$$

Recall by the definition of the winding number

$$\int_\gamma \frac{f(z)}{z - a}\, dz = n(\gamma, a) f(a) 2\pi i$$

Thus we get the integral formula

$$n(\gamma, a) f(a) = \frac{1}{2\pi i} \int \frac{f(z)}{z - a}\, dz$$

(This problem feels like using an intercontinental ballistic missile to swat a fly...)

5. Let γ be a closed rectifiable curve in $\mathbb{C}$ and $a \notin \{\gamma\}$. Show that for $n \geq 2$ $\int_\gamma (z - a)^{-n}\, dz = 0$

Using the following (**Corollary 5.9**)

$$f^{(k)}(a) n(\gamma, a) = \frac{k!}{2\pi i} \int_\gamma \frac{f(z)}{(z - a)^{(k+1)}}\, dz$$

We get that $f(z) \equiv 1$ and thus $f^{(k-1)}(a) \equiv 0$ for $k \geq 2$. Thus we get

$$0 = \frac{(k-1)!}{2\pi i} \int_\gamma \frac{1}{(z - a)^k}\, dz$$

Multiplying out the constants

$$0 = \int_\gamma \frac{1}{(z - a)^k}\, dz$$

6. Let f be analytic on $D = B(1,0)$ and suppose $|f(z)| \leq 1$ for $|z| < 1$. Show that $|f'(0)| \leq 1$.

Use the same corollary

$$f^{(k)}(a)n(\gamma, a) = \frac{k!}{2\pi i} \int_\gamma \frac{f(z)}{(z-a)^{(k+1)}}\, dz$$

Where $k = 1$, $a = 0$, and γ is the path $|z| = r < 1$. Thus $n(\gamma, a) = 1$ and we get

$$f'(0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z^2}\, dz$$

Taking the modulus of both sides

$$|f'(0)| = \left|\frac{1}{2\pi i}\right| \left|\int_\gamma \frac{f(z)}{z^2}\, dz\right|$$

$$\leq \frac{1}{2\pi} \left|\frac{1}{n(\gamma, a)}\right| \int_\gamma \left|\frac{f(z)}{z^2}\right| |dz|$$

$$\leq \frac{1}{2\pi} \sup_{|z|=r} \left|\frac{f(z)}{z^2}\right| \cdot L_\gamma$$

$$\leq \frac{1}{2\pi} \frac{1}{r^2} \sup_{|z|=r} |f(z)| \cdot (2\pi r) \leq \frac{1}{r}$$

Finally sending r to 1 we get that

$$|f'(0)| \leq 1$$

7. Let $\gamma(t) = 1 + e^{it}$ for $0 \leq t \leq 2\pi$. Find $\int_\gamma \left(\frac{z}{z-1}\right)^n dz$ for all positive integers n.

Rewriting the integral we get

$$\int_\gamma \frac{z^n}{(z-1)^{([n-1]+1)}}$$

Using the corollary once again, we have that $f(z) = z^n$, $a = 1$, $n(\gamma, a) = 1$ and $k = n - 1$. Thus

$$f^{(n-1)}(1) \cdot 1 = \frac{(n-1)!}{2\pi i} \int_\gamma \frac{z^n}{(z-1)^{([n-1]+1)}}$$

$$n! = \frac{(n-1)!}{2\pi i} \int_\gamma \frac{z^n}{(z-1)^{([n-1]+1)}}$$

$$2\pi i n = \int_\gamma \frac{z^n}{(z-1)^{([n-1]+1)}}$$

8. Let G be a region and suppose that $f_n : G \to \mathbb{C}$ is analytic for each $n \geq 1$. Suppose that $\{f_n\}$ converges uniformly to a function $f : G \to \mathbb{C}$. Show that f is analytic.

Since f_n converges uniformly to f and f_n is analytic, f must at least be continuous. Choose some arbitrary triangular path $T \subset G$. Since f_n is analytic for all f_n we have

$$\int_T f_n = 0$$

Thus letting $n \to \infty$ we get

$$\int_T f = \lim_{n\to\infty} 0 = 0$$

Since f is continuous, and T was an arbitrary triangular path. We get that f is analytic.

9. Show that if $f : \mathbb{C} \to \mathbb{C}$ is a continuous function such that f is analytic off $[-1, 1]$ then f is an entire function.

It would suffice to show that for any triangle $T \subset \mathbb{C}$, then $\int_T f = 0$.
Case 1 Suppose our arbitrary triangle doesn't touch the interval. Then $\int_T f = 0$ trivially.
Case 2 Next suppose the triangle touches the line at exactly the corner. Then we can cut the base of the triangle so we get a much smaller triangle. suppose we let the triangle go ε small. Since f is continuous then there exists $M > 0$ such that for all $z \in T_\varepsilon$ we have $\sup f(z) < M$ Thus our integral becomes (and using ML inequality)

$$\left| \int_{T_\varepsilon} f \right| < M\varepsilon$$

Letting $\varepsilon \to 0$ we get our triangle is analytic.
Case 3 Suppose that one of the triangle's edges is fully contained in the interval. Then we can cut the triangle from the tip opposite of the interval and thus all those triangles have integrals evaluated at zero

$$\int_{T_n} f = 0 \quad (\forall n \in \mathbb{N}) \implies \lim_{n\to\infty} \int_{T_n} f = \int_T f = 0$$

$$\lim_{n\to\infty} \int_{T_n} f$$

Case 4 Suppose the triangle is merely pierced by the interval. Draw lines from whichever end of the interval is in the interior to the three other vertices of the triangle. We get a triangle is composed of triangles from case 2 and 3 and thus the the integral over all triangles is still zero.

Case 5 Suppose that the triangle is split by the interval. Draw lines from where the interval intersects the boundary of the triangle to the same edge. We get the triangle is composed up triangles from case 2 and 3. Thus our integral over the triangle is still zero.
Thus, from all the cases we get that for every triangle T in $\mathbb{C}$ we get that

$$\int_T f = 0$$

And thus f is entire.

10. Use Cauchy's integral formula to prove the Cayley-Hamilton Theorem: If A is an $n \times n$ matrix over $\mathbb{C}$ and $f(z) = \det(Iz - A)$ is the characteristic polynomial of A then $f(A) = 0$.

Recall the integral formula

$$f(a) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - a}\, dz$$

For the set of all eigenvalues $\{z_A\}$ of A choose $R > \sup |z_A|$ and $\gamma = Re^{it}$ for $0 \leq t \leq 2\pi$. Substituting everything in we get

$$f(A) = \frac{1}{2\pi i} \int_\gamma \frac{\det(Iz - A)}{Iz - A}\, dz$$

By the definition of the adjugate matrix

$$(Iz - A) \cdot \operatorname{adj}(Iz - A) = \det(Iz - A) \cdot I$$

So we get

$$f(A) = \frac{1}{2\pi i} \int_\gamma \frac{\det(Iz - A)}{\frac{\det(Iz - A)}{\operatorname{adj}(Iz - A)}}\, dz$$

$$f(A) = \frac{1}{2\pi i} \int_\gamma \operatorname{adj}(Iz - A)\, dz$$

Since every entry of $\operatorname{adj}(Iz - A)$ is a polynomial in z we get that each entry is analytic and thus is zero. Meaning $f(A)$ is the zero matrix.

4.6 The homotopic version of Cauchy's Theorem and simple connectivity

1. Let G be a region and let $\sigma_1, \sigma_2 : [0, 1] \to G$ be rectifiable curves $\sigma_1(t) \equiv a, \sigma_2(t) \equiv b$. Show that if γ is closed rectifiable curve in G and $\gamma \sim \sigma_1$ then $\gamma \sim \sigma_2$. (Hint: connect a and b by a curve.)

Since $\gamma \sim \sigma_1$, there is a continuous function $\Gamma : [0,1] \times [0,1] \to G$ such that

$$\begin{cases} \Gamma(s,0) = \gamma(s), \quad \text{and} \quad \Gamma(s,1) = \sigma_1(s) & 0 \le s \le 1 \\ \Gamma(0,t) = \Gamma(1,t) & 0 \le t \le 1 \end{cases}$$

Consider the curve $\varphi(t,s) : [s,t] \times [0,1] \to G$ such that

$$\varphi(s,0) = \sigma_1(s) \quad \text{and} \quad \varphi(s,1) = \sigma_2(s)$$
$$\varphi(0,t) = \varphi(1,t)$$

Then we have $\sigma_1 \sim \sigma_2$. And since $\sim$ is an equivalence relation, we get that

$$\gamma \sim \sigma_1 \text{ and } \sigma_1 \sim \sigma_2 \implies \gamma \sim \sigma_2$$

2. Show that if we remove the requirement "$\Gamma(0,t) = \Gamma(1,t)$ for all t" from Definition 6.1 then the curve $\gamma_0 = e^{2\pi i t}$, $0 \le t \le 1$ is homotopic to the constant curve $\gamma_1 \equiv (1)$ in the region $G = \mathbb{C} - \{0\}$.

Consider the curve $\Gamma : [0,1] \times [0,1] \to G$

$$\Gamma(s,t) = e^{2\pi i t(1-s)}$$

We get that

$$\Gamma(s,0) = e^{2\pi i t} = \gamma_0(s)$$
$$\Gamma(s,1) = e^0 = 1 \equiv \gamma_1(s)$$

Thus we get $\Gamma \cap \{0\} = \varnothing$. We unwound the unit circle to show that when we drop the requirement that Γ has to be closed.

3. Let $\mathscr{G}$ = all rectifiable curves in G joining a to b and show that Definition 6.11 gives an equivalence relation on $\mathscr{G}$.

6.11 Definition If $\gamma_0, \gamma_1 : [0,1] \to G$ are two rectifiable curves in G such that $\gamma_0(0) = \gamma_1(0) = a$ and $\gamma_0(1) = \gamma_1(1) = b$ then γ_0 and γ_1 are *fixed-end-point.*

1. **Reflexivity**
   Trivially if $\gamma_0 = \gamma_0$ then γ_0 is fixed-end-point with itself.
2. **Symmetry**
   Again, trivial if $\gamma_0 \sim \gamma_1$ then $\gamma_0(0) = \gamma_1(0) = a$ and $\gamma_0(1) = \gamma_1(1) = b$ and thus $\gamma_1 \sim \gamma_0$
3. Transitivity If $\gamma_0 \sim \gamma_1$ and $\gamma_1 \sim \gamma_2$, then we get

   $$\gamma_0(0) = \gamma_1(0) = a,\ \gamma_0(1) = \gamma_1(1) = b$$
   $$\gamma_1(0) = \gamma_2(0) = a,\ \gamma_1(1) = \gamma_2(1) = b$$

   And thus

   $$\gamma_0(0) = \gamma_2(0) = a,\ \gamma_0(1) = \gamma_2(1) = b$$
   $$\gamma_0 \sim \gamma_2$$

4. Let $G = \mathbb{C} - \{0\}$ and show that every closed curve in G is homotopic to a closed curve whose trace is contained in $\{z : |z| = 1\}$

For some arbitrary curve $\sigma : [0,1] \to \{z : |z| = 1\}$, consider the curve

$$\sigma_\gamma(t) = \frac{\gamma(t)}{|\gamma(t)|}$$

Next, consider the closed curve $\Gamma : [0,1] \times [0,1] \to G$ Such that

$$\begin{cases} \Gamma(s,0) = \gamma(s), \quad \text{and} \quad \Gamma(s,1) = \sigma_\gamma(s) & 0 \le s \le 1 \\ \Gamma(0,t) = \Gamma(1,t) & 0 \le t \le 1 \end{cases}$$

Since $|\Gamma| > 0$ we get $\Gamma \subset G$ and thus γ and σ_γ are homotopic

5. Evaluate the integral.

$$\int_\gamma \frac{dz}{z^2+1}$$

where $\gamma(\theta) = 2|\cos 2\theta| e^{i\theta}$ for $2 \le \theta \le 2\pi$

The function $\frac{1}{z^2+1}$ is analytic in $\mathbb{C} - \{\pm i\}$. Looking at the graph of γ looks like a flower with 4 petals and we see that $n(\gamma, i) = 1$ and $n(\gamma, -i) = 1$. Using partial fraction decomposition we get

$$\frac{1}{2i}\int_\gamma \frac{dz}{z-i} - \frac{1}{2i}\int_\gamma \frac{dz}{z+i}$$

Multiplying my $\dfrac{\pi}{\pi}$ and using the Cauchy integral formula we get

$$\frac{\pi}{2\pi i}\int_\gamma \frac{dz}{z-i} - \frac{\pi}{2\pi i}\int_\gamma \frac{dz}{z+i}$$

$$= \pi\, n(\gamma, i) - \pi\, n(\gamma, -i) = \pi - \pi = 0$$

Thus

$$\int_\gamma \frac{dz}{z^2+1} = 0$$

6. Let $\gamma(\theta) = \theta e^{i\theta}$ for $0 \le t \le 2\pi$ and $\gamma(\theta) = 4\pi - \theta$ for $2\pi \le t \le 4\pi$. Evaluate

$$\int_\gamma \frac{dz}{z^2+\pi^2}$$

The function $\frac{1}{z^2+\pi^2}$ is analytic in $\mathbb{C} - \{\pm \pi i\}$.

Looking at the graph of γ looks like a nautilus and we see that $n(\gamma, i\pi) = 0$ and $n(\gamma, -i\pi) = 1$. Using partial fraction decomposition we get

$$\frac{1}{2\pi i}\int_\gamma \frac{dz}{z - \pi i} - \frac{1}{2\pi i}\int_\gamma \frac{dz}{z + \pi i}$$

Using the Cauchy integral formula we get

$$= n(\gamma, i\pi) - n(\gamma, -i\pi)$$

$$= 0 - 1 = -1$$

Thus

$$\int_\gamma \frac{dz}{z^2 + \pi^2} = -1$$

7. Let $f(z) = [(z - \frac{1}{2} - i) \cdot (z - 1 - \frac{3}{2}i) \cdot (z - 1 - \frac{i}{2}) \cdot (z - \frac{3}{2} - i)]^{-1}$ and let γ be the polygon $[0, 2, 2 + 2i, 2i, 0]$. Find $\int_\gamma f$

The function f is analytic in $\mathbb{C} - \{(\frac{1}{2} + i), (1 + \frac{3}{2}i), (1 + \frac{i}{2}), (\frac{3}{2} + i)\}$. Looking at γ we get the following winding numbers

$$n(\gamma, \tfrac{1}{2} + i) = 1 \quad n(\gamma, 1 + \tfrac{3}{2}i) = 1 \quad n(\gamma, 1 + \tfrac{i}{2}) = 1 \quad n(\gamma, \tfrac{3}{2} + i) = 1$$

By partial fraction decomposition we get (just trust me bro)

$$\int_\gamma \frac{2}{z - (\frac{3}{2} + i)}\, dz - \int_\gamma \frac{2}{z - (\frac{1}{2} + i)}\, dz + \int_\gamma \frac{2i}{z - (1 + \frac{3}{2}i)}\, dz - \int_\gamma \frac{2i}{z - (1 + \frac{1}{2}i)}\, dz$$

Using the winding numbers we get

$$2\pi i(2 - 2 + 2i - 2i) = 0$$

8. Let $G = \mathbb{C} - \{a, b\}$, $a \neq b$, and let γ be the curve in the figure below.

(a) Show that $n(\gamma, a) = n(\gamma, b) = 0$

(b) Convince yourself that γ is not homotopic to zero.

Choose a point w in G that is outside of γ so that the winding number is clearly 0. Then Draw a line from a to w. For every intersection of the line and γ consider the direction γ is heading in. If γ is heading counter clockwise, add one to your count (starting at 0 because $n(\gamma, w) = 0$) and if it's clockwise, subtract one from the total. For any path from w to a we always get a running total of 0. Therefore $n(\gamma, a) = 0$. The same can be shown for b. (If it's hard to interpret if the path is going clockwise or counterclockwise through our path, approximate both paths at each intersection with a vector and put the vectors into a matrix. If the determinant of the matrix is positive, add one to the running total, if it's negative, subtract one.)

a and b look like similar points relative to γ. They both look like if one was added to G then γ could be maneuvered around the other point and be homotopic to 0. Suppose we label the curve as

$$\gamma : AB^{-1}A^{-1}B$$

Meaning the curve goes around a clockwise, b counterclockwise, a counter, b clockwise. Notice that if all A terms were removed then we would get $\gamma = B^{-1}B \cong 0$ (approximately speaking we are assuming a homomorphism from matrix multiplication to winding numbers around points which seems like enough to kickstart a proof but as of now I'm fully convinced).

9. Let G be a region and let γ_0 and γ_1 be two closed smooth curves in G. Suppose $\gamma_0 \sim \gamma_1$ and Γ satisfies (6.2). Also suppose that $\gamma_t(s) = \Gamma(s,t)$ is smooth for each t. If $w \in \mathbb{C} - G$ define $h(t) = n(\gamma_1, w)$ and show that $h : [0,1] \to \mathbb{Z}$ is continuous.

Stating **6.2** explicitly we get that $\Gamma : [0,1] \times [0,1] \to G$ such that

$$\Gamma(s,0) = \gamma_0(s) \text{ and } \Gamma(s,1) = \gamma_1(s)$$

$$\Gamma(0,t) = \Gamma(1,t)$$

Define $h(t)$ as

$$h(t) = \frac{1}{2\pi i} \int_{\gamma_t} \frac{dz}{z-w}$$

Performing substitution $z \to \Gamma(s,t)$ we get

$$\frac{1}{2\pi i} \int_0^1 \frac{\partial_s \Gamma(s,t)}{\Gamma(s,t) - w}\, ds$$

Since $\gamma_t(s)$ is smooth on a compact space $[0,1]^2$, it means it's uniformly continuous and thus for all $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|t_0 - t_1| < \delta \implies |\gamma_{t_0}(s) - \gamma_{t_1}(s)| < \varepsilon$$

for all $s \in [0,1]$. Thus to show h is continuous. Fix $t_0 \in [0,1]$. Then for all t near t_0

$$\begin{aligned} |h(t) - h(t_0)| &\leq \frac{1}{2\pi} \int_0^1 \left| \frac{\partial_s \Gamma(s,t)}{\Gamma(s,t) - w} - \frac{\partial_s \Gamma(s,t_0)}{\Gamma(s,t_0) - w} \right| ds \\ &\leq \frac{1}{2\pi} \sup_{s \in [0,1]} \frac{\partial_s \Gamma(s,t)}{\Gamma(s,t) - w} - \frac{\partial_s \Gamma(s,t_0)}{\Gamma(s,t_0) - w} \underset{t \to t_0}{\to} 0 \end{aligned}$$

Since h thus all t near t_0 get mapped to the same discrete space in $\mathbb{Z}$, it's constant and thus h is continuous.

10. Find all possible values of

$$\int_\gamma \frac{dz}{1+z^2}$$

where γ is any closed rectifiable curve in $\mathbb{C}$ not passing through $\pm i$

Similar to problem 5 we get that our

$$\int_\gamma \frac{dz}{z^2+1} = \pi(n(\gamma, i) - n(\gamma, -i))$$

This the value of the integral depends on the winding numbers about $\pm i$. Thus since $n(\gamma, w) : [0, 1] \to \mathbb{Z}$ for some fixed w we get that

$$\int_\gamma \frac{dz}{z^2+1} = k\pi \quad \text{for some } k \in \mathbb{Z}$$

11. Evaluate

$$\int_\gamma \frac{e^z - e^{-z}}{z^4}$$

were γ is one of the curves depicted below. (Justify your answer)

We must write the integral in terms of winding numbers about 0. Using the integral formula for $f(z) = e^z - e^{-z}$, $a = 0$, $k = 3$ and $f^{(3)}(0) = e^0 + e^{-0} = 2$ we get

$$\frac{2\pi i}{3!}(2)n(\gamma, 0) = \int_\gamma \frac{e^z - e^{-z}}{(z-0)^{3+1}}$$

$$= \frac{2\pi i}{3} n(\gamma, 0)$$

The winding numbers for each curve are 1, 2, and 2 respectively. So we get our integrals are

$$\frac{2\pi i}{3!} \cdot 1, \quad \frac{2\pi i}{3!} \cdot 2, \quad \frac{2\pi i}{3!} \cdot 2$$

Answer justification
Yep... Looks good to me :)

4.7 Counting zeros, the Open Mapping Theorem

1. Show that if $f : G \to \mathbb{C}$ is analytic and γ is a rectifiable curve in G, then $f \circ \gamma$ is also a rectifiable curve.

(Taken from 4.1 problem 18b)
Take any fine δ partition $P = \{t_k\}$ of γ such that our integral is

$$\int_\gamma f = \sum_{k=1}^{m} f(\tau_k)[\gamma(t_k) - \gamma(t_k - 1)]$$

By triangle inequality

$$\left|\int_\gamma f\right| = \left|\sum_{k=1}^{m} f(\tau_k)[\gamma(t_k) - \gamma(t_k - 1)]\right| \leq \sum_{k=1}^{m} |f(\tau_k)||\gamma(t_k) - \gamma(t_k - 1)| = \int_\gamma |f||dz|$$

Passing the limit we get

$$\int_\gamma |f||dz| = \int_a^b |f(\gamma(t))||\gamma'(t)||dt| \leq (\sup|f(z)|)(|\gamma(a)| - |\gamma(b)|) = V(\gamma) \sup_{z\in\gamma} |f(z)|$$

Thus taking the supremum over all partitions we get.

$$\text{len}(f \circ \gamma) \leq M \ \text{len}(\gamma) < \infty$$

And thus it's rectifiable

2. Let G be open and suppose that γ is a closed rectifiable curve in G such that $\gamma \approx 0$. Set $r = d(\{\gamma\}, \partial G)$ and $H = \{z \in \mathbb{C} : n(\gamma, z) = 0\}$. (a) Show that $\{z : d(z, \partial G) < \frac{1}{2}r\} \subset H$. (b) Use part (a) to show that if $f : G \to \mathbb{C}$ is analytic then $f(z) = \alpha$ has at most a finite number of solutions z such that $n(\gamma, z) \neq 0$

(a) Trivially $\text{int}(\gamma) \not\subset H$ thus we only need to consider points outside of γ. Redefine r such that

$$r = \inf_{z\in\{\gamma\}, w\in\partial G} (|z - w|)$$

Thus we get that

$$\gamma \cap \{z : d(z, \partial G) < r\} = \varnothing$$

(b) Consider the set Suppose that $a \in G$ such that $f(a) = \alpha$ and $n(\gamma, a) \neq 0$. We get from a that $d(a, \partial G) \geq \frac{r}{2}$. Assume for contraction that there were infinitely many isolated solutions to $f(z) = \alpha$ there exists a rectifiable curve σ such that $n(\sigma, \alpha) = \infty$. However σ isn't rectifiable. A contradiction. Thus there are only finitely many solutions.

3. Let f be analytic in $B(a, R)$ and suppose that $f(a) = 0$. Show that a is a zero of multiplicity m iff $f^{(m-1)}(a) = ... = f(a) = 0$ and $f^{(m)}(a) \neq 0$

Suppose that a is a zero of multiplicity m. There there's an analytic function g, where $g(a) \neq 0$ such that we we can rewrite f as

$$f(z) = g(z)(z-a)^m$$

Differentiating f $k < m$ times always contains a $(z-a)$ term in the expansion. However evaluating $f^{(m)}(a)$ we get

$$f^{(m)}(a) = m!\, g(a) \neq 0$$

To prove the converse, suppose that all $m-1$ derivatives vanish at $z = 0$, but the mth derivative doesn't. Since f is analytic, then we have the Taylor expansion of f around a.

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(z-a)^n$$

Splitting the sum

$$f(z) = \sum_{n=0}^{m-1} \frac{f^{(n)}(a)}{n!}(z-a)^n + \sum_{n=m}^{\infty} \frac{f^{(n)}(a)}{n!}(z-a)^n$$

The first part of the sum is zero. Re-indexing the sum we get

$$f(z) = (z-a)^m \sum_{j=0}^{\infty} \frac{f^{(m+n)}(a)}{(m+n)!}(z-a)^n$$

The series converges to $\dfrac{f^{(m)}(a)}{m!} \neq 0$ on $B(a, R)$ and thus we have an analytic function g we can write as $f(z) = (z-a)^m g(z)$ for some $g(a) \neq 0$. By definition f has a zero of multiplicity m

4. Suppose that $f : G \to \mathbb{C}$ is analytic and one-to-one, show that $f'(z) \neq 0$ for any z in G

Assume for contradiction there exists $a \in G$ such that $f'(a) = 0$. By the previous problem we get that for some $m \geq 2$ we have an analytic function g such that

$$f(z) = f(a) + (z-a)^m g(z)$$

Where $h(a) \neq 0$. By the open mapping theorem and continuity of h, we see that G is an open set and we can take an open ball $B(a, r) \subset G$ such that $h(z) \neq 0$ for all $z \in B(a, r)$. The mapping $z \mapsto (z-a)^m$ for $m \geq 2$ is not an injective map because it maps a small neighborhood of a as m-to-one. Thus $f(z) = (z-a)^m g(z)$ is not one-to-one. A contradiction. Thus our assumption that there exists a such that $f'(a) = 0$ was wrong.

5. Let X and Ω be metric spaces and suppose that $f : X \to \Omega$ is one-to-one and onto. Show that f is an open map iff f is a closed map (a function f is a closed map if it takes closed sets onto closed sets).

Recall the key bijection lemma such that for any $A \subseteq X$ and that

$$f(X/A) = \Omega/f(A)$$

Suppose that f is an open map. Let $C \subseteq X$ be a closed set. Then X/C is open, thus we get

$$f(X/C) = \Omega/f(C)$$

which is also open. Thus $f(C)$ is closed and f is a closed map. Similarly if we assume f is a closed map and that C is open, then X/C is closed and thus $f(C)$ is open. Thus f is also an open map

6. Let $P : \mathbb{C} \to \mathbb{R}$ be defined by $P(z) = \Re(z)$; show that P is an open map but not a closed map. (Hint: Consider the set $F = \{z : \Im z = (\Re z)^{-1}$ and $\Re(z) \neq 0\}$)

Suppose we have an open set $A \subset \mathbb{C}$. Then for every $z \in A$ there exists $r > 0$ such that $B(z, r) \subset A$. Then $P(A)$ is also an open set because for every point $f(z) \in \mathbb{R}$ we have the interval $(f(z) - r, f(z) + r) \subset f(A)$. Consider the set $F = \{z : \Im z = (\Re z)^{-1}\}$ which is closed because F^C is open. We see that $f(F) = \mathbb{R}/\{0\}$ which is not closed. Thus f is an open map but not a closed one.

7. Use theorem 7.2 to give another proof of the Fundamental Theorem of Algebra

7.2 Theorem Let G be a region and let f be an analytic function on G with zeros $a_1, ...a_m$ (repeated according to multiplicity). If γ is a closed rectifiable curve in G which does not pass through any point a_k and if $\gamma \approx 0$ then

$$\frac{1}{2\pi i} \int_\gamma \frac{f'(z)}{f(z)}\, dz = \sum_{k=1}^{m} n(\gamma, a_k)$$

Suppose $p(z)$ is a polynomial of degree n. Thus we can write

$$p(z) = \sum_{k=1}^{n} a_k z^k$$

Since f has finitely many zeros, we can choose $R > 0$ so large that every no zeros lies on the curve

$$\gamma_R(t) = Re^{it}$$

Factor out the leading term $p(z) = a_n z^n q(z)$ and we get

$$q(z) = 1 + \frac{b_1}{z} + \frac{b_2}{z^2} + ... + \frac{b_n}{z^n} \qquad b_k := \frac{a_{n-k}}{a_n}$$

Thus we get

$$\frac{p'(z)}{p(z)} = \frac{n}{z} + \frac{q'(z)}{q(z)}$$

Then

$$\int_{\gamma_R} \frac{p'(z)}{p(z)}\,dz = \int_{\gamma_R} \frac{n}{z}\,dz + \int_{\gamma_R} \frac{q'(z)}{q(z)}\,dz = 2\pi i n + \int_{\gamma_R} \frac{q'(z)}{q(z)}\,dz$$

To use ML-inequality consider

$$\left|\frac{q'(z)}{q(z)}\right| = \left|\frac{1}{z}\right| \left|\frac{b_1 z^{n-1} + 2b_2 z^{n-2} + ... + nb_n}{z^n + b_1 z^{n-1} + ... + b_n}\right| =: \left|\frac{1}{z}\right| |g(z)|$$

Notice that as $z \to \infty$ we get that $g(z) \to 0$ By ML-inequality

$$\left|\int_{\gamma_R} \frac{q'(z)}{q(z)}\,dz\right| \leq (2\pi R) \sup_{z \in R} \left|\frac{q'(z)}{q(z)}\right| \leq 2\pi R \left|\frac{1}{R}\right| |g(z)| \to 0$$

Thus, letting $R \to \infty$ we get that

$$\frac{1}{2\pi i} \int_{\gamma_R} \frac{p'(z)}{p(z)}\,dz = n$$

And thus

$$\sum_{k=1}^{n} n(\gamma, a_k) = n$$

Thus, for any polynomial $p(z)$ of degree n, there are n solutions (with multiplicity) to $p(z)$

4.8 Goursat's Theorem

Holy fucking shit thank GOD there are no problems assigned to this section

5 Chapter 5: Singularities

5.1 Classification of singularities

1. Each of the following functions f has an isolated singularity at $z = 0$. Determine its nature; if it is a removable singularity define $f(0)$ so that f is analytic at $z = 0$; if it is a pole find the singular part if it is an essential singularity determine $f(\{z : 0 < |z| < \delta\})$ for arbitrarily small values of δ

(a) $f(z) = \frac{\sin(z)}{z}$

The only point of interest is $z = 0$. So we know it's an isolated singularity. Using the Taylor expansion of $\sin z$ we get that

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - ...$$

By definition of the removable singularity we have

$$\lim_{z\to 0}(z - 0)f(0) = 0$$

Thus f is a removable singularity

(b) $f(z) = \frac{\cos z}{z}$

Using the Taylor expansion of cos we get

$$\frac{\cos z}{z} = \frac{1}{z} - \frac{z}{2!} + \frac{z^3}{4!}$$

Which has a simple pole at $z = 0$ with the $1/z$ term

(c) $f(z) = \frac{\cos z - 1}{z}$

Again using the Taylor expansion, we get

$$\frac{\cos z - 1}{z} = -\frac{z}{2!} + \frac{z^3}{4!}$$

Which seems analytic and by the definition of the removable singularity we have

$$\lim_{z\to 0}(z - 0)f(0) = 0$$

Thus f is a removable singularity

(d) $f(z) = \exp(z^{-1})$

Again, Taylor expansion

$$1 + \frac{1}{z} + \frac{1/z^2}{2!} + \frac{1/z^3}{3!} + ...$$

This is not a pole or a removable singularity, so it's an essential singularity. Fix arbitrary $\delta > 0$. We have that for any $z \in \{z : 0 < |z| < \delta\}$ We get

$$e^{1/z} = e^{\overline{z}/|z|}$$

$$|e^{\overline{z}} e^{1/\delta}| < |e^{\overline{z}/|z|}| < |e^{\overline{z}}|$$

Which gives us that $f(\{z : 0 < |z| < \delta\}) = \mathbb{C} - \{0\}$

(e) $f = \frac{\log(z+1)}{z^2}$

Using the Taylor expansion we get

$$\frac{1}{z} - \frac{1}{2} + \frac{z}{3} - \frac{z^2}{4}...$$

Which has a simple pole with the $1/z$ term

(f) $f(z) = \frac{\cos(1/z)}{1/z}$

You know the drill...

$$z - \frac{1/z}{2!} + \frac{1/z}{4!}...$$

Infinitely many terms that are simple poles, not a removable singularity or a pole. Must be essential. Fix $\delta > 0$ and rewrite $f(z)$

$$\frac{e^{i/z} + e^{-i/z}}{2/z} = z\frac{e^{i\overline{z}/|z|} + e^{-i\overline{z}/|z|}}{2}$$

$$z\frac{e^{i\overline{z}} + e^{-i\overline{z}}}{2} \cdot e^{|z|}$$

We get that

$$z\frac{e^{i\overline{z}} + e^{-i\overline{z}}}{2} \cdot e^0 < z\frac{e^{i\overline{z}} + e^{-i\overline{z}}}{2} \cdot e^{|z|} < z\frac{e^{i\overline{z}} + e^{-i\overline{z}}}{2} \cdot e^{\delta}$$

$$z\cos\overline{z} < z\cos\overline{z} \cdot e^{|z|} < z\cos\overline{z} \cdot e^{\delta}$$

Letting $\delta \to 0$ then by squeeze theorem we get that

$$f(\{z : 0 < |z| < \delta\}) = \text{the image of } z\cos\overline{z} \text{ on } \mathbb{C} = \mathbb{C}$$

for every δ

(g) $f(z) = \frac{z^2+1}{z(z-1)}$

Rewrite $f(z)$

$$f(z) = \frac{z^2 + (z-z) + 1}{z^2 - z} = \frac{z^2 - z}{z^2 - z} + \frac{z+1}{z^2 - z} = 1 + \frac{1}{z-1} - \frac{1}{z}$$

With the $1/z$ term, it's a simple pole

(h) $f(z) = (1 - e^z)^{-1}$

I think this one is especially evil since we can just rewrite $f(z)$ as

$$f(z) = -\frac{1}{z}\left(\frac{z}{1-e^z}\right) = -\frac{1}{z}\left(\sum_{n=0}^{\infty} \frac{B_n z^n}{n!}\right) = \sum_{n=0}^{\infty} \frac{B_n z^{n-1}}{n!}$$

$$= -\frac{1}{z} + \text{analytic part}$$

Thus a simple pole at $z = 0$

(i) $f(z) = z \sin \frac{1}{z}$

This time we won't use Taylor series, instead we will use the Maclaurin series centered about 0.

$$f(z) = z \sin \frac{1}{z} = z \sum_{n=0}^{\infty} \frac{(-1)^n}{z^{2n+1}(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n}{z^{2n}(2n+1)!}$$

An essential singularity. By Casorati-Weierstrass, we get that

$$f(\{z : 0 < |z| < \delta\}) = \mathbb{C}$$

(j) $f(z) = z^n \sin \frac{1}{z}$

It's the same as (i). Essential singularity

Give the partial fraction expansion of $r(z) = \dfrac{z^2 - 1}{(z^2 + z + 1)(z-1)^2}$

Rewriting as our preferred format we get

$$r(z) = \frac{Az + B}{z^2 + z + 1} + \frac{C}{z-1} + \frac{D}{(z-1)^2}$$

$$r(z) = \frac{(Az+B)(z-1)^2}{(z^2+z+1)(z-1)^2} + \frac{C(z^2+z+1)(z-1)}{(z^2+z+1)(z-1)^2} + \frac{D(z^2+z+1)}{(z^2+z+1)(z-1)^2}$$

$$r(z) = \frac{(Az+B)(z^2-2z+1)}{(z^2+z+1)(z-1)^2} + \frac{C(z^3-1)}{(z^2+z+1)(z-1)^2} + \frac{D(z^2+z+1)}{(z^2+z+1)(z-1)^2}$$

$$\begin{aligned} z^3: &\quad A + C = 0 \\ z^2: &\quad -2A + B + D = 1 \\ z: &\quad A - 2B + D = 0 \\ c: &\quad B - C + D = 1 \end{aligned}$$
$$\implies A = 0, B = \frac{1}{3}, C = 0, D = \frac{2}{3}$$

Thus we get

$$r(z) = \frac{1}{3(z^2 + z + 1)} + \frac{2}{3(z-1)^2}$$

Give the details of the derivation of (1.17) from (1.16).

Since $g(0) = 0$ we get no constant term in the power series expansion and thus

$$g(z) = \sum_{n=1}^{\infty} B_n z^n \quad \text{for } B_n = \frac{g^{(n)}(0)}{n!}$$

Next we substitute $g(z) = f(a + \frac{1}{z}) \implies g(\frac{1}{z-a}) = f(z)$. Our sum becomes

$$\sum_{n=1}^{\infty} B_n (z-a)^{-n}$$

Evaluating B_n, by the integral formula

$$f_1^{(n)}(a) = -\frac{n!}{2\pi i} \int_{\gamma_1} \frac{f_1(z)}{(z-a)^{n+1}} dz$$

$$\implies B_n = \frac{1}{2\pi i} \int \frac{f_1(z)}{(z-a)^{1-n}} dz = a_{-n}$$

So we get

$$\sum_{n=1}^{\infty} a_{-n} (z-a)^{-n}$$

Let $f(z) = \dfrac{1}{z(z-1)(z-2)}$; give the Laurent expansion of $f(z)$ in each of the following annuli: (a) ann (0; 0, 1); (b) ann (0; 1, 2); (c) ann (0; 2, ∞).

Doing partial fraction decomposition we get

$$f(z) = \frac{1}{z} \left(\frac{1}{z-2} - \frac{1}{z-1} \right)$$

Recall the geometric series

$$\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n \quad \text{for } |w| < 1$$

(a) $0 < |z| < 1$

We have $|z| < 1$ so we can rewrite $f(z)$ as

$$f(z) = \frac{1}{z}\left(\frac{1}{1-z} - \frac{1}{2}\frac{1}{1-\frac{z}{2}}\right)$$

$$= \frac{1}{z}\left(\sum_{n=0}^{\infty} z^n - \frac{1}{2}\sum_{n=0}^{\infty} z^n \left(\frac{1}{2}\right)^n\right)$$

$$= \sum_{n=0}^{\infty} z^{n-1}\left[1 - \left(\frac{1}{2}\right)^{n+1}\right]$$

Re-indexing for $k = n - 1$

$$= \sum_{k=-1}^{\infty} z^k \left[1 - \left(\frac{1}{2}\right)^{k+2}\right]$$

(b) $1 < |z| < 2$

We have $\left|\frac{1}{z}\right| < 1$ and $\left|\frac{z}{2}\right| < 1$ so we can write $f(z)$ as

$$f(z) = -\frac{1}{z}\left(\frac{1}{2}\frac{1}{1-\frac{z}{2}} + \frac{1}{1-\frac{1}{z}}\right)$$

$$f(z) = -\frac{1}{z}\left(\frac{1}{2}\sum_{n=0}^{\infty} z^n \left(\frac{1}{2}\right)^n + \sum_{n=0}^{\infty}\left(\frac{1}{z}\right)^n\right)$$

$$= \frac{1}{2}\sum_{n=0}^{\infty} z^{n-1}\left(\frac{1}{2}\right)^n + \sum_{n=0}^{\infty} z^{-n-1}$$

It's not really feasible to combine these sums so re-indexing them separately we get

$$= -\sum_{k=-1}^{\infty} z^k \left(\frac{1}{2}\right)^{k+2} - \sum_{-\infty}^{k=-2} z^k$$

(c) $2 < |z| < \infty$

We have $\frac{2}{|z|} < 1$ so we can write $f(z)$ as

$$f(z) = \frac{1}{z^2}\left(\frac{1}{1-\frac{2}{z}} - \frac{1}{1-\frac{1}{z}}\right)$$

$$= \frac{1}{z^2}\left(\sum_{n=0}^{\infty}\left(\frac{1}{z}\right)^n 2^n - \sum_{n=0}^{\infty}\left(\frac{1}{z}\right)^n\right)$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^{n-2} 2^n - \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^{n-2}$$

Re-indexing

$$\sum_{-\infty}^{k=-2} \left(2^{2-k} - 1\right) z^k$$

5. Show that $f(z) = \tan z$ is analytic in $\mathbb{C}$ except for simple poles at $z = \frac{\pi}{2} + n\pi$ for each integer n. Determine the singular part of f at each of these poles.

Recall that

$$\tan z = \frac{\sin z}{\cos z}$$

Meaning f is analytic everywhere except for $\cos z = 0$ or when $a_n = \frac{\pi}{2} + n\pi$. These are simple poles because $\frac{d}{dz} \cos z = -\sin z$ and $-\sin a_n \neq 0$ and thus $\tan z$ has simple poles at each a_n with residue -1 so we get

$$\tan z = -\frac{1}{z - a_n} + \text{analytic part}$$

6. If $f : G \to \mathbb{C}$ is analytic except for poles show that the poles of f cannot have a limit point in G.

Assume for contradiction that $a \in G$ is a limit point of the poles of f. By definition of a limit point for all $\varepsilon > 0$ there exists N such that for all $N > n$ we get

$$|a_N - a| < \varepsilon$$

Meaning that the ball $B(a, \varepsilon)$ contains infinitely many other poles and thus a can't be an isolated singularity, meaning f must be analytic at a. But f can't be analytic at a because there is no open ball in G that is analytic around a. Thus there is no limit point of the poles of f.

7. Let f be an isolated singularity at $z = a$ and suppose $f \not\equiv 0$. Show that if either (1.19) or (1.20) holds for some s in $\mathbb{R}$ then there is some integer m such that (1.19) holds if $s > m$ and (1.20) holds if $s < m$

1.19

$$\lim_{z \to a} |z - a|^s |f(z)| = \infty$$

1.20

$$\lim_{z \to a} |z - a|^s |f(z)| = 0$$

By Casorati-Weierstrass we get that the limit is undefined for an essential singularity. Thus it must be a pole or a removable singularity. Suppose (1.20) holds. Then the limit holds for any n larger than s. Recall that for some analytic function $g(z)$ where $g(a) \neq 0$ and for some integer k we get

$$f(z)(z-a)^n = g(z)(z-a)^k$$

Rewriting the original limit we get

$$\lim_{z\to a} f(z)(z-a)^s = \lim_{z\to a} g(z)(z-a)^{s-n+k}$$

Thus we get that $\lim f(z)(z-a)^s = 0$ for $s > n-k$ and ∞ for $s < n-k$
Now suppose (1.19) holds. Similarly the limit holds for any n less than s. Again we can rewrite f such that for some analytic function $h(z)$ where $h(a) \neq 0$ and for some integer k we get

$$f(z)(z-a)^n = h(z)(z-a)^{-k}$$

and the proof follows similarly from the first part.

8. Let $f, a,$ and m be as is in Exercise 7. Show: (a) $m = 0$ iff $z = a$ is a removable singularity and $f(a) \neq 0$; (b) $m < 0$ iff $z = a$ is a removable singularity and f has a zero at $z = a$ of order $-m$; (c) $m > 0$ iff $z = a$ is a pole of f of order m

(a) Suppose $m = 0$, For $s = 1$ we get a removable singularity and evaluating the limit for $s = 1$ we get that $f(a) \neq 0$ by definition of the removable singularity. Next suppose that $f(a) \neq 0$ and $z = a$ is a removable singularity of f. Trivially if we choose $m = 0$ then for $s > m$ the limit is 0 and for $s < m$ the limit is ∞.

(b) Suppose that $m < 0$ then we have for $s = 0$ and $s = 1$ the limit evaluates to 0. So f has a removable singularity at $z = a$. Define a new function

$$g(z) = \frac{f(z)}{(z-a)^m}$$

We see that

$$\lim_{z-a} g(z)(z-a) = 0$$

By Exercise 7 we get that $g(a) \neq 0$ and thus a is a zero of order $-m$. Next suppose the converse, we get an analytic function of $g(z) = f(z)(z-a)^{-m}$ where $g(a) \neq 0$ we get that for $s > -m$ the limit evaluates to 0 and for $s < -m$ the limit evaluates to ∞ meaning that $m < 0$

(c) Suppose that $m > 0$ then we get $\lim_{z\to a} f(z)(z-a)^m \neq 0$. Meaning for some analytic $g(z)$ we get

$$f(z) = \frac{g(z)}{(z-a)^m}$$

Which classifies it as a pole of order m. For the converse we for for some s

$$\lim_{z\to a}(z-a)^s f(z) = \lim_{z\to a}(z-a)^{s-m}(z-a)^m f(z) = \begin{cases} 0, & s > m \\ \infty, & s < m \end{cases}$$

By Exercise 7 we get $m > 0$

9. A function f has an essential singularity at $z = a$ iff neither (1.19) nor (1.20) holds for any real s

Suppose $z = a$ is an essential singularity for f Assume for contradiction that 1.19 and 1.20 held for some essential singularity at f. By Casorati-Weierstrass we get that for some $\delta > 0$ we get that

$$f\{z : 0 < |z-a| < \delta\} = \mathbb{C} \quad \text{(potentially excluding } z = a\text{)}$$

meaning from some direction θ_0 we get

$$\lim_{(z\to a)_{\theta_0}} |f(z)||(z-a)|^s = \infty \cdot 0 \quad \text{(undefined)}$$

Thus 1.19 and 1.20 don't hold for an essential singularity. Cor the converse, suppose $z - a$ is not an essential singularity. Meaning it must be a pole or a removable singularity. By problems 7 and 8 we get that 1.20 and 1.19 hold.

10. Suppose that f has an essential singularity at $z = a$. Prove the following strengthened version of the Casorati-Weierstrass theorem. If $c \in \mathbb{C}$ and $\varepsilon > 0$ we are given then for each $\delta > 0$ there is a number $\alpha : |c-\alpha| < \varepsilon$ such that $f(z) = \alpha$ has infinitely many solutions in $B(a, \delta)$.

Unfortunately I could not solve this problem for the life of me and I couldn't find any satisfying solutions online that either cited the chapter or were less than 8 whole pages... I will come back to this with more incite if I can cite later theorems that don't cite this theorem.

11. Give the Laurent series development of $f(z) = \exp\left(\frac{1}{z}\right)$. Can you generalize this result?

Use the Taylor expansion of e^z to get

$$e^{\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{z^n} \frac{1}{n!} = \sum_{n=-\infty}^{0} \frac{z^n}{(-n)!}$$

Suppose there exists some Taylor expansion of f. We get

$$f(z) = \sum_{n=0}^{\infty} a_n z^n$$

Then the Laurent expansion of $f\left(\frac{1}{z}\right)$ gets us

$$\sum_{n=1}^{\infty} a_n \left(\frac{1}{z}\right)^n$$

12. (a) Let $\lambda \in \mathbb{C}$ and show that

$$\exp\left\{\frac{1}{2}\lambda\left(z + \frac{1}{z}\right)\right\} = a_0 + \sum_{n=1}^{\infty} a_n \left(z^n + \frac{1}{z^n}\right)$$

for $0 < |z| < \infty$, where $n \geq 0$

$$a_n = \frac{1}{\pi} \int_0^{\pi} e^{\lambda \cos t} \cos nt \, dt$$

Call the function f. Since f is analytic in the punctured plane, then the Laurent coefficient is represented as

$$a_n = \frac{1}{2\pi i} \int_{\gamma} \frac{e^{\lambda/2(z+z^{-1})}}{z^{n+1}} \, dz$$

Where γ is the unit circle parameterized as e^{it} for $0 \leq t \leq 2\pi$ and $dz = ie^{it} \, dt$ we get

$$a_n = \frac{1}{2\pi i} \int_0^{2\pi} \frac{e^{\lambda \cos t}}{e^{it(n+1)}} ie^{it} \, dt$$

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} e^{\lambda \cos t} e^{-itn} \, dt$$

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} e^{\lambda \cos t} \left(\cos nt + i \sin nt\right) \, dt$$

Since cos is an even function about π and sin is odd then it disappears when we integrate from 0 to π and we get

$$a_n = \frac{1}{\pi}\int_0^{\pi} e^{\lambda \cos t}\cos nt\, dt$$

so we get

$$f(z) = \sum_{-\infty}^{\infty} a_n z^n$$

Also considering cos is even about 0 we get that $a_n = a_{-n}$. Using linearity of the sum we get

$$f(z) = \sum_{-\infty}^{\infty} a_n z^n = a_0 + \sum_{-1}^{\infty} a_n z^n + \sum_{-\infty}^{1} a_n z^n$$

$$= a_0 + \sum_{n=1}^{\infty} a_n \left(z^n + \frac{1}{z^n}\right)$$

(b) Similarly show

$$\exp\left\{\frac{1}{2}\lambda\left(z - \frac{1}{z}\right)\right\} = b_0 + \sum_{n=1}^{\infty} b_n\left(z^n + \frac{(-1)^n}{z^n}\right)$$

for $0 < |z| < \infty$ **where**

$$b_n m = \frac{1}{\pi}\int_0^{\pi} \cos(nt - \lambda \sin t)$$

Similar to part (a), the coefficient is defined as

$$\frac{1}{2\pi i}\int_{\gamma} \frac{e^{\lambda/2(z-z^{-1})}}{z^{n+1}}\, dz$$

Where γ is the unit circle parameterized as e^{it} for $0 \le t \le 2\pi$ and $dz = ie^{it}\, dt$ we get

$$\frac{1}{2\pi}\int_0^{2\pi} e^{\lambda/2\cdot(\cos t + i\sin t - \cos(-t) - i\sin(-t))} e^{-itn}\, dt$$

$$\frac{1}{2\pi}\int_0^{2\pi} e^{\lambda/2\cdot(\cos t + i\sin t - \cos(t) + i\sin t)} e^{-itn}\, dt$$

$$\frac{1}{2\pi}\int_0^{2\pi} e^{\lambda\cdot i\sin t} e^{-itn}\, dt$$

$$\frac{1}{2\pi}\int_0^{2\pi} e^{i(\lambda\cdot\sin t - tn)}\, dt$$

$$\frac{1}{2\pi}\int_0^{2\pi} \cos(tn - \lambda \sin t) - i \sin(tn - \lambda \sin t)\ dt$$

Since cos is an even function about π and sin is odd then it disappears when we integrate from 0 to π and we get

$$\frac{1}{\pi}\int_0^{\pi} \cos(tn - \lambda \sin t)\, dt$$

Notice that

$$\begin{cases} n \equiv 0 \mod 2, & b_{-n} = b_n \\ n \equiv 1 \mod 2, & b_{-n} = -b_n \end{cases}$$

And thus we get

$$b_0 + \sum_{n=1}^{\infty} b_n \left(z^n + \frac{(-1)^n}{z^n} \right)$$

13. Let $R > 0$ and $G = \{z : |z > R|\}$; a function $f : G \to \mathbb{C}$ has a removable singularity, a pole, or an essential singularity at infinity if $f(z^{-1})$ has, respectively a removable singularity, a pole, or an essential singularity at $z = 0$. If f has a pole at ∞ the the order of the pole is the order of the pole of $f(z^{-1})$ at $z = 0$.

(a) Prove that an entire function has a removable singularity at infinity iff it is a constant.

(b) prove that an entire function has a pole at infinity of order m iff it is a polynomial of degree m

(c) Characterize those rational functions which have a removable singularity at infinity

(d) Characterize those rational functions which have a pole of order m at infinity.